

Almost-everywhere regularity of stationary integral varifolds

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Abstract

We resolve the almost-everywhere regularity conjecture of Brena, Decio, and De Lellis for stationary integral varifolds of arbitrary positive dimension and codimension in Euclidean open sets. The singular set has zero measure in the dimension of the varifold: near almost every support point, the varifold is a constant positive integer multiple of a smooth embedded minimal submanifold. The corresponding statement also holds on round spheres.

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1 Introduction

A stationary integral varifold is a measure-theoretic minimal surface with integer multiplicity. Its tangent plane exists almost everywhere, but a tangent plane alone need not describe an entire neighborhood. The basic regularity question is whether the points without a smooth minimal neighborhood can carry positive measure. We prove that they have zero m -dimensional measure.

Let $m, n \geq 1$, let $U \subset \mathbb{R}^{m+n}$ be open, and let $E \subset U$ be countably m -rectifiable. Let $\theta : E \rightarrow \mathbb{N} \setminus \{0\}$ be measurable and locally integrable with respect to $\mathcal{H}^m$. Write

$$\mu(A) = \int_{E \cap A} \theta \, d\mathcal{H}^m, \quad P(x) = P_{T_x E}.$$

The associated integral varifold V is stationary if

$$\int \operatorname{tr}(P(x)DX(x)) \, d\mu(x) = 0 \quad \text{for every } X \in C_c^1(U; \mathbb{R}^{m+n}). \quad (1.1)$$

A point $p \in \operatorname{spt} \mu \cap U$ is *regular* if there are a neighborhood W of p , a smooth embedded minimal m -submanifold $\Sigma \subset W$, and an integer $q \geq 1$ such that the restriction of the varifold to W is $q|\Sigma|$. The remaining support points form $\operatorname{Sing} V$.

Theorem 1.1. *Every stationary integral m -varifold in an open subset of $\mathbb{R}^{m+n}$ satisfies*

$$\mathcal{H}^m(\operatorname{Sing} V) = 0.$$

Here $m, n \geq 1$ are arbitrary. No stability, minimizing, orientability, density upper bound, or codimension-one hypothesis is required.

Corollary 1.2 (Round spheres). *Let $m, n \geq 1$ and let V be an integral m -varifold on the full unit round sphere S^{m+n} , stationary for variations tangent to the sphere. A support point is regular if on a spherical neighborhood $V = q|\Sigma|$ for an integer $q \geq 1$ and a smooth embedded minimal m -submanifold Σ of the sphere. If $\operatorname{Sing}_S V$ denotes the remaining support points, then*

$$\mathcal{H}_{\text{ground}}^m(\operatorname{Sing}_S V) = 0.$$

No fixed upper bound on mass, density, or multiplicity is required, nor are stability, orientability, or mod-two cyclicity. The same conclusion holds on every round sphere of positive radius.

Section 9 derives this corollary from Theorem 1.1 by a separate cone argument. That proof adapts the hypersurface first-variation calculation in [17, Lemma 6.1] and rederives it in the present dimensions.

1.1 Background and significance

Allard's regularity theory gives a relatively open dense regular set and, in particular, regularity at density-one points [1]; see also [2, Theorem 23] and [19, Chapter 5, Theorem 6.1 and Corollary 6.3]. Higher integer density presents a different issue: dividing a varifold by its density at a point need not preserve the almost-everywhere lower density bound required by that theorem in a neighborhood. Standard accounts of the first variation, monotonicity, and integral varifolds may be found in [18].

Simon's 2018 notes already record the question whether the singular set must be $\mathcal{H}^m$ -null even when the generalized mean curvature is zero [19, Chapter 5, Remark 6.4]. The measure-zero singular-set assertion is Conjecture 1.2 of Brena, Decio, and De Lellis [5]. Theorem 1.1 proves this conjecture in its full Euclidean scope. They isolate a local criterion in Conjecture 1.3:

a support point with a unique tangent varifold $Q|\Pi|$, where Π is an m -plane and Q a positive integer, should be regular when the mass of $\{\theta \neq Q\}$ in a ball of radius r about it is $o(r^m)$. They then propose approximating the varifold by a smooth minimal graph with the squared-height integral vanishing to every order, followed by a unique-continuation step that upgrades this contact to local support containment [5, Conjectures 1.3–1.5]. Their analysis of multiplicity and height in flat cylinders also provides useful precedents for the local estimates developed here [5, Theorem 1.9 and Lemma 2.3].

Related progress concerns the differentiability of a countable covering. Menne proved second-order rectifiability for integral varifolds of locally bounded first variation [15, Theorem 3.6]. For stationary integral varifolds of dimension at least two, Brena, De Lellis, and Franceschini proved smooth rectifiability and infinite-order approximation at μ -almost every point by one smooth graph [4, Theorems 1.2 and 1.3]. The approximating graph in the latter statement is not asserted to be minimal; Brena–Decio–De Lellis present this as a weaker form of their approximation conjecture [5, Theorem 1.8]. Kolasiński obtains rectifiability of every finite order and an alternative proof of the smooth-rectifiability conclusion, following an architecture he credits to Menne [14, Theorem 1.1, Corollary 10.8, and the historical note in Section 1]. Smooth rectifiability gives a countable smooth covering outside a μ -null set; Theorem 1.1 requires equality with one smooth minimal submanifold throughout a neighborhood of almost every point.

In codimension one, Hirsch and Spolaor prove almost-everywhere regularity for integral currents that are stationary as varifolds and have density at most two at every support point [12, Theorem 1.1]. These current and multiplicity hypotheses distinguish that result from the arbitrary-codimension statement here.

De Lellis, Hirsch, Lihn, and Spolaor construct, in a smooth Riemannian metric, a stationary varifold with a double-plane tangent at an exceptional point and topology accumulating there [8, Theorem 1.1].¹

1.2 The argument

The main estimate compares projected mass and tilt with their leading terms kept signed. In product coordinates $\mathbb{R}^m \times \mathbb{R}^n$, let π be projection onto the first factor, fix a positive integer Q , and let P_{vv} be the vertical block of the tangent projection. With lengths and mass normalized to the working radius, define

$$M = \pi_{\#}\mu - Q \, dy, \quad D = \pi_{\#}(\mathrm{tr} \, P_{vv} \mu).$$

Here $|M|$ denotes the total variation of the signed measure M , and $\tau = |M| + D$ is the positive excess measure in this flat setting. When τ has mass at most d on a fixed enlarged ball, we prove an interior bound of the form

$$\int w(2M - D) \leq d \exp \left[-F_Q \left(\log \frac{1}{d} \right) \right] \quad (1.2)$$

for nonnegative Lipschitz weights with fixed bounds. The explicit function F_Q grows faster than every fixed power of $\log T$ as $T \rightarrow \infty$. Theorem 3.1 gives the precise statement, including the curvature error for a reference surface of sufficiently high finite regularity. Retaining the sign permits a substantially smaller error than an absolute-value estimate.

The proof of (1.2) is an induction on Q . A heat-potential construction turns a positive violation into a positive measure on selected centers and scales. At each selected point it gives

¹In their respective disclosures, Kolasiński reports using Claude AI and Claude Code, mostly Opus 5; the abstract of De Lellis–Hirsch–Lihn–Spolaor credits a crucial suggestion to `gpt-astra-6`. These attributions follow the cited papers’ own descriptions.

an initial sign and a nonnegative cumulative integral along larger scales; the selection includes vertical sides of the stopping boundary. For configurations whose normal heights separate into groups, a stopping argument reduces the estimate to stationary restrictions with smaller integer counts. The resulting comparison controls tilt away from the groups.

Along a selected path, we take a supremum of damped signed-mass averages over larger scales and compare it with a clipped squared-distance moment from an unordered list of at most Q representative normal heights. A weighted combination of this moment and the cumulative contact integral stays below the supremum even when representative heights merge. On an interval without a merger, this comparison forces a quantitative amount of tilt at many scales. Normal stationarity turns that tilt into a flux pairing height displacement with radial tilt. A higher height moment prevents all of that flux from concentrating on sets that have small base volume but large local excess density. At each selected scale this yields a base cube in which a quantitative fraction of the τ -mass remains below a density threshold at all smaller dyadic scales. Packing these nested cubes shows that the union of their full innermost members has small τ -mass. A final charge bounds the contact measure by the mass of that union, contradicting the original violation.

For regularity, fix a point where dilated varifolds converge to a multiplicity- Q plane and the mass of other multiplicities has relative density zero. At each nearby base point, we fit minimal graphs at successive scales, stopping when a prescribed bound on the support height fails. The correction of each fit is closely related to the Jacobi regularization and nonlinear correction of Brena–De Lellis–Franceschini [4, Sections 2 and 6]. A centered first-moment estimate, combined with elliptic regularity, makes neighboring fits agree to second order in their height error. The signed estimate is first used in a contradiction argument for a sequence of failed fits normalized by their height errors. Together with the limiting first-variation identities, its one-sided inequality shows that the limiting squared-height measure is nonzero in every interior subball. Strong control of the centered first moment, together with the integer count Q , then forces that second moment to come from separated projection entries of lower multiplicity on a definite portion of each such ball. The regions influenced by failures therefore have vanishing relative volume.

The compatible fits can now be glued into one fixed center of sufficiently high finite regularity. The weighted integral of the squared mean curvature and the square of its first derivative is bounded by the corresponding squared-height integral. The support height is zero outside the regions influenced by failures. Almgren developed the center-manifold approach to higher-multiplicity regularity [3]; De Lellis–Spadaro construct their center by gluing smoothed local averages [9]. For the inward step we use the comparison of height and energy in the frequency method of this program [11, Section 3.4], including its cutoff form on a center manifold [10, Section 3]. The minimizing-current and Dirichlet-minimizing hypotheses in the Almgren–De Lellis–Spadaro regularity results are unavailable here, so the needed inequalities are derived from stationarity and the signed estimate.

This second use of signed excess takes place at finite radii relative to the fixed center. Interior balls retain the leading signed comparison between projected mass and tilt. A separate radial test controls the thin outer collar by a small multiple of the tilt in the radial direction. It thereby preserves the leading tilt coefficient when the comparison is inserted into the frequency identities. The resulting inward comparison supplies height doubling without assuming it initially. Along a sequence of radii with controlled frequency, the local mass estimate transfers the vanishing relative volume of the failure regions to vanishing relative projected mass. A higher height moment then excludes nonzero height carried by those regions. The support lies on the center in a neighborhood, and stationarity gives constant multiplicity and makes the center a smooth minimal graph.

Organization. Section 2 establishes the first-variation, integer-count, and height estimates. Section 3 states the signed-excess theorem and fixes its parameter hierarchy. Section 4 constructs the heat contacts and reduces separated configurations to lower multiplicities; Section 5 develops the outward comparison and controls mergers; Section 6 closes the integer induction by packing witness scales. Section 7 constructs the center from minimal fits and multiplicity defects. Section 8 derives inward doubling, forces the height to vanish, and proves the Hausdorff-measure conclusion on the entire support. Section 9 proves the round-sphere consequence by the separate cone argument.

1.3 Conventions

All estimates are interior and local. Constants may depend on m, n , an upper bound for the particular integer count Q , and the fixed coordinate geometry. They are independent of the small scale and excess unless stated otherwise. Fixed changes of concentric radii are allowed when positive interior room is retained. A dot denotes differentiation with respect to the logarithm of the radius. Every projection count includes the integer multiplicity. We distinguish a positive tensor from the measure obtained by multiplying it by μ ; projections of the latter to the base are stated when introduced. Section 3 specifies the quantitative hierarchy used in the induction.

2 Local first variation and height estimates

We first collect the local estimates used in the signed comparison. Their constants depend on the dimensions, an upper bound for the integer under consideration, fixed interior enlargement factors, and the indicated bounds for the coordinates. Counts in a normal fiber always include integer multiplicity. A configuration of size Q is therefore an unordered list of Q vectors, with repetitions allowed.

2.1 First variation and integer counts

We begin with the standard stationary monotonicity and mean-value arguments; compare [2, Theorem 9] for monotonicity. We include the derivation to retain the lower mass bound at every point of the support.

Lemma 2.1 (Monotonicity and mean value). *Let V be stationary and integral in an open Euclidean set. If $p \in \text{spt } \mu$ and $\overline{\mathbf{B}_r(p)}$ lies in that set, then*

$$\mu(\mathbf{B}_r(p)) \geq \omega_m r^m. \quad (2.1)$$

If $u \geq 0$ is a C^2 ambient function with $\text{tr}(P_T D^2 u) \geq 0$ for V -almost every (x, T) in the ball, then

$$u(p) \leq \frac{1}{\omega_m r^m} \int_{\mathbf{B}_r(p)} u \, d\mu. \quad (2.2)$$

The same assertion holds for continuous functions satisfying the corresponding weak subharmonicity inequality.

Proof. Put $t = |x - p|$ and let $f \geq 0$ be a smooth decreasing radial cutoff. Testing stationarity with $(x - p)f(t/r)$ and differentiating under the integral gives

$$\frac{d}{d \log r} \left(r^{-m} \int f(t/r) \, d\mu \right) = r^{-m} \int \frac{-f'(t/r)}{rt} |(I - P_T)(x - p)|^2 \, d\mu \geq 0.$$

Approximation of the indicator of $[0, 1)$ proves ball monotonicity. At a point where the rectifiable density is an integer, its limit at zero is at least ω_m . Such points are dense in the

support. Applying the conclusion at points tending to p , with radii increasing to r , proves (2.1) at every support point.

Stationarity with $\varphi \nabla u$ shows $\int \nabla_T u \cdot \nabla_T \varphi \, d\mu \leq 0$ for every nonnegative compactly supported φ . The preceding radial calculation, with its field multiplied by u , has the additional term

$$r^{-m} \int f(t/r) \nabla_T u \cdot (x - p) \, d\mu.$$

It is nonnegative: use the nonnegative radial primitive $\varphi(t) = \int_t^\infty a f(a/r) \, da$ in the weak inequality. Thus $r^{-m} \int_{\mathbf{B}_r(p)} u \, d\mu$ is nondecreasing. At integer-density points its limit is at least $\omega_m u(p)$. Continuity and the same approximation of the center prove (2.2). This argument uses only the weak inequality, and therefore proves the last assertion as well. $\square$

We use a reference disk S of class C^J , where the fixed finite order $J \geq 6$ is chosen sufficiently large for all derivatives below, including the dimension-dependent test order in Theorem 2.4. The disk has a fixed normal tube, small graph slope, and bounded coordinate derivatives through order J . All pulled-back first-variation fields have at least class C^1 ; Gaussian smoothing takes place in the base variables. The same fixed derivative order suffices under rescaling and throughout the induction. The flat case permits an arbitrary vertical translate. Restrict the varifold to a cylinder whose support is separated from its vertical boundary over every compact subset of the base. All fields used below have compact base support and are multiplied by a vertical cutoff equal to one on that support. Consequently this restriction has no vertical boundary term. Fixed enlargements of a working ball are always assumed to remain in this cylinder.

Write a point of the tube as $q = y + z_{\text{phys}}$, where $y = \pi(q) \in S$ and $z_{\text{phys}} \in N_y S$. The second fundamental form is denoted by Π , its trace by H_S , and its shape operators by

$$\langle A_v a, b \rangle = \langle \Pi(a, b), v \rangle, \quad J_z = I - A_{z_{\text{phys}}}.$$

All norms of H_S and $\nabla^\perp H_S$ below are in the physical Euclidean metric. At q , decompose the tangent projection $P = P_{T_q V}$ relative to $T_y S \oplus N_y S$, and set

$$B = I - P_{hh}, \quad e = \text{tr } B = \text{tr } P_{vv}, \quad \lambda = |z_{\text{phys}}|^2 + |H_S|^2 + |\nabla^\perp H_S|^2. \quad (2.3)$$

Thus B is a tensor at q , not a measure on the base. It is positive semidefinite, $0 \leq e \leq m$, and the projection identities imply

$$P_{hv} P_{vh} = B - B^2, \quad |P_{vh} a|^2 \leq B[a, a], \quad |P_{vh}| \leq C\sqrt{e}. \quad (2.4)$$

In physical coordinates the measures on the base are

$$\nu = \pi_\# \mu, \quad M = \nu - Q \text{vol}_S, \quad D = \pi_\#(e\mu), \quad \Lambda = \pi_\#(\lambda\mu), \quad L = 2M - D, \quad \tau = |M| + D + \Lambda. \quad (2.5)$$

For a flat reference we set $\lambda = 0$ and $\Lambda = 0$. In particular a constant sheet offset is not charged as a geometric error in that case.

At the center of the original physical chart, normalize the base coordinates so that their metric is Euclidean there. For the coordinate immersion F , let $g = (DF)^\top DF$, choose the orthonormal horizontal frame $E = DF g^{-1/2}$ using the positive square root, and write $C_y = g^{1/2}$; thus $DF X = EC_y X$. Choose the normal frame by parallel transport along radial geodesics from the same center. This normal frame is the radial gauge for the original chart; its regularity and connection bounds follow from the fixed geometry above. When a physical base radius l is scaled to one, we use base coordinates y/l and height coordinates $\zeta = z_{\text{phys}}/l$

in this fixed normal frame. The fixed coordinate bounds and the normalization at the center give

$$|C_y - I| + |D_{y/l}C_y| + |D_{y/l}E| \leq Cl$$

throughout the original chart. Measures, including the reference volume, are divided by l^m and pushed forward to these coordinates. We retain the symbols in (2.5) for the resulting measures. The function λ is still evaluated in physical units; it is not replaced by $|\zeta|^2$ or by rescaled curvature. A further coordinate ball of radius s consequently has physical radius ls . Define

$$\gamma = l + \sup |z_{\text{phys}}|. \quad (2.6)$$

The reference disks used below have a uniformly small tube, so that J_z and J_z^{-1} are uniformly bounded.

Lemma 2.2 (Curved horizontal first variation). *For a compactly supported tangent field X on S ,*

$$\begin{aligned} & \left| \int \operatorname{div}_S X \, dM - \int B : \nabla^S X \, d\mu \right| \\ & \leq C \int \left\{ |z_{\text{phys}}| e |\nabla^S X| + [(|\nabla^\perp H_S| + |z_{\text{phys}}|) |z_{\text{phys}}| + \sqrt{e} |z_{\text{phys}}|] |X| \right\} d\mu. \end{aligned} \quad (2.7)$$

The integrands containing tensors are evaluated at q , with the base field pulled back by π . In coordinates scaled by l , the right side is bounded by

$$C\gamma \int (e + \lambda)(|\nabla X| + |X|) \, d\mu. \quad (2.8)$$

Here and below a dimensionless scaled test field is used in a scaled identity.

Proof. The nearest-point derivative is $D\pi = J_z^{-1}P_h$. Lift X to the ambient field $\bar{X}(q) = J_z X(y)$. This is the velocity obtained by moving y with velocity X and transporting z_{phys} parallel in the normal connection. Compute at a point in tangent and normal frames whose respective connections vanish there. With $K_z(a) = (\nabla_a A)_{z_{\text{phys}}} X$, the four blocks of its derivative are

$$\begin{aligned} (D\bar{X})_{hh} &= J_z(\nabla X)J_z^{-1} - K_z J_z^{-1}, & (D\bar{X})_{hv} &= -A_v X, \\ (D\bar{X})_{vh} &= \Pi(J_z^{-1}a, J_z X), & (D\bar{X})_{vv} &= 0. \end{aligned}$$

The first term of the horizontal block has trace $\operatorname{div}_S X$. Codazzi in Euclidean space gives

$$\operatorname{tr} K_z = \langle \nabla_X^\perp H_S, z_{\text{phys}} \rangle.$$

For completeness, Codazzi follows by writing $\Pi_{ij} = (\partial_i \partial_j F)^\perp$ in local coordinates and commuting the three derivatives of the immersion; at the chosen point the connection terms are precisely those in the covariant derivative. Since $K_z = O(|z_{\text{phys}}||X|)$ and $J_z^{-1} - I = O(|z_{\text{phys}}|)$,

$$\operatorname{tr}(D\bar{X})_{hh} = \operatorname{div}_S X - \langle \nabla_X^\perp H_S, z_{\text{phys}} \rangle + O(|z_{\text{phys}}|^2 |X|).$$

The symmetric sum of the off-diagonal blocks, paired with $a \in T_y S$ and $v \in N_y S$, is exactly

$$\langle a, J_z^{-1}[A_{z_{\text{phys}}}, A_v]X \rangle.$$

Indeed it equals $\langle a, (J_z^{-1}A_v J_z - A_v)X \rangle$. Its size is $O(|z_{\text{phys}}||a||v||X|)$, including in arbitrary codimension. Contracting it with P_{vh} costs $C\sqrt{e}|z_{\text{phys}}||X|$ by (2.4). Contracting the horizontal block with $I - B$ instead of the identity gives $-B : \nabla X$ and an error at most $C|z_{\text{phys}}|e(|\nabla X| +$

$|X|$). Absorb its last summand in the preceding $\sqrt{e}$ term. Integrating $P : D\bar{X} = 0$ and using $\int_S \operatorname{div}_S X \, d\operatorname{vol}_S = 0$ proves (2.7).

After scaling, the coefficients of $|\nabla X|$ and $|X|$ are respectively

$$|z_{\text{phys}}|e, \quad l[(|\nabla^\perp H_S| + |z_{\text{phys}}|)|z_{\text{phys}}| + \sqrt{e}|z_{\text{phys}}|].$$

Young's inequality bounds them by $C\gamma(e + \lambda)$. Bounded coordinate changes preserve this estimate. $\square$

Lemma 2.3 (Projection and integer counts). *The tangential Jacobian of π is*

$$J_\pi = \frac{\sqrt{\det P_{hh}}}{\det J_z}, \quad |J_\pi - 1| \leq C(e + \lambda). \quad (2.9)$$

There is a nonnegative integer-valued function N on the base, finite almost everywhere, such that

$$\pi_\#(J_\pi \mu) = N \operatorname{vol}_S, \quad |\nu - N \operatorname{vol}_S| \leq C(D + \Lambda). \quad (2.10)$$

The inequalities are measure inequalities on every relatively compact base set. Moreover, for sufficiently small e ,

$$2(1 - J_\pi) - e \leq Ce^2 + C(|z_{\text{phys}}|^2 + |H_S|^2). \quad (2.11)$$

Proof. The squared Jacobian of the orthogonal projection of T onto $T_y S$ is $\det P_{hh}$. Composition with J_z^{-1} proves the formula. If the eigenvalues of B are $b_i \in [0, 1]$, then $1 - \sqrt{\prod_i (1 - b_i)} \leq \sum_i b_i = e$. Also

$$\det J_z = 1 - \langle H_S, z_{\text{phys}} \rangle + O(|z_{\text{phys}}|^2).$$

These facts and Young's inequality prove the bound in (2.9). Expanding the square root one order further gives $\sqrt{\det(I - B)} = 1 - e/2 + O(e^2)$, which proves (2.11). The rectifiable area formula [19, Chapter 3, equation (2.5)] gives the first equality in (2.10), where N counts points of nonzero projection Jacobian with their multiplicities. Its second assertion follows by pushing forward $(1 - J_\pi)\mu$. The part on which the projection has deficient rank has $e \geq 1$ and is included in this error; no absolute-continuity assertion about its pushforward is needed. $\square$

The next elementary analytic observation explains why a scalar weak- L^1 estimate is sufficient here. Both its distributional estimate and the integer count are essential. Its proof uses the Calderón–Zygmund decomposition and cancellation argument [7, Chapter I, Lemmas 1–2], with a localized potential and a harmonic remainder controlled modulo constants.

Lemma 2.4 (Scalar potential estimate). *Suppose a finite scalar measure v on $B_4 \subset \mathbb{R}^m$ satisfies*

$$\partial_j v = \sum_i \partial_i F_{ij} + g_j, \quad \sum_{ij} |F_{ij}|(B_4) + \sum_j |g_j|(B_4) \leq \epsilon. \quad (2.12)$$

Then there is a real constant c such that the density v_{ac} obeys

$$|\{y \in B_2 : |v_{\text{ac}}(y) - c| > t\}| \leq C\epsilon/t \quad (t > 0), \quad (2.13)$$

and, for every smooth test supported in B_2 ,

$$|\langle v - c \, dy, \varphi \rangle| \leq C\epsilon \|\varphi\|_{C^k}, \quad (2.14)$$

where k is a fixed integer depending only on m . Fixed changes of the radii give the same conclusion.

Proof. First mollify on a slightly smaller domain. Choose a cutoff equal to one on B_3 and supported in $B_{7/2}$, and let Φ be the fundamental solution of the Laplacian. The potential

$$u = \sum_{ij} \partial_i \partial_j \Phi * (\chi F_{ij}) + \sum_j \partial_j \Phi * (\chi g_j)$$

has the same Laplacian as v on B_3 , so $h = v - u$ is harmonic there. The second-derivative kernels have the weak-type estimate $|\{|u| > t\} \cap B_3| \leq C\epsilon/t$; the first-derivative kernels are integrable on bounded sets. We recall the kernel argument. Decompose a mollified finite measure at level t into its bounded part and zero-mean pieces on maximal dyadic cubes. The bounded part has squared L^2 norm at most Ct times the original mass norm. The Fourier multiplier $\xi_i \xi_j / |\xi|^2$ has L^2 norm at most one. The bad cubes have total volume at most $C\epsilon/t$; off their fixed enlargements, cancellation and the bound $|\nabla \partial_i \partial_j \Phi(x)| \leq C|x|^{-m-1}$ bound the L^1 norm of their contributions by $C\epsilon$. Chebyshev's inequality proves the stated weak estimate, uniformly in mollification. Pairing the same potentials against a smooth test, moving the derivatives to the test, also gives $|\langle u, \varphi \rangle| \leq C\epsilon \|\varphi\|_{C^k}$ on any fixed bounded region.

The harmonic remainder is controlled modulo a constant by the full gradient equation. For $x \in B_{5/2}$ choose a fixed radial unit-integral bump η_x supported in a small ball about x . Harmonic mean value and (2.12) give

$$\partial_j h(x) = \langle \partial_j h, \eta_x \rangle = - \sum_i \langle F_{ij}, \partial_i \eta_x \rangle + \langle g_j, \eta_x \rangle + \langle u, \partial_j \eta_x \rangle.$$

Consequently $\sup_{B_{5/2}} |\nabla h| \leq C\epsilon$. Taking $c = h(0)$ proves both estimates for the mollified quantities. The constants c are bounded by pairing against one fixed unit-integral bump and the finite mass of v . Pass to a subsequence as the mollification tends to zero; the densities converge almost everywhere at Lebesgue points. This proves (2.13), and weak convergence against tests proves (2.14). $\square$

Lemma 2.5 (Scalar mass and propagation of the integer). *Fix a bound for Q and fixed concentric interior enlargements of a ball B_s . Suppose the normalized projected mass on the outer enlargement is bounded, and*

$$\epsilon = s^{-m}(D + \Lambda)(B_{cs})$$

is sufficiently small; $c > 1$ is a fixed enlargement with room for the argument. The count selects a unique integer $q \geq 0$ on most of an intermediate ball. If either a fixed smooth mass average differs from Q by less than $1/4$, or $N = Q$ on a fixed majority of that ball, then $q = Q$. In that case

$$|M|(B_s) + \text{vol}_S\{y \in B_s : N(y) \neq Q\} \leq C(D + \Lambda)(B_{cs}). \quad (2.15)$$

In the smooth-average condition the test is nonnegative, has reference integral one, and has fixed scaled derivative bounds. The same integer propagates along overlapping comparable balls for which these small-energy and mass bounds hold and whose overlap has a fixed positive relative volume.

Proof. Scale to $s = 1$. Write $d \text{vol}_S = a(y) dy$ and put the factor a^{-1} in each coordinate component of the tangent test field. Then Theorem 2.2 gives an equation of the form (2.12) for the scalar measure $a^{-1}\nu$. The divergence data have mass at most $C(D + \Lambda)$, as do the zeroth-order data; derivatives of the fixed coordinate factors have bounded size. In particular, no derivative of the unknown scalar mass is estimated separately. Apply Theorem 2.4. By Theorem 2.3, the difference between $a^{-1}\nu$ and $N dy$ has mass at most $C\epsilon$. The density version of this difference and (2.13) give

$$|\{|N - c| > t\} \cap B_2| \leq C\epsilon/t.$$

If $\epsilon = 0$, this estimate for every $t > 0$ gives $N = c$ almost everywhere. Since N is integer-valued, c is an integer, and the distributional bound is exact; the conclusions follow directly. Assume henceforth that $\epsilon > 0$. Taking $t = C_0\epsilon$ with C_0 large proves that, on a fixed large majority, N is within $C_0\epsilon$ of c . For small ϵ there is at most one integer in this interval; call it q . Thus $|c - q| \leq C\epsilon$. The distributional bound against the specified smooth average makes its value $q + O(\epsilon)$, so the smooth-average hypothesis forces $q = Q$. The majority hypothesis has the same consequence by intersecting the two majority sets.

Take now t to be a fixed fraction of one. Since $|c - Q| \leq C\epsilon$,

$$|\{N \neq Q\} \cap B_2| \leq C\epsilon. \quad (2.16)$$

The nonnegativity of N gives $\int_{B_2} (N - Q)_- dy \leq Q|\{N \neq Q\} \cap B_2|$. The small measure error between ν and $N \text{ vol}_S$ therefore bounds $M_-(B_2)$ by $C\epsilon$. Choose $0 \leq \varphi \leq 1$, equal to one on B_1 and supported in B_2 . Equation (2.14), applied also to the bounded volume factor, gives $|\int \varphi dM| \leq C\epsilon$. Hence

$$M_+(B_1) \leq \int \varphi dM + \int \varphi dM_- \leq C\epsilon.$$

This proves total variation and the exceptional-volume assertion, and rescaling proves (2.15). Finally, on an overlap of fixed relative volume, the same fixed-threshold argument gives $|\{N \neq q_i\}| \leq C\epsilon_i$ for each ball's selected integer q_i , before any identification with Q . These exceptional sets cannot fill the overlap. At a remaining point both selected integers equal N . Induction along a finite chain proves propagation. $\square$

Remark 2.6. The enlargement constant in (2.15) can be any prescribed $c > 1$, at the cost of constants depending on the interior room. Applying the argument on a finite cover gives, in particular, the notation $|M|(B) \leq C(D + \Lambda)(2B)$. It is never an estimate on a ball with no interior enlargement or with an unspecified integer.

2.2 Configurations and height bounds

Lemma 2.7 (Configurations, higher moments, and support heights). *Fix $Q \geq 1$, and choose a sufficiently large fixed enlargement $c > 1$ depending only on the dimensions, Q , and the fixed coordinate and cylinder-room data. For every prescribed exceptional proportion $\delta_0 > 0$ there is a threshold $d_0 = d_0(\delta_0) > 0$, depending otherwise only on these fixed data, such that the following holds. Suppose Q is selected on the enlarged ball $B_{cs}(x)$ and*

$$s^{-m}\tau(B_{cs}(x)) \leq d \leq d_0,$$

with the cylinder and interior-room conventions above. There is a configuration $\mathcal{A} = \{a_1, \dots, a_Q\}$ of scaled heights such that, outside a subset of $B_s(x)$ of relative volume at most δ_0 , the fiber consists of exactly Q projection entries and can be matched to $\mathcal{A}$ within $C_{\delta_0}s\sqrt{d}$. Configurations on comparable overlapping balls can be chosen, and any such majority configurations can be matched, within the sum of these bounds, provided the exceptional proportions are smaller than a fixed fraction of the overlap.

For every fixed finite $p > 2$ satisfying

$$m(1/2 - 1/p) < 1, \quad (2.17)$$

and every fixed $0 < \beta < \min\{1/2, 1/m\}$, this same configuration satisfies

$$\left(s^{-m} \int_{\pi^{-1}(B_s(x))} \text{dist}(\zeta, \mathcal{A})^p d\mu \right)^{1/p} \leq C_p s \sqrt{d}, \quad (2.18)$$

$$\sup_{\text{spt } \mu \cap \pi^{-1}(B_s(x))} \text{dist}(\zeta, \mathcal{A}) \leq C_\beta s d^\beta. \quad (2.19)$$

The enlargement c , threshold d_0 , and configuration do not depend on p or β . The constants C_p, C_β may depend on their indicated exponents and the fixed data, but not on δ_0 ; the majority constant C_{δ_0} may depend on it. The support estimate holds for all support points, including points with zero projection Jacobian.

Proof. It is enough to prove the estimates for $d > 0$. If the actual energy is zero, fix $0 < \delta' < \min\{\delta_0, 1/2\}$ and positive upper bounds $d_j \downarrow 0$ below $d_0(\delta')$. Let G_j be the measurable good sets from the positive case. Fatou's lemma applied to their complements gives

$$\text{vol}_S(\limsup_j G_j) \geq (1 - \delta') \text{vol}_S(B_s(x)).$$

Choose a defined fiber $\mathcal{F}_0$ in this limsup and retain the indices j_k for which it is good. Its Q entries match the corresponding configurations within $C_{\delta'} s \sqrt{d_{j_k}} \rightarrow 0$, so those configurations converge to $\mathcal{F}_0$ up to permutations. The limsup of the selected G_{j_k} has the same volume bound. On each fiber in that limsup, the matching triangle inequality along its good subsequence gives exact matching to $\mathcal{F}_0$. For one admissible $\beta_0 > 0$, the support estimate and the matching convergence put every support height in $\mathcal{F}_0$. All support distances and hence all finite moments are zero, proving the assertion for $d = 0$. We give the finite selection argument before the iteration that controls exceptional fibers. If f is a bounded Lipschitz function of height with $|f| \leq 1$, use $f(\zeta)\bar{X}$ in the horizontal first variation. In flat coordinates the additional term is $\nabla f \cdot P_{vh}X$. In normal coordinates its additional frame term is bounded by $C \text{Lip}(f)l|z_{\text{phys}}||X|$. In the curved case, as in [Theorem 2.5](#), write $d \text{vol}_S = a \, dy$ and take the coordinate components $X = a^{-1}Y$. Then $\text{div}_S X = a^{-1} \text{div} Y$ exactly; derivatives of a^{-1} remain in the tensor and geometric terms. Cauchy–Schwarz, the normalized mass bound, and [Theorems 2.3](#) and [2.4](#) show that the fiber function

$$N_f(y) = \sum_{q \in \pi^{-1}(y), J_\pi(q) > 0} \theta(q) f(\zeta(q))$$

differs from its fixed smooth average $\bar{N}_f$ by a weak- L^1 quantity whose normalized size is at most

$$C\{d + s \text{Lip}(f)\sqrt{d}\}. \quad (2.20)$$

The same bound controls distributional pairings. Changing between Jacobian-weighted and ordinary mass in this statement costs Cd .

For this finite selection prescribe $0 < \delta \leq 1/4$; a larger requested exceptional proportion is covered by using $\delta = 1/4$. Normalize the smoothly averaged vertical counting measure to a probability measure $\mathcal{P}$; its total before normalization is $Q + O(d)$ by [Theorem 2.5](#). Fix a mass mesh η_0 much smaller than $1/Q$. For each of the n coordinates choose its quantiles at levels $j\eta_0$, and test smoothed half-lines immediately to both sides of these quantiles, with transition width $w = Ks\sqrt{d}$. Repeated quantiles cause no problem: retain both one-sided tests. There are at most $C(n)/\eta_0$ tests. For each, the right side of (2.20) is $C(d + K^{-1})$. First choose η_0 , then choose K large enough for these tests and the joint-count tests below, and finally choose a positive energy threshold small. The union of the exceptional sets where any test differs from its average by more than η_0 then has relative volume at most $\delta/3$.

Also discard the set where $N \neq Q$. By (2.16) its relative volume is $O(d)$, so it is at most $\delta/3$ after decreasing the same threshold. Every remaining fiber therefore has exactly Q projection entries, counted with multiplicity.

Let F_y be the empirical coordinate distribution on a remaining fiber (normalized by Q), and let F be that of $\mathcal{P}$. Bracketing an arbitrary real number by adjacent quantiles gives

$$F(t - 2w) - 3\eta_0 \leq F_y(t) \leq F(t + 2w) + 3\eta_0.$$

If t is in a quantile jump, the left and right tests at that quantile give the same inequalities; thus atoms require no continuity assumption. Comparison of two good fibers gives this estimate with $4w$ and $6\eta_0$. For $6\eta_0 < 1/(2Q)$, comparison at the levels $(j - 1/2)/Q$ shows that their ordered coordinate lists differ by at most Cw . Choose one such fiber as a temporary reference. In each coordinate group its at most Q levels by joining consecutive levels whose gap is at most a sufficiently large fixed multiple of w . Each resulting interval has length at most CQw , and distinct intervals have an empty gap in which a cutoff can change from zero to one.

Coordinate lists alone do not determine vector configurations. To determine the joint counts, test products of the interval cutoffs just constructed. There are at most Q^n such tests and their Lipschitz constants are at most C/w . On the good coordinate fibers every product equals an exact indicator. After decreasing the threshold if necessary and discarding a further relative volume at most $\delta/3$, each joint count is within less than $1/4$ of its smooth average. It is an integer, so it is the same on all remaining fibers. Taking this many copies of one point in each occupied coordinate box gives $\mathcal{A}$. The boxes have diameter Cw , proving the majority assertion. Intersecting the good sets of two overlapping balls proves the matching assertion. Write $d_{\text{sel}}(\delta) > 0$ for a threshold sufficient for this finite selection, and use $d_{\text{sel}}(1/4)$ when $\delta > 1/4$.

There is also an exact finite selection when the energy on this ball is zero. Fix $0 < \delta' < \min\{\delta, 1/2\}$, apply only the finite selection just proved with positive upper bounds $d_j \downarrow 0$ below $d_{\text{sel}}(\delta')$, and use the limsup construction and matching through one common fiber from the start of this proof. That part uses only the good sets and their matching errors, not the moment or support estimates. It yields an exact majority configuration with exceptional proportion at most $\delta' < \delta$.

The dyadic covering geometry below, including its enlargement and overlap ratios, is fixed from the data at the outset and determines c in the statement. Choose an internal proportion $0 < \delta_{\text{int}} < 1/4$ so that two exceptional sets cannot fill any comparison overlap, and small enough for the support argument. Then choose a fixed threshold $d_{\text{int}} \leq d_{\text{sel}}(\delta_{\text{int}})$ small enough for that argument and the iteration, and set

$$d_0(\delta_0) = \min\{d_{\text{int}}, d_{\text{sel}}(\delta_0)\}.$$

For every selection ball in the descent, including the root, use the actual normalized mass of τ on the associated fixed enlargement as the selection upper bound, using the exact finite selection when that mass is zero. Use the root configuration $\mathcal{A}$ selected with δ_{int} throughout the descent. If $\delta_0 < \delta_{\text{int}}$, apply the finite selection once more at the root with exceptional proportion δ_0 and the same actual energy. A common good fiber matches the two root configurations within $C_{\delta_0} s\sqrt{d}$, after enlarging that constant. On the good fibers of the second selection, the triangle inequality then gives the prescribed majority bound for $\mathcal{A}$. If $\delta_0 \geq \delta_{\text{int}}$, the original good set suffices. Thus the descent and its root configuration use only the fixed internal choices; the finer selection affects only the majority constant and input threshold.

We next record how to include every support point. On a continuing ball of radius r with normalized τ at most d_{int} , the majority assertion and [Theorems 2.3](#) and [2.5](#) bound the mass outside the configuration's height neighborhoods by

$$C(\delta_{\text{int}} + d_{\text{int}})r^m.$$

Indeed the mass above bad fibers is at most their Q -fold base volume plus $|M|$, and the projection error also has mass at most $C(D + \Lambda)$. Because the restricted support is separated from the artificial vertical boundary, its extension by zero across that boundary remains stationary for fields supported away from the lateral boundary; multiply such a field by a vertical cutoff equal to one near the restricted support to verify this assertion. The monotonicity ball is taken in this extension and remains in the lateral working region; it need not lie inside

the original vertical cutoff. If a support point over an inner base ball were at distance at least $c_1 r$ from these height neighborhoods, its ambient ball of radius $c_2 r$ would lie outside them. For fixed small $c_2 > 0$, this contradicts (2.1) when $\delta_{\text{int}}, d_{\text{int}}$ are sufficiently small. The same reasoning works in the curved tube since its coordinate map and inverse are uniformly Lipschitz. This proves an $O(r)$ bound for every support point on an interior ball. In particular it applies to a first stopped child, using its continuing parent.

For the quantitative iteration, subdivide the inner base region into dyadic cubes, using around each cube a ball of a fixed large multiple of its side. Start with sufficiently generous interior room. Continue down a branch while the normalized τ on these balls is at most the fixed threshold d_{int} . Otherwise stop at the first such cube. On continuing balls the normalized mass is bounded by $Q + C d_{\text{int}}$, the integer is Q , and the majority configurations at a child and its parent match with increment

$$Cr\sqrt{r^{-m}\tau(\text{fixed enlargement of the parent})}.$$

The fixed parent enlargement here contains both selection enlargements. If its τ -mass is zero, both actual selection energies vanish, their exact majority configurations match on the overlap, and the child cannot stop. At a first stopped cube the all-support argument gives instead Cr for the distance from each support height to the parent's configuration. Its stopping condition charges r^m to a fixed multiple of τ on that enlargement. Configurations are regarded with their optimal matching, so their nearest-point distances obey the triangle inequality even at coincident entries.

Write $r_j = 2^{-j}s$, and let $b_j(q)$ be the corresponding increment for a point above a continuing cube, or its final error at its first stopped cube; put $b_j = 0$ after a stop. Fixed enlargements of cubes have bounded overlap at each generation. The preceding charge, the bounded mass of a continuing parent, and $\tau(B_{cs}) \leq ds^m$ give

$$s^{-m} \int b_j^2 d\mu \leq Cr_j^2 d, \quad (2.21)$$

$$\|b_j\|_\infty \leq Cr_j \min\{1, 2^{jm/2}\sqrt{d}\}. \quad (2.22)$$

For the second bound on stopped cubes, stopping itself implies $2^{jm}d \geq c > 0$, so its Cr_j error has the required form. For a branch which never stops, the all-support Cr_j bound tends to zero; thus its configurations approach the actual support height in distance. It follows in both cases that $\text{dist}(\zeta(q), \mathcal{A}) \leq \sum_j b_j(q)$.

The covering, stopping decisions, and configuration were fixed before either exponent was chosen. The following summations therefore apply to this same $\mathcal{A}$ for every exponent in the stated ranges. Interpolate (2.21) with the second part of (2.22), without the minimum, to obtain

$$\left(s^{-m} \int b_j^p d\mu\right)^{1/p} \leq Cs\sqrt{d} 2^{-j[1-m(1/2-1/p)]}.$$

Minkowski's inequality and (2.17) prove (2.18). Summing (2.22) above and below $2^{-j} \simeq d^{1/m}$ gives $Cs\sqrt{d}$ when $m = 1$, $Cs\sqrt{d}(1 + |\log d|)$ when $m = 2$, and $Csd^{1/m}$ when $m > 2$. Each is bounded by $C_\beta sd^\beta$ for the stated range of β , proving (2.19). $\square$

2.3 Gaussian and height identities

Corollary 2.8 (Gaussian weights and polynomial dilations). *Define*

$$G_s(y) = (2\pi s^2)^{-m/2} \exp(-|y|^2/(2s^2))$$

and fix a center x . On the region $|y - x| \leq Rs$, cover the base by balls of radii comparable to $s/(1 + |y - x|/s)$, with fixed enlarged balls having bounded overlap. Suppose each enlarged ball meets the hypotheses of [Theorem 2.5](#) with the same propagated integer Q . Then

$$\int_{|y-x| \leq Rs} G_s(y-x) d|M|(y) \leq C \int_{|y-x| \leq (R+C)s} G_s(y-x) d(D + \Lambda)(y). \quad (2.23)$$

The constant is independent of R . The part outside the indicated region is retained as a tail error.

For the configuration conclusion, fix one exceptional proportion $\delta_{\text{cmp}} > 0$ smaller than the fixed comparison-overlap proportions. If, in addition, on all admissible dilations $1 \leq t \leq R_*$ one has

$$(ts)^{-m} \tau(B_{cts}(x)) \leq Ct^J d, \quad (2.24)$$

with $Ct^J d \leq d_0(\delta_{\text{cmp}})$ throughout this range and the same integer propagated, then configurations matched from the scale s have moment and support estimates on a dilation t equal to those in [Theorem 2.7](#), with an additional fixed power of t . Gaussian integrals of these estimates have constants depending on J and the fixed moment order, once the tail beyond R_* is accounted for.

Proof. The logarithmic derivative of G_s is $-(y - x)/s^2$. Its values on each fixed enlargement of a ball of the stated radius differ by bounded factors. Such a cover can be obtained by a maximal disjoint family of sufficiently small balls for this radius function; nearby radii are comparable. Apply (2.15) on each ball and sum. This proves (2.23) without a factor depending on the number of balls. Apply [Theorem 2.7](#) with the fixed proportion δ_{cmp} successively to concentric dyadic dilations and to overlapping comparable balls. Every increment is bounded by a fixed power of t times $s\sqrt{d}$, and each local moment estimate has the same property by (2.24). Summation of the geometric increments gives the claimed polynomial bounds. Finally $\sum_{j \geq 0} 2^{jC} \exp(-c4^j) < \infty$ absorbs any fixed polynomial loss in the Gaussian integral. $\square$

We specify the cutoff convention for every Gaussian identity. Choose a smooth base cutoff χ which is one on the region under consideration and has compact support in the chart. A subscript s denotes convolution at x with G_s of the extended measure; on the chart it is the original measure, and outside it any stated extension is used. In a first-variation identity the test field is multiplied by χ . Contributions where $\chi \neq 1$, including the difference from the chosen extension, are denoted by $\mathcal{T}_s$. If the cutoff annulus has distance at least h from x and the relevant total measure is at most K_0 , decrease h to the available transition width if necessary, and choose $0 \leq \chi \leq 1$ with $\|\nabla \chi\|_\infty \leq C(1 + h^{-1})$. The first-variation product rule differentiates this fixed cutoff once; scale derivatives leave it fixed. For $0 < s \leq 1$, direct differentiation of the Gaussian bounds each such term by

$$CK_0 s^{-C} (1 + h^{-1})^C \exp(-h^2/(4s^2)), \quad (2.25)$$

times the stated bounds for the height multiplier and its derivatives. One may use a larger fixed exponent C for the finite set of Gaussian and height-multiplier derivatives used below. When a polynomial dilation bound such as (2.24) is available through R_* , annular summation bounds the contribution from $Rs < |y - x| \leq R_*s$ by $C \text{poly}(R) e^{-cR^2}$ times its reference density. The part beyond R_*s remains a separate tail, including any part still inside the chart, until a further dilation or total-mass bound controls it. The exterior-chart term is also retained. A periodized Gaussian on a fixed large torus obeys the same estimates. These formulas are the meaning of a negligible Gaussian tail; no unrestricted extension is silently discarded.

Lemma 2.9 (Gaussian mass identity). *Set $u = (y - x)/s$, $A(s) = s^2 M_s(x)$, and*

$$R_s(x) = \int G_s(y - x) B[u, u] \, d\mu.$$

In flat coordinates, apart from $\mathcal{T}_s$,

$$\dot{A} = s^2 L_s + s^2 R_s, \quad \dot{\cdot} = \frac{d}{d \log s}. \quad (2.26)$$

In the curved coordinates fixed above,

$$\dot{A} = s^2 L_s + s^2 R_s + \mathcal{E}_s + \mathcal{T}_s, \quad |\mathcal{E}_s| \leq C\gamma s^2 \int G_s(y - x)(1 + |u|^2) \, d\tau(y). \quad (2.27)$$

Here the entries of u in $B[u, u]$ are taken in the chosen orthonormal horizontal frame, and γ retains its definition from the fixed physical coordinates.

Proof. In flat coordinates $\dot{G}_s = -\operatorname{div}_y((y - x)G_s)$. The horizontal first variation says $\int \operatorname{div} X \, dM = \int B : DX \, d\mu$. Take $X = (y - x)G_s$. Since $DX = G_s(I - u \otimes u)$, it follows that $\dot{M}_s = -D_s + R_s$. Differentiating the factor s^2 proves (2.26).

For the curved identity apply Theorem 2.2 to the same coordinate field. Its scaled derivative is bounded by $CG_s(1 + |u|^2)$ and its size by $CsG_s|u|$. On a physical chart of radius l , coordinate coefficients and their scaled first derivatives differ from the centered Euclidean ones by $O(l)$. The divergence comparison is made after subtracting $Q \operatorname{vol}_g$; its coefficient error therefore multiplies M , not the background mass. The tensor coefficient errors multiply B and are bounded by $C\gamma e$. Equation (2.8) controls the remaining geometric terms. Multiplying by s^2 proves (2.27). Differentiation of χ produces precisely the tails described in (2.25). $\square$

Lemma 2.10 (Height tests and geometric errors). *In flat coordinates, for a base test φ , a scalar height function f , a horizontal field X , and a normal vector function v , stationarity gives*

$$\int f \operatorname{div} X \, d\mu = \int f B : DX \, d\mu - \int Df \cdot P_{vh} X \, d\mu, \quad (2.28)$$

$$0 = \int \{\varphi Dv : P_{vv} + v \cdot P_{vh} \nabla \varphi\} \, d\mu. \quad (2.29)$$

All height functions here use the scaled variable ζ . For a scale-independent C^2 height function Γ , these formulas give

$$\frac{d}{d \log s} \int G_s \Gamma \, d\mu = \int G_s \Gamma(B[u, u] - e) \, d\mu + s^2 \int G_s D^2 \Gamma : P_{vv} \, d\mu, \quad (2.30)$$

apart from the specified chart tails. If Γ depends on s , add $\int G_s \dot{\Gamma} \, d\mu$.

In the curved tube, use the original radial-gauge normal frame fixed above. The extra trace terms in a normal test $\varphi v(\zeta)$ without a derivative of φ are bounded pointwise by $C|\varphi|$ times

$$|Dv|l|z_{\text{phys}}|\sqrt{e} + l|v|(|H_S| + |z_{\text{phys}}| + e + \sqrt{e}). \quad (2.31)$$

The terms with a derivative of φ have the flat tensor expression in (2.29), with a coefficient error at most $C\gamma|P_{vh}|$. The curved horizontal identity is covariant, with $\operatorname{div}_S X$ and $\nabla^S X$ in place of the flat base operators. Its geometric error is bounded by the right side of (2.8) with the integrand multiplied by $|f|$. The additional flux error from the height gradient is bounded pointwise by

$$C\gamma|Df||X|(\sqrt{e} + \sqrt{\lambda}). \quad (2.32)$$

In its coordinate form the divergence comparison also contributes an error bounded by $Cl \int |f||X| d\mu$; the coordinate changes in the B term are included in the preceding geometric error. Under a further interior rescaling, retain the original horizontal and normal frames and the original γ in (2.6). Transform the displayed estimates by change of variables. The inherited physical connection remains $O(l)$; no improvement to $O(ls)$ is asserted on an off-center ball of coordinate radius s . In particular first-order flux errors have size at most $C\gamma(\sqrt{e} + \sqrt{\lambda})$ times the relevant multiplier, whereas quadratic errors have size at most $C\gamma(e + \lambda)$. The factor $|v|$ in (2.31) must be retained when estimating a normal test.

Proof. In a product Euclidean chart, differentiate the ambient fields $f(\zeta)X(y)$ and $\varphi(y)v(\zeta)$. Their horizontal and vertical blocks give (2.28) and (2.29). Apply the first with $f = \Gamma$ and $X = (y - x)G_s$. It yields the first term of (2.30) and the flux $s \int G_s D\Gamma \cdot P_{vh} u d\mu$. Apply the normal formula with $v = D\Gamma$ and $\varphi = G_s$; since $\nabla G_s = -G_s u/s$, this flux is exactly $s^2 \int G_s D^2\Gamma : P_{vv} d\mu$. This proves the identity and fixes both its sign and its factors of s .

We give the geometric error calculation. In radial gauge the normal connection one-form vanishes at the chart center and has physical norm $O(l)$ throughout the chart, by its bounded curvature and the radial parallel-transport equation. Its ambient derivative still has the tangential component $-A_v$; that component is not part of the small connection one-form. A horizontal derivative in scaled ambient coordinates of $\zeta = z_{\text{phys}}/l$ is

$$D_h \zeta = -\omega(J_z^{-1}h)z_{\text{phys}}, \quad |D_h \zeta| \leq Cl|z_{\text{phys}}||h|,$$

while the vertical derivative is unchanged. Differentiating $v(\zeta)$ therefore gives the first summand of (2.31) on contraction with the mixed block. Differentiating its normal frame gives a horizontal trace whose identity part is $-l\langle H_S, v \rangle$, with error $Cl|z_{\text{phys}}||v|$ from $J_z^{-1} - I$; its B part is bounded by $Cle|v|$. The remaining mixed-frame term is bounded by $Cl\sqrt{e}|v|$. This proves (2.31). A base derivative uses $D\pi = J_z^{-1}P_h$; its coefficient on a coordinate gradient in the chosen frame is $J_z^{-1}C_y^{-\top}$, which differs from I by at most $C\gamma$. This gives the stated coefficient error on the mixed block.

For the horizontal test, first use the calculation in Theorem 2.2, and then differentiate $f(\zeta)$. If X_{coord} is the coordinate column of the scaled field, its lift has column $J_z C_y X_{\text{coord}}$ in the horizontal frame. Subtracting the flat mixed flux leaves

$$Df \cdot P_{vh}(J_z C_y - I)X_{\text{coord}}, \quad |J_z C_y - I| \leq C\gamma.$$

The bound $|P_{vh}| \leq C\sqrt{e}$ gives its first contribution to (2.32). The separate connection derivative is bounded by $Cl|z_{\text{phys}}||Df||X_{\text{coord}}|$, which gives the second contribution because $l \leq \gamma$ and $|z_{\text{phys}}| \leq \sqrt{\lambda}$. For the coordinate form, in the original centered coordinates put $a_l(\xi) = \sqrt{\det g(l\xi)}$. For a scaled coordinate field $X(\xi)$,

$$\text{div}_S(l DF(l\xi)X) = \text{div}_\xi X + D_\xi(\log a_l) \cdot X, \quad |D_\xi \log a_l| \leq Cl.$$

This weighted identity has no subtracted reference volume, so this last term gives the stated $Cl \int |f||X| d\mu$ error. The remaining coordinate coefficient changes are contracted with B and cost $Cle|f|(|\nabla X| + |X|)$, already included in the geometric error. At a further radius s , put $y' = (y - x)/s$, $\zeta' = \zeta/s$, $T_{x,s}(y, \zeta) = (y', \zeta')$, $\mu' = s^{-m}(T_{x,s})_\# \mu$, and $v'(\zeta') = v(s\zeta')/s$. Then $D'v' = Dv$, while λ remains in physical units. Keeping the same frame retains the coefficient l in $l|Dv||z_{\text{phys}}|\sqrt{e}$; the terms involving v become $(ls)|v'|(|H_S| + |z_{\text{phys}}| + e + \sqrt{e})$. The resulting estimates use the original γ . All configuration comparisons likewise use the original height frame, so there is no change of height coordinates from a new gauge. Young's inequality changes products of two first-order errors into the stated quadratic bound. Applying Cauchy-Schwarz to (2.31) retains the actual L^2 size of v , which is why this form of the bound is useful for displacement and clipped-height tests. Finally multiply each field by the chart cutoff; its derivative terms obey (2.25). $\square$

3 The signed-excess estimate

We now state the estimate that drives the proof. The definitions of M, B, D, L, Λ , and τ are those of [Section 2](#); in particular, $L = 2M - D$ and $\tau = |M| + D + \Lambda$ are measures on the base. In the curved case the reference graph has small gradient, a fixed normal tube, and fixed bounds through a sufficiently large fixed finite order J , as specified in [Section 2](#). All applications below use C^J reference graphs with these uniform bounds.

For a positive integer j , let $\exp_j$ and $\log_j$ denote j -fold composition of the exponential and logarithm. For large T set

$$F_Q(T) = \exp_{2Q} \left(\sqrt{\log_{2Q} T} \right). \quad (3.1)$$

Only sufficiently large arguments are used. Thus

$$(\log T)^p = o(F_Q(T)) \quad (p < \infty), \quad F_Q(T) = T^{o(1)}. \quad (3.2)$$

Theorem 3.1 (Signed excess). *Fix a positive integer Q and $\kappa > 0$. Work in the interior normal-cylinder setting of [Section 2](#), with base coordinates scaled by a physical radius l so that B_4 is available. Suppose*

$$\tau(B_4) \leq d = e^{-T}, \quad l + \sup |z_{\text{phys}}| \leq e^{-\kappa T}. \quad (3.3)$$

The supremum is over the restricted support above the available chart. The statement also holds after an interior rescaling in the inherited frames of [Theorem 2.10](#), provided in the curved case the second smallness condition is imposed on the inherited parameter γ from (2.6). There are a constant C_Q and a threshold T_0 , depending only on the stated fixed data, such that for $T \geq T_0$,

$$\int w(L - C_Q \Lambda) \leq e^{-F_Q(T)} d \quad (3.4)$$

whenever w is supported in B_1 , $0 \leq w \leq 1$, and $\text{Lip } w \leq 1$. For a flat reference plane, take $\Lambda = 0$ and omit the second condition in (3.3); constant sheet offsets need not be bounded.

The integer Q is the reference count subtracted in M ; the theorem does not assume the pointwise bound $\theta \leq Q$. The proof occupies the next three sections and proceeds by induction on this reference count. At the step $Q > 1$, the hypothesis is the theorem for every integer $1 \leq q < Q$. If the asserted bound fails, [Section 4](#) subtracts a small margin from the weighted signed source and assigns its remaining positive mass to selected centers and scales. Let h denote the positive spatial projection of this contact measure. At a contact, the smoothed difference between the source and h is nonnegative. At larger scales only its cumulative integral with weight $v^2 d \log v$ is nonnegative. This remains true for contacts on vertical sides of the stopping boundary. The construction itself uses no smaller-integer assertion.

Induction enters through a configuration of Q heights. If its entries split into separated nonempty groups, sufficiently small working balls contain actual stationary restrictions with positive reference counts $q < Q$. The separated estimate in [Section 4](#) applies [Theorem 3.1](#) at every such lower count and also gains a small positive multiple of the tilt far from the configuration. In the main induction step, [Section 5](#) retains contacts whose outward vertical contains a threshold equality and constructs an envelope of future signed mass. Along the selected path beginning at that equality, the cumulative sign makes the envelope control unsigned averages at larger radii up to the chosen outer scale, with a fixed power of the radius ratio. This permits a comparison with squared distance from finitely many height centers. The separated gain pays for the regions where that distance function changes between centers. The centers may merge, but at most $Q - 1$ times, so one of Q fixed ranges of excess levels is traversed without a merger.

Section 6 extracts many scales with a definite normal flux on that traversal. The higher height moment prevents this flux from lying entirely where the excess measure has large density. A dyadic packing argument then makes the union carrying all these scales too small to support the original contact mass, closing the induction at Q . For $Q = 1$ every step involving separated groups or several wells is omitted.

Order of choices. Dimensions, a bound on Q , coordinate geometry, and κ are fixed first. Interior radii, covering constants, moment exponents, and the finite number of derivative bounds are then fixed. Polynomial error exponents may be made large enough for these data. For $Q > 1$, the separated estimate uses a function $b = b_Q$ with

$$b(T) \gg (\log T)^{200}, \quad F_{Q-1}(b(T)/2) \gg (\log T)^2, \quad b(T) = T^{o(1)}. \quad (3.5)$$

For example, a sufficiently large fixed multiple of

$$1 + (\log T)^{201} + F_{Q-1}^{-1}((\log T)^3)$$

suffices. Fixed increases of this choice will be made before fixing T_0 . The separation factors are exponentials of fixed multiples of b . Theorem 5.5 verifies the additional rate comparisons required by the finitely many mergers. Finally T is taken large. A fixed weakening of κ changes only the threshold; this permits all the rescaled lower-integer applications.

Margins and tails. Whenever a compactly supported chart is replaced by a Gaussian convolution, the source includes a negative margin proportional to $\eta(\tau + d \, dy)$, where $\eta > 0$ is a chosen small parameter. Contact scales lie below a chosen outer scale ρ , with $|\log \rho| = o(T)$ in the main application. On the continuation region, bounds at every larger scale control the Gaussian tails. Theorem 4.2 gives the precise localization statement. Curved first-variation errors contain the factor $e^{-\kappa T}$, which is smaller than the margins used below. These conventions are retained at recursive nodes, with lengths normalized by the node radius.

4 Heat-potential contacts and separated configurations

The contact construction replaces a positive signed integral by a positive measure carried by selected space-scale points. Its useful property is an initial sign and a cumulative sign on each outward vertical. We first give this construction without any geometric assumptions. We then use it, together with the elementary estimates of Section 2, to reduce a separated configuration to smaller multiplicities.

The extension viewpoint of Caffarelli and Silvestre [6, Sections 1–2] motivated an earlier potential-based approach to the contact construction. The present construction uses a heat potential and is proved locally; their fractional-Laplacian extension theorem is not invoked as a varifold regularity input.

4.1 The entrance boundary and its contact measure

Let X be a flat torus, with its volume measure denoted by dy . Gaussian kernels on X are periodized, and preserve constants. Fix a bottom scale $a > 0$. A *continuation set* is a closed subset $\mathcal{D} \subset X \times [a, \infty)$ such that

$$(x, s) \in \mathcal{D} \implies (x, v) \in \mathcal{D} \quad (v \geq s), \quad X \times [s_{\text{top}}, \infty) \subset \mathcal{D}$$

for some $s_{\text{top}} > a$. Its entrance boundary is $K = \mathcal{D} \setminus \text{int } \mathcal{D}$, where the interior is taken in $X \times (0, \infty)$. Thus K is compact. No artificial upper time boundary belongs to K .

Here are the two types of continuation sets used below. For a finite mesh $t_j \in [a, s_{\text{top}})$ and continuous tests q_j , the inequalities $q_j(x) \leq \lambda_j$ are required only at strictly larger mesh scales:

$$\mathcal{D} = \{s \geq a\} \cap \bigcap_j (\{s \geq t_j\} \cup \{q_j(x) \leq \lambda_j\}). \quad (4.1)$$

Each factor is closed and upward closed. At $s = t_j$ the current test is omitted. With tests continuous in the scale, impose the non-strict continuation inequality at every scale $v \geq s$ at which a test is defined, and extend by the full cylinder above the last test. In our applications all tests hold strictly near that last scale, so this extension is closed. Equivalently one can first impose the inequalities at $v > s$ and pass to $v = s$ by continuity. Taking a limit of points of $\mathcal{D}$ proves closedness: any fixed $v > s$ is eventually larger than the approximating scales.

The entrance boundary includes bottom points, current hits, and vertical sides below an equality at a larger scale. These sides are essential: a tangential equality can prevent a spatial neighborhood from belonging to the continuation set even though the outward vertical itself remains in that set. Upward closure implies that a backward space-time cylinder contained in $\mathcal{D}$ provides forward room as well; hence the above definition agrees with the parabolic entrance boundary.

Lemma 4.1 (Contact measure). *Let f be a finite signed Borel measure on X , with $M_f = f(X) > 0$, and let K be the entrance boundary of a continuation set. There is a probability measure σ on K such that the positive spatial measure*

$$h = M_f(\pi_X)_\# \sigma$$

has mass M_f and, at every $(x, s_0) \in \text{spt } \sigma$,

$$(f - h)_{s_0}(x) \geq 0, \quad \int_{s_0}^s v^2 (f - h)_v(x) d \log v \geq 0 \quad (s \geq s_0). \quad (4.2)$$

The second inequality concerns the integral; it does not assert a pointwise sign at scales larger than s_0 .

Proof. For $\sigma \in \mathcal{P}(K)$ put $g_\sigma = f - M_f(\pi_X)_\# \sigma$. This measure has zero mass. Define

$$b_\sigma(x, s) = G_s * (-\Delta_X)^{-1} g_\sigma(x),$$

using the inverse with zero spatial mean. If λ_k are the nonzero eigenvalues of $-\Delta_X$, its kernel has Fourier coefficients $e^{-\lambda_k s^2/2}/\lambda_k$. Since $s \geq a > 0$, the kernel and all its derivatives converge uniformly on bounded scale intervals. Thus b_σ is smooth, and

$$\frac{db_\sigma}{d \log s} = -s^2 (g_\sigma)_s, \quad \partial_t b_\sigma = \Delta_X b_\sigma, \quad t = s^2/2. \quad (4.3)$$

Weak convergence of probabilities implies uniform convergence of their potentials on K : the kernel is jointly continuous on the relevant compact product, and can be approximated uniformly by finite sums of products of continuous functions of its two variables.

Choose finite $1/j$ -nets $K_j = \{z_{j1}, \dots, z_{jN_j}\}$ in K . For a probability vector p on K_j and $\beta > 0$, the map

$$p \mapsto \left(\frac{\exp(\beta b_p(z_{ji}))}{\sum_{k=1}^{N_j} \exp(\beta b_p(z_{jk}))} \right)_{i=1}^{N_j}$$

is continuous from the probability simplex to itself. Brouwer's Fixed Point Theorem gives a fixed point [16, Theorem 2]. First let $\beta \rightarrow \infty$ on a fixed net and take a convergent subsequence. If a point's limiting payoff is at least δ below the maximum, its fixed-point weight is at most

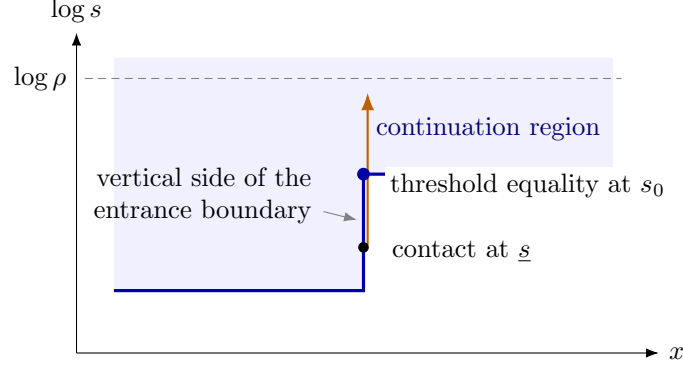


Figure 1: A schematic entrance boundary. In this illustration the contact scale is denoted by $\underline{s}$ and a larger threshold scale by s_0 . A contact on a vertical side can have derivative scale $\underline{s} < s_0$, where s_0 is a larger threshold scale. The contact rule controls the initial derivative and the cumulative integral along the outward vertical. It does not impose a pointwise sign at every later scale. The two scales remain distinct in the final mass charge.

$e^{-\beta\delta/2}$ for all sufficiently large β . The limiting probability is therefore carried by maximizers of its own potential on K_j . Next pass to a weak limit as $j \rightarrow \infty$. Uniform convergence of the potentials and density of the nets show that the resulting probability σ satisfies

$$\text{spt } \sigma \subset \arg \max_{z \in K} b_\sigma(z). \quad (4.4)$$

Indeed a compact subset separated from the limiting maximum by δ receives no mass from sufficiently late net probabilities; a nearby net point witnesses the strictly larger payoff. Exhaust the complement of the closed maximizer set by such compact subsets.

Write $b = b_\sigma$. Its maximum on the whole continuation set is bounded by $\max_K b$. To verify this on an irregular boundary, truncate at any scale $S > s_{\text{top}}$ and apply the elementary maximum argument to $b - \epsilon t$. At a maximum outside K below the artificial top, $\partial_t(b - \epsilon t) - \Delta(b - \epsilon t) = -\epsilon$ contradicts the derivative conditions. At a maximum at the artificial top, the one-sided time derivative is nonnegative and the spatial Laplacian is nonpositive, giving the same contradiction. Let $\epsilon \downarrow 0$. This argument uses only that points outside K have an open neighborhood in the continuation set, and hence also covers arbitrary vertical-side geometry.

For $(x, s_0) \in \text{spt } \sigma$, (4.4) and upward closure now give $b(x, s_0) \geq b(x, s)$ for every $s \geq s_0$. The right derivative at s_0 is nonpositive. Applying (4.3) proves the first inequality in (4.2); integrating that identity proves the second. $\square$

4.2 Localization of the signed source

All extensions in the next statement are made consistently: if L' is restricted to a working chart, τ is restricted to the same or a larger chart, so that $|L'| \leq C_0\tau$ survives extension by zero. The constant density $d \, dy$ is placed on the entire torus.

Lemma 4.2 (Uniform weight localization). *Let $|L'| \leq C_0\tau$, where τ is a finite positive measure on a torus of diameter comparable to R , and let $0 \leq w \leq 1$, $\text{Lip } w \leq R^{-1}$. Put*

$$\nu = \tau + d \, dy, \quad f = wL' - \eta\nu, \quad 0 < \eta < 1.$$

Suppose, for some fixed C_1 , that $\tau(X)/(dR^m) \leq e^{C_1T}$. For all sufficiently large T , if

$$0 < \rho \leq c\eta^2 R/T^2, \quad (4.5)$$

then, uniformly for every $x \in X$ and $0 < s \leq \rho$,

$$f_s(x) \leq w(x)L'_s(x) - \frac{\eta}{2}(\tau_s(x) + d). \quad (4.6)$$

Here $c > 0$ depends only on C_0, C_1 and the fixed torus shape. If σ, h are furnished by [Theorem 4.1](#) with contacts at scales $s_0 \leq \rho$, then $w(x) > 0$ at each contact, and

$$L'_{s_0}(x) \geq h_{s_0}(x) + \frac{\eta}{2}\nu_{s_0}(x), \quad (4.7)$$

$$\int_{s_0}^s v^2 \left(L' - \frac{\eta}{2}\nu \right)_v(x) d\log v \geq 0 \quad (s_0 \leq s \leq \rho). \quad (4.8)$$

In particular, contacts lie in the interior region containing $\{w > 0\}$. These conclusions are uniform as the bottom scale tends to zero.

Proof. The only commutator is

$$f_s(x) = w(x)L'_s(x) - \eta\nu_s(x) + E_s(x), \quad |E_s(x)| \leq C_0 \int G_s(x-y)|w(y) - w(x)| d\tau(y).$$

Split at torus distance $r_* = c_0\eta R$. For sufficiently small fixed c_0 , the near contribution is at most $\eta\tau_s(x)/4$. The far contribution is at most

$$C\tau(X)s^{-m} \exp(-c_2\eta^2 R^2/s^2).$$

Writing $x_* = s/R$, its ratio to d is bounded by $Ce^{C_1T}x_*^{-m}e^{-c_2\eta^2/x_*^2}$. On $0 < x_* \leq c\eta^2/T^2$ this is $o(\eta)$ uniformly: the negative exponent is at least a fixed multiple of T^4/η^2 , which dominates $C_1T, m|\log x_*|$, and $|\log \eta|$. The same estimate applies to the nonnearest periodized copies. This proves (4.6).

At a contact $f_{s_0} \geq h_{s_0} \geq 0$. If $w(x) = 0$, (4.6) would give a strictly negative value. Division by $w(x) > 0$, followed by $w(x) \leq 1$, proves (4.7). Integrate (4.6), use the cumulative contact inequality and the nonnegativity of the corresponding integral of h , and again divide by the fixed number $w(x)$. This proves (4.8). $\square$

The elementary proof just given requires no positive lower bound on s . In applications of first variation we also cut off the Gaussian test field within the original chart. If $s \leq \rho \ll R$ and its center lies in the fixed interior region, this introduces a factor $\exp(-cR^2/s^2)$ times the bounded chart mass and fixed powers of s^{-1} . The same reasoning makes these errors smaller than any prescribed part of the subtractive margin. At a node of the recursion below, $s \geq r_b$ and $\rho/R = T^{-4N}$, so this remains true even after the factor $s^{-m} \leq \exp(O(T))$ is included. All subsequent Gaussian identities use this convention.

4.3 The estimate for separated configurations

We use the signed estimate [Theorem 3.1](#) only at smaller integers in this subsection. Separation refers to the scaled normal coordinates; the integrands defining Λ continue to use physical units. In this subsection write $z = \zeta = z_{\text{phys}}/l$, in the root chart; these height coordinates remain fixed at all recursive nodes.

For two lists $\mathcal{A} = \{a_1, \dots, a_Q\}$ and $\mathcal{B} = \{b_1, \dots, b_Q\}$, write

$$d_\infty(\mathcal{A}, \mathcal{B}) = \min_{\sigma \in \mathfrak{S}_Q} \max_{1 \leq j \leq Q} |a_j - b_{\sigma(j)}|.$$

Thus repetitions are distinct entries in a matching. Choose once $0 < r_{\text{maj}} < 1$ so that the fixed enlargements needed first to select the integer by [Theorem 2.5](#) and then to apply [Theorem 2.7](#) on the inner ball $B_{r_{\text{maj}}}$ lie in B_4 . The corresponding inner ball of a work ball $B_r(y)$ is $B_{r_{\text{maj}}r}(y)$.

We use comparison chains whose work radii change by factors at most two and whose center steps are at most one quarter of the smaller inner radius. For adjacent inner balls B, B' , these choices give a fixed $\vartheta_{\text{ov}} > 0$ such that

$$\text{vol}_S(B \cap B') \geq \vartheta_{\text{ov}}(\text{vol}_S(B) + \text{vol}_S(B')). \quad (4.9)$$

Indeed their intersection contains a ball of a fixed fraction of the smaller radius, and their radii are comparable. Fix $0 < \delta_{\text{maj}} < \vartheta_{\text{ov}}$, and let C_{maj} be a fixed constant valid in [Theorem 2.7](#) for this exceptional proportion and the finitely many fixed enlargements used below. Every auxiliary configuration below uses this same exceptional proportion. The corresponding smallness threshold in [Theorem 2.7](#) is therefore fixed before T ; the local bounds used to select these configurations tend to zero with T .

Suppose $s^{-m}\tau(B_{cs}(x)) \leq d_{\text{loc}}$ on a ball and its required fixed enlargement. A *majority configuration* on $B_s(x)$ is a list of Q heights such that, outside a set of base volume at most $\delta_{\text{maj}} \text{vol}_S(B_s(x))$, each fiber has exactly Q projection entries and these entries match the list within $C_{\text{maj}}s\sqrt{d_{\text{loc}}}$. The projection entries repeat each spatial point according to its integer multiplicity. If majority configurations $\mathcal{A}, \mathcal{B}$ on adjacent balls of radii s, t have local upper bounds d_s, d_t , then

$$d_{\infty}(\mathcal{A}, \mathcal{B}) \leq C_{\text{maj}}(s\sqrt{d_s} + t\sqrt{d_t}). \quad (4.10)$$

In fact, (4.9) leaves a fiber outside both exceptional sets. Composing its two matchings proves the estimate, including repetitions.

At the root the inner ball is $B_{r_{\text{maj}}}$ and $d_{\text{loc}} = r_{\text{maj}}^{-m}d$, so its matching error is a fixed multiple of $\sqrt{d}$. The majority property itself imposes no condition on exceptional fibers; their contribution remains in the global τ budget. A partition of a list's entries into groups is separated by δ if different groups have mutual distance at least δ . Identical list entries therefore belong to the same group when $\delta > 0$.

Proposition 4.3 (Separated configurations). *Fix $Q > 1$ and assume [Theorem 3.1](#) for every integer $1 \leq q < Q$. For each fixed $P > 0$ there are constants C_Q, C_S and a function $b = b_Q$ satisfying*

$$b(T) \gg (\log T)^{200}, \quad b(T) = T^{o(1)}, \quad F_{Q-1}(b(T)/2) \geq (\log T)^3 \quad (4.11)$$

such that the following holds. Under the hypotheses of [Theorem 3.1](#), let $\mathcal{A} = \{a_1, \dots, a_Q\}$ be any majority configuration on $B_{r_{\text{maj}}}$ with local upper bound $d_{\text{loc}} = r_{\text{maj}}^{-m}d$, and suppose it has a partition into at least two nonempty groups separated by $10S(T)\sqrt{d}$, where

$$S(T) = e^{C_S b(T)}, \quad c(T) = \frac{(\log T)^2}{T}.$$

Let D_{far} be the projection of tilt restricted to $\{\text{dist}(z, \mathcal{A}) > S(T)\sqrt{d}\}$. Then every nonnegative weight $w \leq 1$, supported in B_1 with $\text{Lip } w \leq 1$, satisfies

$$\int w \, d(L - C_Q \Lambda + c(T) D_{\text{far}}) \leq T^{-P} d. \quad (4.12)$$

The statement holds on other fixed interior balls after rescaling. Polynomial factors in the Lipschitz norm or the enlargement of a Gaussian working ball can be allowed by increasing P, C_S and the fixed constants.

Proof. The proof is a finite recursion over working balls, with the root configuration and its far-tilt region kept fixed. On a sufficiently small ball, the support lies in disjoint neighborhoods of the root groups. Cutting through the gaps gives stationary pieces with positive counts below Q , so the lower-integer theorem makes the signed integral small on that ball.

On a larger ball, if the signed integral exceeds a small margin, the remaining positive source gives a contact measure. The cumulative contact inequality and the Gaussian mass identity exclude a contact whose derivative scale is close to a threshold hit or a side equality: it would force too much signed mass at the outer scale. Bottom contacts are excluded by the small-ball reduction. Each remaining contact lies on a vertical side, far below its equality scale. Its signed Gaussian integral is partitioned over smaller working balls, while a choice of thresholds made before constructing the contact measure confines the associated charging balls to a small part of the measure on the working ball. Thus a large generation pays a small margin and at most half of the next-generation bound. The choices below enforce these alternatives and keep the configuration motion smaller than the original separation.

For example, after extending F_{Q-1} monotonically over a fixed initial interval, one can use

$$b(T) = 1 + (\log T)^{201} + 4F_{Q-1}^{-1}((\log T)^3).$$

The definition of the iterated exponentials gives $F_{Q-1}^{-1}((\log T)^3) = T^{o(1)}$, and this choice has (4.11). All changes of b below are by fixed constants or an additional fixed power of $\log T$.

Choose a fixed integer $N > P + 10$, increasing it later if necessary, and then choose a fixed J_0 sufficiently large in terms of m, n, Q, N . In particular require

$$J_0 > 4N(m + 2) + 100, \quad (4.13)$$

and enlarge J_0 to exceed all fixed polynomial density losses in the cell constructions below. Set

$$\eta = T^{-N}, \quad \varepsilon = \frac{(\log T)^{60}}{T}, \quad r_b = \sqrt{d}e^{3b}, \quad n_0 = \lceil 10N \log T \rceil, \quad E_j(t) = T^{J_0(j+1)} d t^{-2-\varepsilon}. \quad (4.14)$$

The parameter T , the original configuration $\mathcal{A}$, and the measure D_{far} remain fixed throughout the finite recursion. They are never recomputed at a child node.

Two elementary parameter bounds will be used repeatedly. With $\ell_T = \log T$, direct substitution gives

$$\log E_{2n_0+4}(r_b/T^4) = -6b + \frac{1}{2}\ell_T^{60} - 3\varepsilon b + O_{J_0,N}(\ell_T^2) < -2b. \quad (4.15)$$

Also, for $i \leq n_0$,

$$v\sqrt{E_{2i+1}(v)} = \sqrt{d}T^{J_0(i+1)}v^{-\varepsilon/2}. \quad (4.16)$$

For $v \geq cr_b/T^2$, the logarithm of the factor multiplying $\sqrt{d}$ is $O_{J_0,N}((\log T)^2) + O((\log T)^{60})$. Consequently $O(T)$ such increments, even with any fixed polynomial loss in T , sum to less than $e^b\sqrt{d}$. This is why the much larger separation $e^{Csb}\sqrt{d}$ survives every generation.

At generation i a node consists of a ball $B_{4R}(x_*)$ with

$$\tau(B_{4R}(x_*)) \leq CR^m E_{2i}(R), \quad R \geq cr_b/T^2, \quad (4.17)$$

and a majority configuration $\mathcal{A}_R$ on $B_{r_{\text{maj}}R}(x_*)$, with local upper bound $Cr_{\text{maj}}^{-m}E_{2i}(R)$, satisfying

$$d_\infty(\mathcal{A}_R, \mathcal{A}) \leq (i+1)e^b\sqrt{d}.$$

The integer is Q on the balls constructing the node; the small-density and overlap checks below establish this before their configurations are selected. We estimate all weights $0 \leq w \leq 1$, $\text{Lip } w \leq R^{-1}$, supported in $B_R(x_*)$. Write

$$L' = L - C_Q\Lambda + c(T)D_{\text{far}}, \quad \mathcal{N}_R = \tau(B_{4R}(x_*)) + dR^m.$$

We prove, by downward induction on i , that

$$\int w \, dL' \leq p_i \mathcal{N}_R, \quad p_{n_0} = C_*, \quad p_i = C\eta + \frac{1}{2}p_{i+1}. \quad (4.18)$$

Here C_*, C are fixed, independent of i and T . The last-generation bound follows at once from $|L'| \leq C\tau$. Initialization at $R = 1$ follows from $\tau(B_4) \leq d$ and the given configuration.

Small nodes and actual stationary pieces. Suppose first that $R \leq T^{10N} r_b$, and put $E = E_{2i}(R) < 1$, as follows from (4.15). Cover $B_{2R}(x_*)$ by a fixed finite number of balls of radius σR , including the concentric ball, where the fixed $\sigma > 0$ is small enough that every fixed enlargement needed for scalar selection and then for the support estimate in Theorem 2.7 lies in $B_{4R}(x_*)$, and the scalar enlargement of the concentric ball lies in $B_{2R}(x_*)$. Their normalized τ is at most CE . The normalized smooth mass averages therefore differ from Q by $O(E) < 1/4$, so Theorem 2.5 selects Q on these balls. On each choose the joint configuration in Theorem 2.7 with exceptional proportion δ_{maj} , retaining its majority property and using its all-support bound with a fixed $0 < \beta < \min\{1/2, 1/m\}$. Its local upper bound is $CE = o(1)$. Connect it to $\mathcal{A}_R$ by a fixed chain satisfying (4.9). The intermediate balls have the same $CE = o(1)$ upper bound, and the same smooth mass test selects Q there. Equation (4.10) bounds this change by $CR\sqrt{E}$, while the support bound is CRE^β . Consequently all support heights over $B_{2R}(x_*)$ satisfy

$$\begin{aligned} \text{dist}(z, \mathcal{A}) &\leq (i+1)e^b\sqrt{d} + CR\sqrt{E} + CRE^\beta \\ &\leq (i+1)e^b\sqrt{d} + CRE_{2i}(R)^\beta \leq e^{C'b}\sqrt{d} \end{aligned} \tag{4.19}$$

for a fixed C' . The second line uses $\beta < 1/2$ and $E < 1$. To see the last bound without a hidden scale restriction, write $R = \sqrt{d}e^{3b}T^\theta$, where $-3 \leq \theta \leq 10N$, and substitute in $RE_{2i}(R)^\beta$; its logarithm after division by $\sqrt{d}$ is bounded above by $C'b$ using (4.15). Choose $C_S > C' + 10$. For a root group $\mathcal{A}_g$, put

$$\mathcal{O}_g = \{z : \text{dist}(z, \mathcal{A}_g) < 2S(T)\sqrt{d}\}.$$

For large T , (4.19) puts all the support over B_{2R} inside the smaller regions with radius $S(T)\sqrt{d}$. The regions $\mathcal{O}_g$ are disjoint, since the root group gap is $10S(T)\sqrt{d}$. Thus there is an open height gap around each group's support, and no far energy in this interior region.

The use of disjoint height regions to obtain stationary restrictions has a precedent in [5, Section 3.1]. For each group, restrict the varifold over $B_{2R}(x_*)$ to $\{z \in \mathcal{O}_g\}$. Choose a smooth height cutoff equal to one near the smaller region of radius $S(T)\sqrt{d}$ around $\mathcal{A}_g$ and zero outside $\mathcal{O}_g$. Its derivative vanishes near the entire support over $B_{2R}(x_*)$. Multiplying any ambient test field with compact base support there by this cutoff proves stationarity of the restriction. It keeps every included spatial point with its entire integer multiplicity, including on exceptional fibers. On a majority fiber of one of the local configurations, its matching to the fixed root list moves each entry by at most $(i+1)e^b\sqrt{d} + CR\sqrt{E} = o(S(T)\sqrt{d})$. An entry matched to $\mathcal{A}_g$ therefore lies in $\mathcal{O}_g$. If two copies of one spatial point were matched to different root groups, their two root entries would be separated by at most twice this error, less than $10S(T)\sqrt{d}$ for large T , contrary to the separation. Each group's projection count on that fiber is thus the number q_g of entries in its root group, including repetitions. Since there are at least two nonempty groups, $1 \leq q_g < Q$ and $\sum_g q_g = Q$. To specify the individual energy budgets, take a finite nonnegative partition of the weight on balls $B_{r_\alpha}(y_\alpha)$, with $r_\alpha = cR$ for a sufficiently small fixed c , and $B_{8r_\alpha}(y_\alpha) \subset B_{2R}(x_*)$. These enlarged balls have bounded overlap. For each stationary restriction, its normalized projected mass is bounded by the normalized total mass, and its normalized $D + \Lambda$ is $O(E) = o(1)$. Apply Theorem 2.5 with its initial intermediate ball equal to the concentric majority ball above. Its selected integer is q_g by intersection with the group majority set there. Connect this ball to the intermediate balls for the scalar applications on $B_{4r_\alpha}(y_\alpha)$ by a fixed finite chain of comparable interior balls with the positive overlap in (4.9). The scalar count exceptions on each intermediate ball have relative volume $O(E)$, smaller than the fixed overlap for large T ; on a remaining fiber the two selected integers coincide. Thus q_g is identified and propagated before the mass defect of the restriction is estimated.

Write μ_g for the mass measure of the restriction and D_g, Λ_g for its projected energy measures. Now set $M_g = \pi_{\#}\mu_g - q_g \text{vol}_S$ and $\tau_g = |M_g| + D_g + \Lambda_g$. The scalar estimate with interior room now gives

$$\tau_g(B_{4r_\alpha}(y_\alpha)) \leq C_s(D_g + \Lambda_g)(B_{8r_\alpha}(y_\alpha)).$$

Define the parameter for the lower-integer application of [Theorem 3.1](#) by

$$d_{g\alpha} = \max \{d, C_s r_\alpha^{-m}(D_g + \Lambda_g)(B_{8r_\alpha}(y_\alpha))\}, \quad T_{g\alpha} = -\log d_{g\alpha}. \quad (4.20)$$

Increasing the fixed C_s if necessary supplies the required bound for τ_g on each rescaled B_4 . Moreover

$$\sum_{g,\alpha} d_{g\alpha} r_\alpha^m \leq C((D + \Lambda)(B_{4R}) + dR^m) \leq C\mathcal{N}_R. \quad (4.21)$$

Here the group energies add exactly on their separated region, and the enlarged base balls have bounded overlap. Individual mass defects have been estimated by [Theorem 2.5](#), rather than by splitting $|M|$. The parameters satisfy

$$b/2 \leq T_{g\alpha} \leq T. \quad (4.22)$$

Indeed $d \leq d_{g\alpha} \leq d + CE_{2i}(R)$, and (4.15) absorbs the fixed constants. Physical radii only decrease, and physical heights are unchanged, so the curvature hypothesis follows from the original one. Apply [Theorem 3.1](#) at q_g to the partition weights normalized by a fixed constant. Taking C_Q at least as large as the finitely many constants for $q < Q$, summing the lower-integer conclusions with (4.21) gives an $o(\eta)\mathcal{N}_R$ bound. Indeed $F_q(T_{g\alpha}) \geq F_{Q-1}(b/2) \geq (\log T)^3$ for large T . This argument also applies to the small cells used at a bottom contact.

Stopping tests at a large node. Now assume $R > T^{10N}r_b$. Use a torus of period CR , identifying the node and a fixed surrounding chart with its central portion. Put

$$\nu = \tau|_{B_{4R}(x_*)} + d \, dy, \quad \rho = T^{-4N}R.$$

Its total mass is comparable to $\mathcal{N}_R$. If $f = wL' - \eta\nu$ has nonpositive integral there is nothing to prove. Otherwise apply [Theorem 4.1](#) with bottom r_b and dyadic test scales $t \in [r_b, \rho]$, using

$$q_t(x) = \frac{(\tau|_{B_{4R}(x_*)})_t(x)}{E_{2i+1}(t)}, \quad q_t(x) \leq \lambda_t, \quad 1 \leq \lambda_t \leq 2. \quad (4.23)$$

The thresholds will be chosen later. There are $O(T)$ test scales. The torus masses satisfy the growth condition in [Theorem 4.2](#), because $\mathcal{N}_R/(dR^m) \leq \exp(O(T))$ by (4.17). Also $\rho/R = T^{-4N} \ll \eta^2/T^2$.

For every contact (x, s_0) , localization places x in the interior support region. Gaussian comparison between successive dyadic scales, and the non-strict tests at larger scales, give

$$\tau_v(x) \leq CE_{2i+1}(v) \quad (s_0 \leq v \leq \rho). \quad (4.24)$$

Here and below the measure in this display is the node restriction. At the current mesh scale its value is bounded by the next larger scale. At the top use the node's total mass. In fact, for $v \geq \rho/16$,

$$q_v(x) \leq CT^{-J_0}(R/v)^{m-2-\varepsilon} \leq CT^{-J_0+4N \max(m-2,0)}.$$

Thus all top tests hold strictly, and every scale with $q_v \geq 1$ lies below $\rho/16$, leaving a fixed outward interval below ρ .

If $s_0 > r_b$, the boundary description (4.1) supplies a mesh scale $v_1 \geq s_0$ with $q_{v_1}(x) \geq 1$. Either it is a current hit, or it is a larger scale with $q_{v_1}(x) = \lambda_{v_1}$. In the latter case every still larger test satisfies its continuation inequality. This follows directly from the finite intersection: away from the bottom, if there were no current obstruction and every relevant larger test were strict, a space-scale neighborhood would belong to $\mathcal{D}$.

Exclusion of a nearby hit or side equality. We show that no such v_1 can satisfy $v_1/s_0 \leq (\log T)^{20}$. All the following estimates are on this one outward vertical. Put

$$I_1 = v_1^2 E_{2i+1}(v_1), \quad A(v) = v^2 M_v(x), \quad Z(s) = \int_{s_0}^s v^2 (L' - \frac{\eta}{4} \nu)_v(x) d \log v.$$

By [Theorem 4.2](#), $Z \geq 0$ and $A(s_0) \geq 0$; the latter uses $c(T) < 1$ and $L' = 2M - D - C_Q \Lambda + c(T) D_{\text{far}}$. The envelope gives

$$\int_{s_0}^{\rho} v^2 D_v d \log v \leq \frac{C I_1}{\varepsilon} (v_1/s_0)^{\varepsilon} \leq \frac{C I_1}{\varepsilon}, \quad \frac{c(T)}{\varepsilon} = (\log T)^{-58}. \quad (4.25)$$

In particular the adverse $c(T) D_{\text{far}}$ term is small enough for the logarithmic losses below.

At v_1 , and on its fixed outward dilations, the scalar estimate is available in weighted form:

$$|M|_{v_1} \leq C(D + \Lambda)_{v_1} + o(E_{2i+1}(v_1)). \quad (4.26)$$

Here are the raw density and integer checks, also needed for the children below. For $s \in [s_0, \rho]$ and $U = 1 + |y - x|/s \leq C\sqrt{T}$, put $r = cs/U$ with a small fixed c . All required fixed enlargements $B_{Lr}(y)$ lie in $B_{2R}(x_*)$: $x \in B_R(x_*)$ and $C\sqrt{T}\rho = o(R)$. Choose $a = C_1 U s$ so that $G_a(x - z) \geq c_1 a^{-m}$ on these enlarged balls. If $a \leq \rho$, positivity and the envelope give

$$r^{-m} \tau(B_{Lr}(y)) \leq C(a/r)^m E_{2i+1}(a) \leq C U^{2m-2-\varepsilon} E_{2i+1}(s).$$

If $a > \rho$, use the node total instead. Dividing by $E_{2i+1}(s)$ gives the bound

$$C T^{-J_0} U^m (R/s)^{m-2-\varepsilon} \leq C T^{-J_0+4N \max(m-2,0)} U^{m+\max(m-2,0)} \leq C.$$

In the middle inequality a negative exponent of R/s is discarded; for a positive exponent use $R/s \leq C U T^{4N}$. Consequently, in both cases,

$$r^{-m} \tau(B_{Lr}(y)) \leq C T^{\max(m-1,0)} E_{2i+1}(s) = o(1). \quad (4.27)$$

Since τ includes $|M| = |\pi_{\#} \mu - Q \text{vol}_S|$, choose a fixed nonnegative smooth mass test on this enlarged ball, normalized to reference integral one and with bounded scaled derivatives. Its average differs from Q by at most the right side times a fixed constant, hence by less than $1/4$. This identifies Q directly in [Theorem 2.5](#), and also provides its mass and energy hypotheses. Apply [Theorem 2.5](#) at $s = v_1$ on a cover by these cells and sum with the comparable Gaussian weights, as in [Theorem 2.8](#). The cutoff at $C\sqrt{T} v_1$ has arbitrarily small exponential tail for a large fixed C , using the node total and $v_1 \geq r_b$.

The identity in [Theorem 2.9](#), integrated using the nonnegative bank Z , gives

$$A(s) \geq A(s_0) + \int_{s_0}^s v^2 R_v d \log v + Z(s) - C(\log T)^{-58} I_1, \quad s_0 \leq s \leq \rho. \quad (4.28)$$

The geometric errors are absorbed by part of the remaining margin $\eta\nu/4$: on $|u| \leq T$ they carry the factor $C e^{-\kappa T} (1 + T^2) = o(\eta)$, and the tails are negligible under (4.24) and the node total. The nonnegative $C_Q \Lambda$ term has been discarded.

There is a fixed $c_1 > 0$ such that $A(v_1) \geq c_1 I_1$. This includes a current hit with $q_{v_1} > \lambda_{v_1}$: although its current test is omitted, the next larger mesh scale $t_+ \leq 2v_1$ obeys $\tau_{t_+} \leq 2E_{2i+1}(t_+)$, and $G_{v_1} \leq (t_+/v_1)^m G_{t_+}$ bounds its current average. Only $q_{v_1} \geq 1$ is needed for the lower bound.

For an explicit cumulative-sign argument, fix $C_Q \geq 1$ first and let C_s be the constant in (4.26). Write $E_1 = E_{2i+1}(v_1)$ and take T large enough that its tail is at most $E_1/2$ and $c(T) \leq 1/2$. Then

$$(D + \Lambda)_{v_1} \geq \alpha E_1, \quad \alpha = \frac{1}{2(C_s + 1)}.$$

If $A(v_1) \leq c_1 I_1$ with $c_1 \leq \alpha/8$, then $L'_{v_1} \leq 2c_1 E_1 - (D + \Lambda)_{v_1}/2 \leq -\beta E_1$, where $\beta = \alpha/4$. For $v \in [v_1, 2^{1/4} v_1]$, differentiating the fixed source L' and dominating $|\dot{G}_v|$ by $C G_{2v}$ gives $|\dot{L}'_v| \leq K_1 E_1$. The constant K_1 is fixed after C_Q ; the envelope at $2v$ follows from strictly larger mesh tests, and $v_1 < \rho/16$ leaves room. In particular this estimate does not use the omitted current test or differentiate the fixed far cutoff. Choose

$$\theta = \min\{\frac{1}{4} \log 2, \beta/(2K_1)\}, \quad \gamma = \beta\theta/2, \quad 0 < c_1 \leq \min\{\alpha/8, \gamma/4\}.$$

On $[v_1, e^\theta v_1]$ one has $L'_v \leq -\beta E_1/2$, hence $\dot{Z} \leq -\beta I_1/2$. Nonnegativity of Z at the right endpoint gives $Z(v_1) \geq \gamma I_1$. Finally take T large enough that the error in (4.28) is at most $\gamma I_1/2$. That inequality at v_1 yields $A(v_1) \geq \gamma I_1/2 > c_1 I_1$, a contradiction. All constants were fixed before this last largeness threshold; only the cumulative sign of Z was used at later scales.

The upper differential inequality $\dot{A} \leq 2A + v^2 R_v +$ absorbed errors, again from Theorem 2.9, and $v_1/s_0 \leq (\log T)^{20}$ now yield

$$A(s_0) + \int_{s_0}^{v_1} v^2 R_v \, d \log v \geq c(\log T)^{-40} I_1. \quad (4.29)$$

One may retain the geometric error explicitly in the integrating factor formula: divided by I_1 it is $O(e^{-\kappa T} T^C / \varepsilon)$, hence smaller than every fixed inverse power of $\log T$. Inserting (4.29) in (4.28) at ρ gives $A(\rho) \geq c(\log T)^{-40} I_1$. On the other hand the node mass gives

$$A(\rho) \leq C \rho^2 (R/\rho)^m E_{2i}(R) = C T^{J_0(2i+1)+4N(m-2)} dR^{-\varepsilon}, \quad I_1 = T^{J_0(2i+2)} dv_1^{-\varepsilon}.$$

The ratio of the upper bound to I_1 is at most $CT^{-J_0+4N(m-2)}$, contradicting (4.13). This excludes current hits and all side equalities within the stated logarithmic factor.

Gaussian children and the bottom. For any remaining contact, configurations on the Gaussian cells at s_0 match the node configuration with movement at most $e^b \sqrt{d}$. Here is the comparison chain for a cell with work radius comparable to s_0/U , where $U = 1 + |y - x|/s_0 \leq C\sqrt{T}$. First shrink concentrically at x_* to a small fixed multiple of R , move the center to x at that radius, and then shrink concentrically to a work radius comparable to Us_0 . Move the center to y at that radius and finally shrink to the cell radius. At each stage change radii by factors at most two and make center steps at most one quarter of the smaller inner majority radius. All these inner balls satisfy (4.9), so (4.10) bounds the changes by the sum of their local matching errors. The initial fixed chain stays inside $B_{4R}(x_*)$; the off-center chain at scale s_0 lies in $B_{CU s_0}(x) \subset B_{2R}(x_*)$. The envelope bounds these increments by (4.16), with fixed polynomial losses. More explicitly, above ρ the node total gives, for $\rho \leq t \leq cR$,

$$t \sqrt{t^{-m} \tau(B_{ct}(x))} \leq CR \sqrt{E_{2i}(R)} T^{2N \max(m-2, 0)}.$$

Off-center balls nested from scale Us_0 down to s_0/U have the same increment bound as $s_0 \sqrt{E_{2i+1}(s_0)}$, with another fixed power of $U \leq C\sqrt{T}$, by the raw estimates proving (4.27). All normalized bounds used for intermediate configurations are at most a fixed power of T times $E_{2i}(R)$ or $E_{2i+1}(t)$ for $t \geq cr_b/T^2$. Hence they tend to zero by (4.15). Their $|M|$ terms give the same smooth mass test selecting Q , and Theorem 2.7 is applied with the fixed δ_{maj} before each matching. The logarithms of all these factors are $O_{J_0, N, m}((\log T)^2) + O((\log T)^{60})$ after division by $\sqrt{d}$. Since there are $O(T)$ steps, (4.16) proves the claimed $e^b \sqrt{d}$ allowance.

Here is the precise normalization of the cells. A smooth partition of unity on $|u| \leq C\sqrt{T}$, where $u = (y - x)/s_0$, can be chosen with bounded overlap on balls $B_{r_j}(y_j)$, with

$$r_j \asymp \frac{s_0}{1 + |y_j - x|/s_0}, \quad |\nabla \chi_j| \leq C/r_j. \quad (4.30)$$

Use a sufficiently small fixed constant in $\asymp$ to leave room for $B_{4r_j}(y_j)$. This follows, for example, by a maximal disjoint selection for the Lipschitz radius function in (4.30), followed by normalized smooth bumps. On the enlarged ball, $G_{s_0}(x - y)$ differs from $G_{s_0}(x - y_j)$ by a fixed factor, and its logarithmic gradient is $O(r_j^{-1})$. Thus

$$w_j(y) = \frac{\chi_j(y)G_{s_0}(x - y)}{CG_{s_0}(x - y_j)}$$

is an admissible child weight after a fixed normalization. In particular, this normalization has no power of T .

The child's unweighted density was estimated in (4.27). The identity

$$E_{2i+2}(r_j) = T^{J_0} E_{2i+1}(s_0)(s_0/r_j)^{2+\varepsilon}$$

then gives

$$r_j^{-m} \tau(B_{4r_j}(y_j)) \leq CT^{\max(m-1, 0)} E_{2i+1}(s_0) \leq CE_{2i+2}(r_j),$$

by the choice of J_0 , with C independent of i, T . Also $r_j \geq cr_b/\sqrt{T} \geq cr_b/T^2$. The smooth mass test following (4.27) identifies Q on each child. After scaling its work ball to one, choose from Theorem 2.7 a majority configuration on $B_{r_{\text{maj}}}$ with local upper bound $Cr_{\text{maj}}^{-m} E_{2i+2}(r_j)$. Using the same normal frame, multiply its heights by r_j to return to the root coordinates. The matching argument above gives its allowance relative to $\mathcal{A}$, so these are admissible next-generation nodes. Although the raw density estimate loses a polynomial factor, summation of the resulting estimates does not: bounded overlap and Gaussian comparability give

$$\sum_j G_{s_0}(x - y_j) [\tau(B_{4r_j}(y_j)) + dr_j^m] \leq C(\tau_{s_0}(x) + d). \quad (4.31)$$

The fixed d floors therefore sum to the correct constant-density Gaussian average.

At a bottom contact $s_0 = r_b$, these cells are small nodes. The stationary-piece argument above, including (4.22), bounds L'_{s_0} by $o(\eta)\nu_{s_0}$. The Gaussian tail outside $C\sqrt{T}$ is $o(\eta)d$ for a sufficiently large fixed C : combine its $e^{-C^2 T/4}$ factor with total node mass and $s_0^{-m} \leq \exp(O(T))$. This contradicts (4.7); there are no bottom contacts. At any other remaining contact the next-generation bound and (4.31) give

$$h_{s_0}(x) \leq C_0 p_{i+1} \nu_{s_0}(x). \quad (4.32)$$

All omitted tails are absorbed by the subtractive margin in (4.7). In particular, every remaining contact is a vertical-side contact with some equality scale

$$v_1/s_0 > (\log T)^{20}. \quad (4.33)$$

First charging step: replace a Gaussian by a nearby ball. Put $J = \log T$ and $p = p_{i+1}$. Call a contact (x, s) good if some closed ball of radius $0 < r \leq Js$ satisfies

$$h(\overline{B}_r(x)) \leq A_0 p \nu(\overline{B}_r(x)), \quad (4.34)$$

where $A_0 \geq 2C_0$ is a fixed constant. Otherwise call it bad. We first show that all bad centers together carry only $o(\eta)\mathcal{N}_R$ mass of h .

For the Euclidean Gaussian, radial layer cake gives

$$\nu_s(x) = c_m s^{-m} \int_0^\infty t e^{-t^2/2} \nu(B_{ts}(x)) dt.$$

At a bad contact the failure of (4.34) for every $r \leq Js$ bounds the part with $t \leq J$ by $h_s/(A_0p)$. Define the remaining layer-cake tail by

$$\mathcal{T}_{J,s}\nu(x) = c_ms^{-m} \int_J^\infty te^{-t^2/2}\nu(B_{ts}(x)) dt.$$

This is a tail in the layer-cake radius: it includes the residual weight $c_ms^{-m}e^{-J^2/2}\nu(B_{Js}(x))$ as well as mass farther away. Combining $\nu_s \leq h_s/(A_0p) + \mathcal{T}_{J,s}\nu$ with (4.32) gives

$$h(B_{5s}(x)) \leq C_ms^m h_s(x) \leq C_m p s^m \mathcal{T}_{J,s}\nu(x). \quad (4.35)$$

Let E_{bad} consist of spatial contact centers that admit no good representative; for each of them assign one contact scale. The elementary $5r$ covering lemma selects disjoint assigned balls whose fivefold enlargements cover these centers. Their scales lie in $[r_b, \rho]$, hence in $O(T)$ dyadic bins. In a bin $[a, 2a)$, disjointness implies, for every y and $t \geq 1$,

$$\sum_j \mathbf{1}_{B_{ts_j}(x_j)}(y) \leq C_m(1+t)^m :$$

the disjoint balls of radius at least a have centers in a ball of radius Cta . Sum (4.35), use Tonelli, and then sum the bins to obtain

$$h^*(E_{\text{bad}}) \leq C_m p T(1+J)^{m+1} e^{-J^2/2} \nu(X) = o(\eta) \mathcal{N}_R. \quad (4.36)$$

Here h^* denotes outer measure, so no measurable choice of a contact above each center is needed. The same proof on the torus uses periodic lifts of h, ν : the small selected balls have disjoint lifts, the packing estimate is unchanged, and the periodic Gaussian is exactly the sum over those lifts. Alternatively the nonnearest copies can be absorbed in the exponentially smaller chart tail.

Second charging step: localize the union of good balls. Every good ball lies in a set determined only by the node's tests and thresholds. To see this, let (x, s) be its contact and choose the equality scale v_1 in (4.33). For every mesh scale $v \geq v_1$ and every $|y - x| \leq Js$, Gaussian differentiation gives

$$|\nabla q_v(y)| \leq C/v. \quad (4.37)$$

Indeed $|\nabla G_v(y - z)|$ is bounded by $Cv^{-1}G_{2v}(x - z)$ when $|y - x| \leq v/10$; the envelope at $2v$ proves the assertion when $2v \leq \rho$. For $2v > \rho$ the node total and the top slack following (4.23) give the same bound. Since $Js/v_1 < J^{-19}$, for sufficiently large T every point of the good ball satisfies

$$q_v(y) \leq \lambda_v + \delta \quad (v > v_1), \quad |q_{v_1}(y) - \lambda_{v_1}| \leq \delta, \quad \delta = J^{-15}. \quad (4.38)$$

Order the mesh scales $v'_1, \dots, v'_K$ from largest to smallest, and define the Borel set

$$\mathcal{E}_\lambda = \bigcup_{k=1}^K \left[\{|q_{v'_k} - \lambda_{v'_k}| \leq \delta\} \cap \bigcap_{j < k} \{q_{v'_j} \leq \lambda_{v'_j} + \delta\} \right]. \quad (4.39)$$

Equation (4.38) says that every original good ball, not just its center, lies in $\mathcal{E}_\lambda$.

We use the following form of the Besicovitch covering theorem [19, Chapter 1, Lemma 4.5]. For a bounded set with one assigned positive-radius ball at each point and bounded assigned radii, there is a countable subfamily covering the set by the original balls, with multiplicity bounded by a dimensional constant N_m . Arbitrarily small assigned radii are allowed. One proof groups the radii into dyadic bins and, in decreasing bins, greedily selects uncovered

centers. The selected centers in each bin are separated by that bin's minimum radius, so only finitely many are needed in a bounded region. For the overlap bound at a point, divide directions into finitely many cones of angular diameter less than $\pi/3$. In a cone take the first selected center x_i among balls containing the point y . For a later such center x_j , the cosine law gives $|x_i - x_j| < \max(|x_i - y|, |x_j - y|) \leq \max(r_i, r_j)$. Since x_j was not covered by the earlier ball, $|x_i - x_j| > r_i$; hence $r_j > r_i$. The decreasing order of the dyadic bins forces these two radii into the same bin. Packing in that bin bounds the number. A selected center equal to y contributes at most one additional ball. This proves the dimensional multiplicity bound.

Choose one good contact and one good ball above each center that has such a representative, and apply this covering fact. Since the original balls lie in $\mathcal{E}_\lambda$, (4.34) gives

$$h^*(E_{\text{good}}) \leq A_0 p \sum_j \nu(B_j) \leq A_0 N_m p \nu(\mathcal{E}_\lambda). \quad (4.40)$$

This is the reason for using original balls in this second covering; the inequality (4.34) need not hold on an enlargement.

Choosing the thresholds. The measure ν and the functions q_v are already fixed by the node. Choose independent thresholds at the mesh scales with distribution function

$$F(t) = \begin{cases} 0, & t \leq 1, \\ \exp(1 - 1/(t - 1)), & 1 < t < 2, \\ 1, & t \geq 2. \end{cases} \quad (4.41)$$

For every fixed $K_0 > 0$, uniformly in real z ,

$$\mathbb{P}\{|\lambda - z| \leq \delta\} \leq T^{-K_0} + C_{K_0}(\log T)^{-13} F(z - \delta). \quad (4.42)$$

If $F(z + \delta) \leq T^{-K_0}$ this is immediate. Otherwise, put $a_* = (1 + K_0 \log T)^{-1}$. One has $z + \delta - 1 \geq a_*$, also when $z + \delta \geq 2$, and consequently $z - \delta - 1 \geq a_* - 2\delta \geq a_*/2$ for large T . Thus this case does not meet the lower endpoint 1. On the part of $[z - \delta, z + \delta]$ below 2 one has $(\log F)' = (t - 1)^{-2} \leq 4/a_*^2 \leq C_{K_0}(\log T)^2$; above 2, $\log F = 0$. Integration proves $F(z + \delta) - F(z - \delta) \leq C_{K_0}(\log T)^{-13} F(z - \delta)$. Truncation at 2 only decreases this difference. This proves (4.42), including intervals meeting either endpoint.

Fix y , and write $z_k = q_{v'_k}(y)$ and $r_k = F(z_k - \delta)$. At the k th event in (4.39), all preceding thresholds have survived, namely $\lambda_{v'_j} \geq z_j - \delta$. Independence, (4.42), and a union bound give

$$\begin{aligned} \mathbb{P}\{y \in \mathcal{E}_\lambda\} &\leq K T^{-K_0} + C_{K_0}(\log T)^{-13} \sum_{k=1}^K r_k \prod_{j < k} (1 - r_j) \\ &\leq K T^{-K_0} + C_{K_0}(\log T)^{-13} = o(1). \end{aligned}$$

The sum telescopes to $1 - \prod_k (1 - r_k) \leq 1$; no disjointness of the candidate events has been assumed. Since $K = O(T)$, any fixed $K_0 > 2$ suffices. Integrating against the fixed measure ν and using Tonelli supplies thresholds with

$$\nu(\mathcal{E}_\lambda) \leq o(1) \nu(X).$$

Choose such thresholds first, and only then construct the contact measure. This order makes independence explicit: the contact measure may depend on the chosen thresholds, but neither ν nor q_v does. For large T , the last bound pays for A_0, N_m and the fixed factor between $\nu(X)$ and $\mathcal{N}_R$. Equations (4.36) and (4.40) imply

$$h(X) \leq \frac{1}{2} p_{i+1} \mathcal{N}_R + o(\eta) \mathcal{N}_R.$$

Because $h(X) = \int w \, dL' - \eta\nu(X)$, this is (4.18) after increasing the fixed C .

Finally solve the recurrence:

$$p_0 \leq 2C\eta + 2^{-n_0}C_*, \quad 2^{-n_0} \leq T^{-10N \log 2}.$$

At the initial node $\mathcal{N}_1 \leq 2d$. Increasing the fixed N relative to the requested P gives (4.12). There have been at most n_0 configuration changes, costing $(n_0 + 1)e^b\sqrt{d} \ll S(T)\sqrt{d}$ in total. Thus the fixed root separation and far cutoff used at every step are valid. A polynomial weight or dilation loss is handled by starting with a correspondingly larger P and using the same Gaussian cell normalization. This proves the final assertion as well. $\square$

5 The outward envelope and the low-frequency barrier

We continue the induction for Theorem 3.1 at the integer Q . Thus Theorem 4.3 is available when $Q > 1$, using only the assertion at smaller integers. We shall construct a positive envelope $I(s)$ of future signed masses $A(s) = s^2 M_s$ and follow its logarithmic level $a(s) = \log(s^2/I(s))$. The output in Theorem 5.5 is a traversal from $a = B$ to $a = 2B$ in one of Q fixed bands, with no merger of its height centers and with $k \geq B^{-\varepsilon_0}$ for the coefficient k of the height moment, where $\varepsilon_0 > 0$ is fixed and small. This holds for every contact selected below whose outward vertical contains a threshold equality; Section 6 will turn the traversal into many scales with normal flux.

All heights in this section are in the scaled coordinates of Theorem 3.1; the quantities defining Λ retain their physical units. Gaussian averages are centered at the contact under consideration. The dot denotes differentiation with respect to $t = \log s$.

Fix, in this order, a sufficiently large Gaussian exponent $K > 2m + 100$, a sufficiently large inverse-polynomial exponent P_0 , and the constants in the separated estimate. They may depend on m, n, Q and the fixed coordinate bounds. Increasing these fixed constants only increases the lower threshold for T . Put

$$F = F_Q(T), \quad \eta = e^{-2F}, \quad a_0 = F^2, \quad d_* = e^{-a_0}, \quad \rho = \eta^4 T^{-2}. \quad (5.1)$$

The elementary rate comparisons proved in Theorem 5.5 include $F = T^{o(1)}$, $a_0 = o(T)$, and $|\log \rho| = o(T)$.

5.1 Selection and the bottom boundary

Extend the localized measures to the fixed torus of Theorem 4.2, and write $\nu = \tau + d \, dy$, with τ restricted to a fixed interior enlargement of the working set. For a weight w in Theorem 3.1, use the source

$$f = w(L - C_Q \Lambda) - \eta\nu. \quad (5.2)$$

To prove the required upper bound it is enough to exclude $\int f \geq \eta d$: the total mass of ν is at most Cd , and $(C + 1)\eta \leq e^{-F}$ for large T .

Start at ρ , continue downwards while $\tau_s(x) \leq d_*$, and stop at a threshold equality or at a positive bottom σ . Use the continuous entrance boundary, including its vertical sides, from Theorem 4.1. At the top the continuation inequality is strict, since $\tau_\rho \leq Cd\rho^{-m} = o(d_*)$. Let K_σ denote this compact entrance boundary. Write $\hat{h}_\sigma$ for its contact measure, $h_\sigma = (\pi_X)_\# \hat{h}_\sigma$ for the spatial projection, and $\mathcal{S}_\sigma = \text{spt } \hat{h}_\sigma$. Both measures have mass $\int f$. In this section a contact means a point of $\mathcal{S}_\sigma$, where Theorem 4.1 supplies the pointwise and cumulative contact inequalities. At every contact $(x, \underline{s})$,

$$(h_\sigma)_{\underline{s}}(x) \leq f_{\underline{s}}(x) \leq Cd_*, \quad \tau_v(x) \leq d_* \quad (\underline{s} \leq v \leq \rho). \quad (5.3)$$

Here and below source localization first places the contact in the fixed interior region and gives $w(x) > 0$. The constants in (5.3) are independent of σ .

Define the threshold event

$$\mathcal{G}_\sigma = \{(x, u) \in K_\sigma : \tau_v(x) = d_* \text{ for some } v \in [u, \rho]\}.$$

The corresponding set of $((x, u), v)$ is closed in $K_\sigma \times [\sigma, \rho]$, by continuity of the Gaussian averages. Its projection $\mathcal{G}_\sigma$ is therefore compact and Borel. The strict top inequality excludes $v = \rho$. If a boundary point above the bottom had no equality, the continuation inequalities would be uniformly strict on its compact outward interval; continuity would then make it an interior point. Hence $K_\sigma \setminus \mathcal{G}_\sigma$ consists precisely of bottom points with no threshold equality.

Lemma 5.1 (Vanishing contribution from the bottom). *The mass $\widehat{h}_\sigma(K_\sigma \setminus \mathcal{G}_\sigma)$ is at most*

$$o_{\sigma \downarrow 0}(1) + Cd_*d.$$

Every contact in $\mathcal{G}_\sigma$ has a scale $s_0 \in [\underline{s}, \rho)$ with $\tau_{s_0}(x) = d_$. The inequalities (5.3) hold also when $\underline{s} \leq s_0$.*

Proof. Only the first assertion needs proof. Put

$$\beta_\sigma = (\pi_X)_\#(\widehat{h}_\sigma \upharpoonright_{K_\sigma \setminus \mathcal{G}_\sigma}) \leq h_\sigma,$$

and let $\mathcal{C}_\sigma$ be the centers of contacts in $K_\sigma \setminus \mathcal{G}_\sigma$. Assign these contacts to grid cubes of side σ . If such a cube J contains a contact x_J , the Gaussian lower bound on J , together with the contact inequality, gives

$$h_\sigma(J) \leq C\sigma^m(f_\sigma(x_J))_+ \leq Cd_*\sigma^m.$$

Since $\beta_\sigma \leq h_\sigma$, this also bounds $\beta_\sigma(J)$. On an occupied half-open grid cube define

$$g_\sigma|_J = C \sup_{x \in \mathcal{C}_\sigma \cap J} (f_\sigma(x))_+, \quad \bar{\beta}_\sigma = \sum_J \frac{\beta_\sigma(J)}{|J|} \mathbf{1}_J \, dy,$$

and put $g_\sigma = 0$ on unoccupied cubes. Then $\bar{\beta}_\sigma \leq g_\sigma \, dy$ and $0 \leq g_\sigma \leq Cd_*$. For every continuous test φ ,

$$\left| \int \varphi \, d\beta_\sigma - \int \varphi \, d\bar{\beta}_\sigma \right| \leq \omega_\varphi(C\sigma)\beta_\sigma(\text{chart}) \longrightarrow 0,$$

where ω_φ is its modulus of continuity.

For a finite measure λ , at almost every Lebesgue point of its absolutely continuous density one has

$$\sup_{|x-y| \leq C\sigma} |(G_\sigma * \lambda)(x) - \lambda_{\text{ac}}(y)| \longrightarrow 0.$$

For the singular part the corresponding limit is zero almost everywhere. To check the assertion, decompose into dyadic annuli about y ; the differentiation bound on the mass in each sufficiently small ball is uniform in the indicated displacement, and the remaining annuli have an exponentially small Gaussian weight. Apply this first to smooth approximations of the absolutely continuous part and then to its L^1 remainder. Apply this assertion simultaneously to τ and f . If $\tau_{\text{ac}}(y) > d_*$, the cube containing y is eventually unoccupied, since all its contacts satisfy $\tau_\sigma \leq d_*$. At every other simultaneous differentiation point the uniform displacement bound controls the supremum defining g_σ . Therefore

$$\limsup_{\sigma \downarrow 0} g_\sigma(y) \leq C \mathbf{1}_{\{\tau_{\text{ac}} \leq d_*\}}(f_{\text{ac}})_+(y) \quad \text{almost everywhere.}$$

Reverse Fatou, using the common majorant Cd_* on the fixed chart, and the weak-test estimate above show that every bottom limit has density bounded by

$$C \mathbf{1}_{\{\tau_{\text{ac}} \leq Cd_*\}}(f_{\text{ac}})_+. \quad (5.4)$$

The value of the fixed constant in the threshold is immaterial.

At a point in this set, the area formula and [Theorem 2.3](#) give exactly Q projection entries, counted with their integer multiplicities. Indeed the difference between their integer count and Q is bounded by $C\tau_{\text{ac}} < 1$. Every entry has $e \leq Cd_*$; an entry with larger tilt would itself contribute more to the projected density of D . The part on which the projection has deficient rank projects to a Lebesgue null set, by the area formula for rectifiable sets. The expansion of the projection Jacobian at each remaining entry is

$$2(1 - J_\pi) - e \leq Ce^2 + C(|z_{\text{phys}}|^2 + |H_S|^2).$$

Sum this inequality with weight θ/J_π . Since the count is Q , its left side is exactly the density of L , with respect to base volume. Taking C_Q large enough absorbs the displayed geometry term and gives $(f_{\text{ac}})_+ \leq Cd_*D_{\text{ac}}$ on the set in (5.4). Its integral is at most Cd_*d . Compactness and (5.4) prove the stated uniform limsup bound, hence the assertion. $\square$

5.2 The future envelope

Fix a contact in $\mathcal{G}_\sigma$, and suppress x and σ from the notation. Define

$$\begin{aligned} f_* &= L - C_Q \Lambda - \frac{\eta}{4}(\tau + d \, dy), & A(s) &= s^2 M_s, \\ Z(s) &= \int_{\underline{s}}^s v^2 (f_*)_v \, d \log v, & J(s) &= C_Q s^2 \Lambda_s + \frac{\eta}{4} s^2 (\tau_s + d). \end{aligned} \quad (5.5)$$

The localization inequality and the contact inequalities imply

$$Z(s) \geq 0, \quad (f_*)_{\underline{s}} \geq 0, \quad A(s) \geq Z(s), \quad \dot{A} \geq \dot{Z}. \quad (5.6)$$

Here $A(\underline{s}) \geq 0$. To verify the last two assertions, use [Theorem 2.9](#) and integrate the nonnegative radial term and the margin. The geometric error is absorbed in, for example, half of the $\eta\nu/4$ margin. Uniformly down to the bottom, split its Gaussian integral at $|y - x|/s = T$. On the body its coefficient is $Ce^{-\kappa T}(1 + T^2) \ll \eta$; on the tail use (5.3) at larger Gaussian scales and the total mass above ρ . The tail is $o(\eta d)$. Thus this absorption uses no lower bound on $\underline{s}$. Only the first contact scale has a pointwise sign; (5.6) is the sign information used at subsequent scales.

To compare configurations at different scales, we need one quantity that controls the unsigned excess averages at larger scales up to ρ , allowing a fixed power of the radius ratio. The next lemma obtains it from future signed mass. At a maximizing scale v for a weighted future τ -average, a value of $A(v)$ that is a sufficiently small multiple of $v^2\tau_v$ would make f_* negative on a short outward interval, spending more of Z than $0 \leq Z \leq A$ allows. The resulting envelope also controls the Gaussian tails needed for the height comparison.

Lemma 5.2 (Future envelope). *Set*

$$I(s) = \max \left\{ \sup_{s \leq v \leq \rho} (s/v)^K A(v), (s/\rho)^K \rho^2 d \rho^{-2m-4} \right\}, \quad a(s) = \log \frac{s^2}{I(s)}. \quad (5.7)$$

Starting at a threshold scale s_0 , stop at the first scale s_ with $a(s_*) = T/2$. This scale exists, and on the path $[s_0, s_*]$ one has*

$$a(s_0) = a_0 + O(1), \quad a(s) \geq a_0 - O(1), \quad v^2\tau_v \leq C(v/s)^K I(s) \quad (s \leq v \leq \rho). \quad (5.8)$$

The functions I, a are locally Lipschitz in $\log s$. Almost everywhere they satisfy

$$\begin{cases} I = A, & \dot{I} = \dot{A}, & \text{on the reset set } \{I = A\}, \\ \dot{I} = KI, & & \text{on its complement,} \end{cases} \quad |\dot{a}| + \frac{|\dot{I}| + |\dot{Z}|}{I} \leq C. \quad (5.9)$$

Moreover $\dot{Z}/I$ is locally Lipschitz, with derivative bounded by C . For every fixed P , weighted scalar control gives

$$|M|_s \leq C(D_s + \Lambda_s) + a^{-P} I(s)/s^2. \quad (5.10)$$

All constants are uniform in the contact and the bottom.

Proof. Write I_{end} for the artificial term in (5.7). The definition makes I positive and continuous. Let

$$J_\tau(s) = \sup_{s \leq v \leq \rho} (s/v)^K v^2 \tau_v.$$

We apply the following argument first at $s = s_0$, and then after initialization up to the first crossing of $a = T/2$. At s_0 the threshold and $K > 2$ give the exact value $J_\tau(s_0) = s_0^2 d_*$, realized at $v = s_0$, whereas $I_{\text{end}}(s_0) = o(s_0^2 d_*)$. At any later scale before that crossing, $I(s)/s^2 \geq e^{-T/2}$ and $I_{\text{end}}(s)/s^2 \leq e^{-T+o(T)}$, so I is realized by an actual future value of A . Since $\tau \geq |M|$, $J_\tau \geq I > I_{\text{end}}$. At a maximizing scale v , writing $\delta = \tau_v$, these observations give

$$e^{-T/2} \leq \delta \leq d_*, \quad \tau_t \leq (t/v)^{K-2} \delta \quad (v \leq t \leq \rho). \quad (5.11)$$

The first lower bound holds at initialization as well. For any fixed N , a maximizer with $v \geq \rho/T^N$ would, by the root total mass bound, satisfy

$$\frac{J_\tau(s)}{I_{\text{end}}(s)} \leq C \rho^{m+4} T^{N(K+m-2)} = o(1),$$

which is impossible. Thus all fixed polynomial dilations needed below, including the body $C\sqrt{|\log \delta|}v$, lie within ρ .

Here is the raw density estimate on that body. A cell centered at y , with $|y - x| \leq Rv$ and radius $r = cv/(1 + R)$, lies with its fixed enlargement inside a ball of radius $C(1 + R)v$ about x . The Gaussian lower bound at that radius and (5.11) imply

$$r^{-m} \tau(B_{4r}(y)) \leq C(1 + R)^{2m+K-2} \delta. \quad (5.12)$$

For $R \leq C\sqrt{|\log \delta|}$ this is small, uniformly in $\delta \leq d_*$. Moreover τ includes $|M| = |\pi_\# \mu - Q \text{vol}_S|$, so the normalized smooth mass average on each cell is close to that same Q . The integer in Theorem 2.5 is therefore Q on every cell. The Gaussian has a bounded ratio on a fixed enlargement of such a cell, and Theorem 2.8 sums the estimates with bounded overlap.

Annular summation of (5.11) bounds the Gaussian tail beyond Rv , including any fixed derivative polynomial, by $C\delta \text{poly}(R)e^{-cR^2}$ up to ρ . Above ρ the root mass bound gives a remainder at most $Cdv^{-m} \text{poly}(\rho/v)e^{-c(\rho/v)^2}$. Since $v < \rho/T^N$, $\delta \geq e^{-T/2}$, and $|\log \rho| = o(T)$, this is $o(\delta)$, uniformly even as v tends to zero. Increasing the fixed body constant makes the full tail $o(\delta)$. Consequently

$$|M|_v \leq C(D_v + \Lambda_v) + o(\tau_v), \quad \tau_v \leq C(D_v + \Lambda_v).$$

Suppose $A(v) \leq \varepsilon v^2 \tau_v$. For C_Q sufficiently large and then ε sufficiently small,

$$(f_*)_v \leq -c\tau_v.$$

Gaussian differentiation and the maximizing dilation bound show that the derivative, in logarithmic scale, of $2M_r - D_r - C_Q \Lambda_r$ is bounded by $C_K \tau_v$ on a fixed short outward interval.

Thus, for a fixed $\vartheta > 0$, $(f_*)_r \leq -c\tau_v/2$ on $[v, e^\vartheta v]$. The negative margin may simply be omitted in obtaining this upper bound. By (5.6),

$$Z(e^\vartheta v) \leq Z(v) - c'\vartheta v^2 \tau_v \leq \varepsilon v^2 \tau_v - c'\vartheta v^2 \tau_v < 0$$

if $\varepsilon < c'\vartheta$. This contradicts the cumulative contact sign. Hence $A(v) \geq cv^2 \tau_v$ at an uncontrolled maximizer. At s_0 this first proves $I(s_0) \geq cs_0^2 d_*$; at each subsequent path scale it proves the last inequality in (5.8). This order uses only maximality and the root mass bound when establishing the raw cell estimates. The same cell argument, now using the proved dilation bound, gives (5.10) at every path scale; any prescribed inverse power is obtained by enlarging the Gaussian body constant.

At s_0 , the threshold equality and the just proved bound give $I(s_0) \geq cs_0^2 d_*$. The opposite bound follows from $A(v) \leq v^2 d_*$, $K > 2$, and the negligible artificial endpoint term. For that last point, $d_* \leq Cds_0^{-m}$ implies $s_0 \leq C \exp(-(T - a_0)/m) \ll \rho$. At the outer endpoint the artificial term dominates, and the exact formula is

$$a(\rho) = T + (2m + 4) \log \rho = T - o(T) > T/2.$$

Before the first crossing of $T/2$, the nonendpoint terms have $I(s)/s^2 \leq d_*$, and the endpoint term has $I_{\text{end}}(s)/s^2 \leq d\rho^{-2m-4} = o(d_*)$. This proves the remaining assertions about the levels.

On any compact logarithmic scale interval, A is smooth and has bounded derivative. The supremum of $v^{-K}A(v)$ over $[s, \rho]$, with the positive endpoint floor, is therefore locally Lipschitz. Its derivative vanishes unless its maximum is realized at the moving endpoint $v = s$. On the equality set, the derivatives of I and A agree almost everywhere. This proves the reset rule. Gaussian differentiation, the dilation bound, and Theorem 2.9 give $|\dot{A}| + |\dot{Z}| \leq CI$. The constant d term is negligible, since $s^2 \eta d/I = \eta d e^a \leq e^{-T/2}$. Together with the reset rule these prove (5.9) and the assertion about $\dot{Z}/I$.

Finally all polynomial dilations in a used here lie well inside the chart. Indeed $I(s) \leq Cds^{2-m} + I_{\text{end}}(s)$, whereas $I(s) \geq s^2 e^{-T/2}$ on the path. The endpoint term cannot account for this inequality. Thus $s \leq Ce^{-T/(2m)}$, up to an immaterial fixed weakening of the exponent. This proves the claimed interior room and justifies the uniform Gaussian cutoffs. $\square$

5.3 Height multipliers and initialization

For $Q > 1$, choose an increasing smooth function b with

$$b(x) \geq (\log x)^{201} + 3F_{Q-1}^{-1}((\log x)^3), \quad b(x) = x^{o(1)},$$

using this displayed sum up to fixed factors. Write $S_1(x) = \exp(Cb(x))$. Its constant is large enough to dominate all polynomial and dilation losses in Theorem 4.3, including applications at $x - O(\log x)$. A further fixed power is denoted by $S_h(x)$. When $Q = 1$, S_h denotes only a sufficiently large fixed constant; no function b , separated estimate, or multiple well construction is used.

All configuration lists and joint moment or support witnesses below use the same fixed exceptional proportion δ_{maj} when $Q > 1$; for $Q = 1$, fix one sufficiently small proportion as well. These proportions are chosen independently of T and a , and their corresponding small-density thresholds in Theorem 2.7 are then fixed. The densities used below are bounded by fixed polynomials times e^{-a_0} or e^{-a} , with $a \geq a_0 - O(1)$; hence they lie below those thresholds for large T .

Here is the precise multiplier convention. For a finite collection $\mathcal{W} = \{b_1, \dots, b_N\}$ of distinct wells let $d_{\mathcal{W}}(z) = \text{dist}(z, \mathcal{W})$. There is a smooth function Γ_0 , comparable to $d_{\mathcal{W}}^2$, equal to $|z - b_i|^2$ within a fixed small fraction of the nearest-neighbor gap at each b_i , and satisfying

$$|D\Gamma_0|^2 \leq C\Gamma_0, \quad |D^2\Gamma_0| \leq C.$$

Construct it by the Whitney partition method [20] on the complement of the wells: on a Whitney ball of radius comparable to distance from the wells, the squared distance has size $O(r^2)$, first variation $O(r)$, and the partition derivatives have sizes $O(r^{-1})$ and $O(r^{-2})$. Patch to the exact quadratics on their disjoint near neighborhoods. This gives the asserted uniform derivative bounds. For one well take its exact square.

Choose a fixed smooth nondecreasing clipping function χ , equal to its argument near zero and constant for large arguments, with $\chi(t)$ comparable to $\min(t, 1)$, and set

$$\Gamma_s(z) = s^2 \chi(\Gamma_0(z)/s^2), \quad U(s) = \int G_s(y-x) \Gamma_s(z) \, d\mu(y, z). \quad (5.13)$$

We may arrange a smooth vector multiplier V_b , equal to $z - b_i$ in smaller near neighborhoods and vanishing outside the exact-quadratic neighborhoods and at distances comparable to s , such that

$$|V_b|^2 \leq \Gamma_s, \quad |DV_b| \leq C, \quad |D\Gamma_s|^2 \leq C\Gamma_s, \quad |D^2\Gamma_s| \leq C. \quad (5.14)$$

Use disjoint radial cutoffs of $z - b_i$; their differentiated cutoffs multiply a displacement of the same size as their radius.

The transition set includes every point outside the smaller pure neighborhoods where either $D^2\Gamma_s \neq 2\text{Id}$ or $DV_b \neq \text{Id}$. In particular it includes the cutoff annuli of V_b even if Γ_s is still exactly quadratic there. Its Gaussian tilt is denoted by D_{tr} . We account separately for Gaussian tails on which clipping occurs. In the one-well case all derivative-error regions lie in those tails, and D_{tr} is omitted.

To state the initialization, write N for the number of wells and, for a small positive parameter k , put $q = -k/a^2$. The comparison to be maintained is

$$kU + \alpha Z \leq I, \quad \alpha = \begin{cases} 1 + C_1 ka / \log a, & N > 1, \\ 1 + (1 - 4k)q - 2k, & N = 1, \end{cases} \quad (5.15)$$

where C_1 is a sufficiently large fixed constant. The next lemma chooses the wells and k so that this inequality holds at s_0 .

Lemma 5.3 (Initialization at a threshold or a side). *At s_0 one can choose at most Q wells and a small parameter $k_0 > 0$ so that the barrier of Theorem 5.4 holds initially. If more than one well is used, one may arrange, for any fixed P ,*

$$Z(s_0) \leq a_0^{-P} I(s_0), \quad U(s_0) \leq CI(s_0),$$

before any immediate mergers, with k_0 a small fixed constant. Alternatively there is one common well, with

$$U(s_0) \leq CS_h(a_0)^2 I(s_0), \quad I(s_0) - Z(s_0) \geq cS_h(a_0)^{-2} I(s_0), \quad k_0 = cS_h(a_0)^{-6}.$$

For $Q = 1$, k_0 is a small fixed constant.

Proof. The bound on s_0 in the envelope proof gives $s_0 \text{poly}(a_0) \ll \rho$. Threshold control on fixed enlargements therefore selects Q by Theorem 2.5. Choose a joint majority-and-moment witness $\mathcal{A}_0 = \{a_1, \dots, a_Q\}$ from Theorem 2.7 at scale s_0 , using $B_{r_{\text{maj}}s_0}(x)$ and the fixed majority proportion when $Q > 1$, and a fixed concentric inner ball when $Q = 1$. Use its distinct entries as the initial wells. Put $R_0 = C_b \sqrt{a_0}$, with a sufficiently large fixed $C_b > 2$. All fixed enlargements needed on $|u| \leq R_0$, where $u = (y - x)/s_0$, lie below ρ . Threshold control gives normalized density at most Cd_* on these dilations. The dilation part of Theorem 2.8, applied only on this body, and $\Gamma_{s_0} \leq C \text{dist}(z, \mathcal{A}_0)^2$ give

$$\int_{|u| \leq R_0} G_{s_0} \Gamma_{s_0} \, d\mu \leq Cs_0^2 d_*.$$

The root mass identity and threshold control at $\sqrt{2}s_0 < \rho$ give

$$(\pi_{\#}\mu)_{\sqrt{2}s_0} \leq C + \tau_{\sqrt{2}s_0} + Cs_0^{-m}e^{-c/s_0^2} \leq C.$$

The last term accounts for the fixed exterior-chart and periodic tails. Since $\Gamma_{s_0} \leq Cs_0^2$, Gaussian comparison yields

$$\int_{|u|>R_0} G_{s_0}\Gamma_{s_0} d\mu \leq Cs_0^2e^{-R_0^2/4}(\pi_{\#}\mu)_{\sqrt{2}s_0} \leq Cs_0^2e^{-C_b^2a_0/4} = o(s_0^2d_*).$$

For the periodized kernel the split and comparison are made termwise in its lifts. Consequently $U(s_0) \leq Cs_0^2d_* = O(I(s_0))$. If the contact itself is a threshold equality, then $Z(s_0) = 0$, which gives the first initialization, with one well when $Q = 1$.

Assume now that $\underline{s} < s_0$. For $\underline{s} \leq v \leq s_0$, put $R_y = |y - x|/v$ and consider the Gaussian work cell $B_r(y)$ with $r = cv/(1 + R_y)$, where $R_y \leq C\sqrt{a_0}$. For any fixed required enlargement B_{Lr} , the Gaussian at $C_L(1 + R_y)v$ is bounded below by a fixed multiple of $((1 + R_y)v)^{-m}$ on that enlargement. This scale lies in $[\underline{s}, \rho]$, so the threshold bound gives, in particular,

$$\begin{aligned} r^{-m}\tau(B_{4r}(y)) &\leq d_c := C(1 + R_y)^{2m}d_*, \\ T_c := -\log d_c &= a_0 - 2m\log(1 + R_y) - O(1), \\ r\sqrt{d_c} &\leq Cv(1 + R_y)^{m-1}\sqrt{d_*}. \end{aligned} \tag{5.16}$$

The same estimate holds for B_{Lr} after changing its fixed constant. Since $d_c = o(1)$, it supplies small $D + \Lambda$ energy and bounded normalized projected mass. Its $|M|$ part puts the normalized smooth mass average within $Cd_c < 1/4$ of Q , so [Theorem 2.5](#) selects that integer on every cell. The cell radius may be below $\underline{s}$; the threshold is used only at the larger Gaussian scale.

When $Q > 1$, scale $B_r(y)$ to one and choose a length- Q majority list $\mathcal{A}_{v,y}$ on $B_{r_{\text{maj}}}$ with local upper bound $r_{\text{maj}}^{-m}d_c$ and the fixed $\delta_{\text{maj}}, C_{\text{maj}}$, as required by [Theorem 4.3](#). Multiply its heights by r when comparing them in the fixed coordinates of this section. The overlap and matching rules (4.9)–(4.10) match these lists to $\mathcal{A}_0$. Along concentric dyadic scales from s_0 to v , the increments $Cr'\sqrt{d_*}$ at radius r' sum to $Cs_0\sqrt{d_*}$. At the fixed scale v , enlarge to radius $C(1 + R_y)v$, move the center in the prescribed steps, and shrink to r . All fixed enlargements in this part lie inside $B_{C'(1+R_y)v}(x)$; at an intermediate radius r' their normalized densities are at most $C((1 + R_y)v/r')^m d_*$. The matching increments therefore add only $C(1 + R_y)^C v\sqrt{d_*}$. The total motion is bounded by a fixed polynomial in a_0 times $s_0\sqrt{d_*}$, uniformly even as the bottom tends to zero.

Suppose $\mathcal{A}_0$ has at least two nonempty groups separated by more than $S_h(a_0)s_0\sqrt{d_*}$. Group the Q entries of each $\mathcal{A}_{v,y}$ according to the corresponding separated groups of the initial wells. Matching preserves their nonemptiness, and their gaps exceed $10S(T_c)r\sqrt{d_c}$: the fixed power $S_h(a_0)$ absorbs $S(T_c)$, the displayed cell factor, and the matching loss, since $T_c = a_0 - O(\log a_0)$. These length- Q lists are the inputs to [Theorem 4.3](#). The cell tests obtained by partitioning the nonnegative Gaussian are admissible after normalization in the cell coordinates. Bounded overlap and a sufficiently large input exponent absorb their fixed polynomial losses. In the curved case the physical cell radius is lr , and $T_c = o(T)$, so the original physical smallness implies the cell hypothesis. The threshold and root tail bounds from the envelope proof apply with controlling density d_* , uniformly down to $\underline{s}$. Under nonnegative testing, $f_* \leq L - C_Q\Lambda$, and the far term in [Theorem 4.3](#) is nonnegative. The resulting bound, integrated in scale, gives

$$Z(s_0) \leq Ca_0^{-P-1}d_* \int_{\underline{s}}^{s_0} v^2 d\log v \leq a_0^{-P}I(s_0).$$

Entries too close to remain separate are merged immediately by the rule below. Their moment before merging is $O(I(s_0))$.

If there are no such separated groups, a chain through at most Q entries of $\mathcal{A}_0$ shows that its diameter is at most $CQS_h(a_0)s_0\sqrt{d_*}$. Use one common well. Pointwise, the new clipped square is bounded by a fixed multiple of the old one plus the squared diameter. The preceding U -bound and bounded Gaussian mass therefore give the stated estimate for U . It remains to check the surplus of I over Z . If $A(s_0)$ is a sufficiently small fraction of $I(s_0)$, this is immediate from (5.6). Otherwise $A(s_0) \geq cI(s_0)$. When $\underline{s} \geq s_0/2$, the inequality $\dot{A} \leq 2A + s^2R_s + \text{error}$ gives

$$A(\underline{s}) + \int_{\underline{s}}^{s_0} v^2 R_v \, d \log v \geq cI(s_0).$$

These terms are retained in $A(s_0) - Z(s_0)$, after absorbing the error into the positive margin.

When $\underline{s} < s_0/2$, Gaussian differentiation and (5.3) give $|\dot{A}(v)| \leq Cv^2\tau_{2v} \leq Cv^2d_*$ on $[s_0/2, s_0]$. Since $A(s_0) \geq cI(s_0) \asymp s_0^2d_*$, for some fixed $0 < \vartheta_0 < \log 2$ one has $A(v) \geq cI(s_0)$ throughout $[e^{-\vartheta_0}s_0, s_0]$. Apply Theorem 2.8 directly on this earlier interval, using the scalar hypotheses verified in (5.16). The tail implication in the envelope proof applies with controlling density d_* : here $v < \rho/T^N$ for every fixed N , and $d_* \geq e^{-T/2}$. It gives an $o(d_*)$ remainder uniformly in the bottom. Thus $|M|_v \leq C(D_v + \Lambda_v) + o(d_*)$; because $M_v = A(v)/v^2 \geq cd_*$, this yields $D_v + \Lambda_v \geq cd_*$.

The surplus uses further information. Where $\Lambda_v \geq cd_*$, the retained term $C_Qv^2\Lambda_v$ in $J(v)$ pays it directly. On the complementary portion $D_v \geq cd_*$. Let V_v be displacement from the common well, smoothly clipped at distance comparable to v . Apply the moment part of Theorem 2.8 to joint configuration witnesses on the admissible dilations and match their centers to the common well. Gaussian summation absorbs their polynomial losses and gives

$$\int G_v |V_v|^2 \, d\mu \leq Cv^2S_h(a_0)^2d_*.$$

The support estimates for those chosen witnesses put all support heights on $|y - x| \leq C_b\sqrt{a_0}v$ within

$$v[\text{poly}(a_0)e^{-\beta a_0} + S_h(a_0)e^{-a_0/2}] = o(v)$$

of the common well. Clipping is therefore absent on this body. For the clipped tail, $\pi_{\#}\mu = Q \text{vol}_S + M$ controls the bare mass as well as the defect. Since $\log S_h(a_0) = o(a_0)$, taking the fixed C_b sufficiently large makes the Gaussian mass tail $o(d_*S_h(a_0)^{-2})$. The defect and chart tails obey the preceding uniform bounds.

Choose one fixed root-chart cutoff $0 \leq \chi \leq 1$, equal to one on the fixed working region, whose transition is at distance at least h from that region and has width at least ch , decreasing h to the available width. Then $\|\nabla\chi\|_\infty \leq C(1+h^{-1})$; the first-variation product rule differentiates χ once, while all scale derivatives leave it fixed. With $u = (y - x)/v$, normal stationarity in Theorem 2.10 now gives

$$D_v \leq \frac{1}{v} \left| \int G_v V_v \cdot P_{vh} u \, d\mu \right| + o(d_*).$$

The factor v^{-1} is the Gaussian derivative. Retaining $|V_v|$ in the curved normal error and using its moment bound leaves relative error $e^{-\kappa T} \text{poly}(a_0, S_h(a_0)) = o(1)$; the normal identity introduces no further inverse power of v . Thus the flux has size at least cvd_* on the D -portion. Cauchy-Schwarz and $|P_{vh}u|^2 \leq B[u, u]$ yield

$$cv^2d_*^2 \leq \left(\int G_v |V_v|^2 \, d\mu \right) R_v \leq Cv^2S_h(a_0)^2d_*R_v, \quad v^2R_v \geq cS_h(a_0)^{-2}v^2d_*.$$

After the margin absorbs the errors, $\dot{A} - \dot{Z}$ retains $v^2R_v + C_Qv^2\Lambda_v$; at all other scales it is nonnegative by (5.6). Integration over the fixed logarithmic interval therefore proves the

surplus claimed in the statement. Since the one-well coefficient α below is less than one, $k_0 = cS_h(a_0)^{-6}$ pays for the moment while leaving a strict initial inequality. The same proof for $Q = 1$ uses the Gaussian-summed moment with fixed S_h and no separation step. $\square$

5.4 The barrier and its finite mergers

Normal stationarity and square completion bound $s^2 R_s$ from below in terms of D_s and U , supplying the term that offsets growth of kU . With several wells, α makes the coefficient of $\dot{Z}$ in the derivative of $I - kU - \alpha Z$ negative, so that the separated far-tilt gain absorbs the transition-tilt losses. The merger rule preserves this comparison with controlled losses, leaving the traversal required by [Theorem 5.5](#).

Proposition 5.4 (Low-frequency barrier). *Along every path in [Theorem 5.2](#), the wells can be chosen piecewise constantly, with at most $Q - 1$ mergers. There is a positive piecewise smooth parameter k , bounded above by a small fixed constant and satisfying between mergers*

$$\frac{k_a}{k} = q = -\frac{k}{a^2}, \quad (5.17)$$

such that the comparison (5.15) holds. Here C_1 is fixed and sufficiently large. Uniformly along the path, $\log(1/k) = o(a_0)$ as $T \rightarrow \infty$. Before any hypothetical first failure of the barrier, the following estimates hold, with $\mathcal{E} \leq a^{-P_0/2}$:

$$\dot{U} \leq 2s^2 D_s + Cs^2 D_{\text{tr}} + \mathcal{E}I, \quad (5.18)$$

$$s^2 R_s \geq 4k(s^2 D_s - Cs^2 D_{\text{tr}}) - 4k^2 U - \mathcal{E}kI. \quad (5.19)$$

With more than one well, in addition,

$$\dot{Z} + c_2 \frac{(\log a)^2}{a} s^2 D_{\text{tr}} \leq a^{-P_0} I. \quad (5.20)$$

The transition terms are absent with one well. By increasing the fixed exponents in the construction, $\mathcal{E}$ can be smaller than any prescribed inverse power of a .

Proof. We specify the mergers first. At stage j , merge a closest pair as soon as its gap g satisfies

$$g \leq s \min\{C_j e^{-\gamma a}, S_1(a) e^{-a/2} k^{-1}\}. \quad (5.21)$$

Retain one of the two centers. Take $\gamma > 0$ sufficiently small, C_{j+1} a sufficiently large fixed multiple of QC_j , and replace k by

$$k_{\text{new}} = ck^6 S_1(a)^{-6}. \quad (5.22)$$

Perform further mergers immediately if required. For $Q = 1$ there is no merger rule. A common-well initialization when $Q > 1$ may be counted as an initial merger; it uses no more than the available $Q - 1$ losses.

The solution of (5.17) on a stage starting at level a_j is

$$k(a)^{-1} = k(a_j)^{-1} + a_j^{-1} - a^{-1}.$$

Retracing levels therefore causes no accumulated loss. Since $a \geq a_0 - O(1)$, k changes by at most a fixed factor on a stage. There are at most $Q - 1$ mergers, and their levels are at most $T/2$. Thus

$$\log(1/k) \leq C_Q \{1 + b(T) + b(a_0)\} = o(a_0),$$

where [Theorem 5.5](#) verifies the final comparison. In particular e^{-a}/k is exponentially small uniformly along the path.

We next verify that the wells continue to represent the configuration. At a scale s , let $\mathcal{A}_s$ be a length- Q majority list on a fixed inner ball, using the fixed convention above when $Q > 1$. For support control, choose joint majority-and-support witnesses from [Theorem 2.7](#) on the balls under consideration and match their centers to these lists. The support and matching bounds, together with (5.8), bound configuration changes on a ball of scale v , including polynomial enlargements, by $Cve^{-\beta a(v)}$ for some fixed $\beta > 0$. Decrease β if necessary so that $\beta|\dot{a}| \leq 1/2$. Summing on a geometric sequence from s_0 to s gives $Cse^{-\beta a(s)}$. The same estimate tracks each initial configuration entry. The additional displacement caused by previous mergers is bounded by the sum of their gaps. Choose $\gamma \ll \beta$ and $\gamma|\dot{a}| \leq 1/2$. Then $se^{-\gamma a(s)}$ increases outwards, and the successive enlargements of C_j make all earlier capped gaps a small fixed fraction of the current cap. These facts give, in particular,

$$\text{dist}(z, \mathcal{W}) \leq Cse^{-\gamma' a} \quad \text{on } |y - x|/s \leq C\sqrt{a} \quad (5.23)$$

for some fixed $\gamma' > 0$. This estimate does not use the barrier. It also bounds the distance from retained wells to the corresponding tracked configurations. The common-well initialization has the same property, since $\log S_h(a_0) = o(a_0)$.

While the barrier holds, each entry of $\mathcal{A}_s$ is matched on a fixed majority of the inner ball to a projection entry within $Cse^{-a/2}$. The area formula and the uniform upper bound for J_π give that entry a fixed amount of Gaussian mass. The bound $|D\Gamma_s|^2 \leq C\Gamma_s$ and the numerical barrier therefore give

$$\max_{\xi \in \mathcal{A}_s} \Gamma_s(\xi) \leq CU + Cs^2e^{-a} \leq Cs^2e^{-a}/k.$$

Since Γ_s is comparable to $\min\{\text{dist}(z, \mathcal{W})^2, s^2\}$ and $e^{-a}/k \rightarrow 0$, every entry lies within $Cs\sqrt{e^{-a}/k}$ of a well. This distance is a small fraction of $S_1(a)se^{-a/2}k^{-1}$. It is also a small fraction of the cap, using $\log(1/k) = o(a_0)$; alternatively (5.23) gives that conclusion directly. When no merger is triggered, the resulting neighborhoods of the wells are disjoint. The group represented by a well is the sublist of $\mathcal{A}_s$ in its neighborhood. When $Q > 1$, its integer count cannot change: on two sufficiently nearby scales the majority balls obey (4.9), their lists match by (4.10), and the matching error is much smaller than the gap. For $Q = 1$, the sole length-one group remains represented. At initialization each well is represented. At a merger the surviving neighborhood contains the entries of both merged wells after any further immediate mergers. This proves representation at every stage, including a first barrier equality.

With several wells, $Q > 1$, so we may apply [Theorem 4.3](#). For a Gaussian cell with $R_y = |y - x|/s \leq C\sqrt{a}$ and $r = cs/(1 + R_y)$, the dilation bound at $C(1 + R_y)s$ gives

$$\begin{aligned} r^{-m}\tau(B_{4r}(y)) &\leq d_c := C(1 + R_y)^{2m+K-2}e^{-a}, \\ T_c &:= -\log d_c = a - (2m + K - 2)\log(1 + R_y) - O(1), \\ r\sqrt{d_c} &\leq Cs(1 + R_y)^{m+K/2-2}e^{-a/2}. \end{aligned} \quad (5.24)$$

The cell and its fixed enlargements lie in the chart, and $T_c = a - O(\log a) \rightarrow \infty$. As in (5.16), the $|M|$ bound selects Q . In the cell coordinates choose a length- Q majority list $\mathcal{A}_{s,y}$ on $B_{r_{\text{maj}}}$, with local upper bound $r_{\text{maj}}^{-m}d_c$ and the fixed $\delta_{\text{maj}}, C_{\text{maj}}$, and multiply its heights by r for comparison in the section coordinates. The off-center overlap chain from initialization now uses (5.8); matching to $\mathcal{A}_s$ costs at most a fixed polynomial in a times $se^{-a/2}$.

Group the Q entries of $\mathcal{A}_{s,y}$ by the represented wells to which their matches in $\mathcal{A}_s$ belong. The wells index these nonempty groups; the input list still has length Q after mergers. Before a merger, if the second threshold in (5.21) is smaller, the gap is larger than $S_1(a)se^{-a/2}k^{-1}$. This dominates the entry-to-well and cell matching errors and $10S(T_c)r\sqrt{d_c}$, by the choice of

S_1 . If the cap is smaller, the ratio of its gap to the last quantity tends to infinity, since it is bounded below by $\exp((1/2 - \gamma)a - O(b(a)) - O(\log a))$. Thus the groups meet the separation premise in both regimes. Away from the clipping tails, every point of the transition set is a fixed fraction of a gap from the represented groups, and remains in the cell's far region after matching. This includes every extra annulus in D_{tr} .

Apply [Theorem 4.3](#) to the nonnegative normalized cell weights. Their polynomial Lipschitz costs are absorbed by taking its exponent larger than P_0 , while $c(T_c) \asymp (\log a)^2/a$. Bounded overlap and the Gaussian tails give total remainder $O(a^{-P_0}I/s^2)$. The curved hypothesis is inherited because the physical cell radius is lr and $T_c \leq T/2 + O(1)$. This proves [\(5.20\)](#).

We prove the two differential inequalities explicitly. In flat coordinates the horizontal test with multiplier Γ_s gives

$$\dot{U} = \int G_s \{ \Gamma_s(B[u, u] - e) + s D\Gamma_s \cdot P_{vh}u + \dot{\Gamma}_s \} d\mu, \quad u = (y - x)/s.$$

Normal stationarity with $G_s D\Gamma_s$ converts the cross term to $s^2 \int G_s D^2\Gamma_s : P_{vv} d\mu$. On the pure regions $D^2\Gamma_s = 2\text{Id}$, and elsewhere its norm is bounded. Dropping $-\Gamma_s e$, and using [\(5.23\)](#) to bound the $\Gamma_s B[u, u]$ term, gives [\(5.18\)](#). The explicit derivative $\dot{\Gamma}_s = 2s^2[\chi(t) - t\chi'(t)]$, $t = \Gamma_0/s^2$, is supported on the clipping region. That region does not meet the Gaussian body by [\(5.23\)](#).

For the second inequality use $|P_{vh}u|^2 \leq B[u, u]$ and complete the square:

$$s^2 R_s \geq 4ks \int G_s V_b \cdot P_{vh}u d\mu - 4k^2 U.$$

Normal stationarity gives

$$s \int G_s V_b \cdot P_{vh}u d\mu = s^2 \int G_s D V_b : P_{vv} d\mu \geq s^2 D_s - C s^2 D_{\text{tr}}$$

up to the already specified tails. This proves [\(5.19\)](#) in flat coordinates.

For completeness the tails here also include terms involving mass without a tilt factor. Since $\Gamma_s \leq C s^2$, the mass tail beyond $C\sqrt{a}$ is bounded by $C s^2 e^{-cC^2 a}$, plus the dilation-controlled defect. Taking C large makes this smaller than any required multiple of $a^{-P}I$. Terms with tilt use [\(5.8\)](#) directly. No inverse power of k is required for these flat errors, because the derivative bounds in [\(5.14\)](#) are uniform. For the curved errors, keep the original coordinates scaled by the physical radius l , and write $\langle f \rangle_s = \int G_s f d\mu$ and $\gamma_{\text{geom}} = l + \sup |z_{\text{phys}}|$. The dilation bound gives $\langle (1 + |u|^2)(e + \lambda) \rangle_s \leq P(a)I/s^2$, where P is a fixed polynomial and $\lambda = |z_{\text{phys}}|^2 + |H_S|^2 + |\nabla H_S|^2$. For either $v = V_b$ or $v = D\Gamma_s$, $\langle |v|^2 \rangle_s \leq CU \leq CI/k$ and $|Dv| \leq C$. After multiplying the normal identity by s^2 , the errors in [Theorem 2.10](#) satisfy

$$\begin{aligned} s^2 l \langle |z_{\text{phys}}| \sqrt{e} \rangle_s &\leq s^2 l \sqrt{\Lambda_s D_s} \leq CP(a)lI, \\ s^2 l \langle |v| (|H_S| + |z_{\text{phys}}| + e + \sqrt{e}) \rangle_s &\leq C s^2 l \sqrt{U} \sqrt{\Lambda_s + D_s} \leq CP(a)l s k^{-1/2} I, \\ \gamma_{\text{geom}} s \langle |u| |v| \sqrt{e} \rangle_s &\leq \gamma_{\text{geom}} s \sqrt{U} \sqrt{\langle |u|^2 e \rangle_s} \leq CP(a) \gamma_{\text{geom}} k^{-1/2} I. \end{aligned}$$

Here $e \leq m$ controls the term $|v|e$; the last line is the error from differentiating the Gaussian. In the horizontal test, $\Gamma_s \leq C s^2$ bounds the error carrying $\Gamma_s(e + \lambda)$ by $C \gamma_{\text{geom}} P(a)I$, while

$$\gamma_{\text{geom}} s \langle |u| |D\Gamma_s| (\sqrt{e} + \sqrt{\lambda}) \rangle_s \leq C \gamma_{\text{geom}} s \sqrt{U} \sqrt{\langle |u|^2 (e + \lambda) \rangle_s} \leq C \gamma_{\text{geom}} P(a) k^{-1/2} I.$$

Coordinate and base-volume comparison may also produce a term $C \gamma_{\text{geom}} \langle (1 + |u|^2) \Gamma_s \rangle_s$. Split at $C\sqrt{a}$; its body is bounded by $C \gamma_{\text{geom}} P(a)U$, and its clipped tail has the already proved negligible bound. Thus it costs at most $C \gamma_{\text{geom}} P(a) k^{-1} I$. These estimates retain physical

heights and curvature inside Λ until after Cauchy–Schwarz; the displayed factors of s cancel the Gaussian inverse-scale factors. They give total error $Ce^{-\kappa T} \text{poly}(a, k^{-1})I$ in the moment inequality and this bound times k after square completion in the radial inequality. Since $\log(1/k) = o(T)$, these errors are smaller than the stated $\mathcal{E}I$ and $\mathcal{E}kI$. Chart-cutoff errors are still smaller because $s \leq e^{-cT}$. This proves both differential inequalities with their asserted uniform errors.

It remains to prove the comparison, including its signs. Work at an equality $kU + \alpha Z = I$ on a fixed stage and put

$$\delta_\alpha = \alpha_a - q\alpha, \quad h_1 = q + \delta_\alpha Z/I.$$

Differentiating $I - kU - \alpha Z$, and using $\dot{a} = 2 - \dot{I}/I$, gives

$$(I - kU - \alpha Z)' = (1 + h_1)\dot{I} - 2h_1\dot{I} - k\dot{U} - \alpha\dot{Z}. \quad (5.25)$$

At a reset write $\mathcal{D} = s^2 D_s$, $\mathcal{R} = s^2 R_s$. Apart from the absorbed geometric error,

$$\dot{A} = 2I - \mathcal{D} + \mathcal{R}, \quad \mathcal{D} = 2I - \dot{Z} - J.$$

Insert (5.18) and (5.19) into (5.25) and use $kU = I - \alpha Z$. The resulting lower bound is

$$\begin{aligned} (4k - 2)h_1 I + 4k(1 + h_1)\alpha Z + c_Z \dot{Z} - Cks^2 D_{\text{tr}} - O(\mathcal{E}kI) + c_J J, \\ c_Z = 1 + h_1 - 2k - 4kh_1 - \alpha, \quad c_J = 2k - (4k - 1)(1 + h_1). \end{aligned} \quad (5.26)$$

In both cases $h_1 = o(1)$, so $c_J > 0$; discarding the margin therefore has the indicated favorable sign. The first-variation error in $\dot{A}$, which initially need not contain k , is absorbed by a fixed fraction of this positive margin. Equivalently, its remaining contribution divided by kI can be included in $\mathcal{E}$, since $\log(1/k) = o(T)$.

With one well,

$$\delta_\alpha = \frac{k}{a^2} + \frac{2k(1 - 4k)}{a^3} - \frac{4k^3}{a^4} = O(k/a^2), \quad c_Z = (1 - 4k)\delta_\alpha Z/I.$$

Thus the $\dot{Z}$ coefficient cancels exactly at $Z = 0$. The estimate $|\dot{Z}| \leq CI$ bounds its remaining contribution by $O(k/a^2)Z$. The term $(4k - 2)qI$ is at least ckI/a^2 ; the coefficient of Z left in the first two terms of (5.26) is at least ck . Hence the reset derivative is strictly positive for large a .

With several wells there is the useful exact cancellation

$$\delta_\alpha = \frac{k}{a^2} + C_1 k \frac{\log a - 1}{(\log a)^2}.$$

In particular $\delta_\alpha = o(k)$, $|h_1| \leq C/a$, and

$$-CC_1 ka / \log a \leq c_Z \leq -cC_1 ka / \log a.$$

Multiply (5.20) by this negative coefficient. It supplies a favorable multiple of $C_1 k \log a s^2 D_{\text{tr}}$, which absorbs the transition loss in (5.26). Its adverse error is $O(ka^{1-P_0}/\log a)I = o(kI/a^2)$. The remaining Z terms are positive, since their possible loss is $O(\delta_\alpha)Z = o(k)Z$. Again $(4k - 2)qI \geq ckI/a^2$ supplies strict positivity.

Off the reset set $\dot{I} = KI$, and $\mathcal{D} \leq 2I - \dot{Z}$. The coarser consequence $\dot{U} \leq C\mathcal{D} + \mathcal{E}I$ gives

$$(I - kU - \alpha Z)' \geq \{(1 + h_1)K - 2h_1 - 2Ck\}I + (Ck - \alpha)\dot{Z} - O(\mathcal{E}kI).$$

The coefficient $Ck - \alpha$ is negative. A negative $\dot{Z}$ therefore helps. For one well, $\dot{Z} \leq 2I$; for several wells, (5.20) gives $\dot{Z} \leq a^{-P_0}I$, and $\alpha a^{-P_0} = o(1)$. Taking K large and the initial upper

bound for k small makes the derivative strictly positive. All the strict inequalities hold in a neighborhood of equality on each compact stage. The absolutely continuous comparison principle therefore applies despite the reset set having only almost-everywhere derivatives. Initialization is supplied by [Theorem 5.3](#); in its multiple-well case choose $P > 3$, so that the initially large coefficient α still leaves strict room.

Finally consider a merger. Pointwise distance comparison and bounded Gaussian mass imply

$$U_{\text{new}} \leq C(U + g^2), \quad g^2 \leq S_1(a)^2 k^{-2} I.$$

Moreover $\alpha_{\text{new}} \leq (1 - ck)\alpha$, whether several wells remain or the last two merge: in the first case use the formula linear in k . In the second case write $\alpha = 1 + x$, where $x = C_1 ka / \log a$. The fixed upper bound for k and large a give $x/(1 + x) \geq ck$, so $\alpha_{\text{new}} < 1 \leq (1 - ck)\alpha$. Set $\theta = Ck_{\text{new}}/k$ and $\lambda = Ck_{\text{new}}S_1^2 k^{-2}$. By (5.22), $\theta \ll 1$ and $\lambda \ll k$. The old comparison then gives

$$k_{\text{new}}U_{\text{new}} + \alpha_{\text{new}}Z \leq (\theta + \lambda)I + (\alpha_{\text{new}} - \theta\alpha)Z \leq I.$$

For the last inequality take the maximum over $0 \leq Z \leq I/\alpha$; both endpoint values are below I . Thus every merger preserves the barrier, completing its proof. $\square$

5.5 Uniform bands of levels

Lemma 5.5 (Choice of bands). *Fix $\varepsilon_0 > 0$, sufficiently small for the later packing argument, and any fixed loss exponent C_* . For large T , there are Q ordered disjoint bands $[B_i, 2B_i] \subset (2a_0, T/2)$, independent of the contact, such that*

$$B_1^{1/10} \gg F_Q(T)^3, \quad S_h(a_0)^{C_*} \ll B_1^{\varepsilon_0}, \quad S_1(2B_i)^{C_*} \ll B_{i+1}^{\varepsilon_0}. \quad (5.27)$$

For $Q = 1$, only the first condition is needed, and a single band of size comparable to T suffices. Every contact path has a traversal contained in one band, from $a = B$ to $a = 2B$, which contains no merger and on which $k \geq B^{-\varepsilon_0}$. The estimates (5.6), (5.8), (5.9), (5.10), and (5.15)–(5.20) hold on that traversal. If it has several wells, the bound $\dot{Z} \leq a^{-P_0}I$ holds on every preceding stage with several wells as well, and the initial value of Z is the arbitrarily small quantity in [Theorem 5.3](#).

Proof. We spell out the rate comparison to keep track of the inverse. Let $Q \geq 2$ and $j = 2Q - 2$. Above the iterated-log threshold,

$$F_{Q-1}^{-1}(u) = \exp_j((\log_j u)^2). \quad (5.28)$$

For b chosen as above, inversion of any fixed power of $S_1(x) = e^{Cb(x)}$ first takes a logarithm, then applies F_{Q-1} , and then inverts $(\log x)^3$. In particular the inverse retains the outer exponential from this last step. For example when $Q = 2$ its leading expression, up to fixed losses, is

$$\exp\left([F_1(c \log u)]^{1/3}\right).$$

It follows directly from (5.28) that, for some fixed $\varsigma > 0$, every such inverse dominates

$$\exp_{2Q-1}((\log_{2Q-1} u)^\varsigma).$$

Any fixed number of inverse compositions, including fixed powers of the argument and value, retains this lower bound with a smaller positive ς . To compare with $F_Q(T)^C$, take $2Q - 1$ logarithms: the lower bound has last value $\exp(\varsigma \log_{2Q} T)$, whereas the latter has last value $\exp(\sqrt{\log_{2Q} T})$, up to lower order terms. The former is larger for every fixed C .

The same comparison shows $b(T) = o(F_Q(T)^c)$ for each fixed $c > 0$, and hence $b(T) + b(a_0) = o(a_0)$, as used above. It also shows that after a fixed number of the inverse operations, $\log B_1 \gg b(a_0)$. This stronger comparison absorbs the initial factor $S_h(a_0)$, including any fixed power of it. Enlarge C_* , if necessary, to exceed the finitely many powers arising from $Q - 1$ applications of (5.22); proving the stronger separation also proves the originally requested one. Now choose B_Q to be a small fixed multiple of T , and choose the earlier bands successively by inverse operations with enough fixed slack to ensure (5.27). The comparisons just proved give all the lower-end requirements. For $Q = 1$, $F_1(T) = T^{o(1)}$, so the claimed single band works and no inverse construction is required.

Consider the path from initialization to its first hit of each successive level $2B_i$. There are Q resulting consecutive time intervals and fewer than Q merger events, counting a common-well initialization if needed. At least one interval has no event. On it take the last preceding hit of B_i before the first hit of $2B_i$; this is a traversal staying in that band. All earlier mergers have levels at most $2B_{i-1}$ (or are initialization events). Iterating (5.22), with at most $Q - 1$ steps, bounds their total loss by a fixed power of $S_1(2B_{i-1})$, the initial $S_h(a_0)$, and fixed constants. Equation (5.27) and the explicit solution of (5.17) then give $k \geq B_i^{-\varepsilon_0}$ throughout the traversal. The remaining assertions are the estimates already proved. The number of wells only decreases, so a traversal with several wells has had several wells at every earlier stage. $\square$

6 Witness scales and the signed estimate

We finish the argument at the integer Q . In this section the induction hypothesis is needed only through Theorem 4.3, as used in the multiple-well initialization and barrier. Put $t = \log s$; a prime below means differentiation with respect to t . All measures, balls, heights, and Gaussian kernels have the normalization of Theorem 5.2. In particular $I = s^2 e^{-a}$, whereas the quantities inside Λ retain their physical units. Write $\nu = \tau$ for the positive projected measure in the working chart.

The purpose of Theorems 6.1 to 6.3 is to turn a path of contact scales into a small set for ν . First we find many scales with a definite normal flux. The height moment with exponent $p > 2$ then prevents this flux from concentrating entirely where the density of ν is large. Finally a dyadic stopping argument makes repeated occurrences of this property expensive in ν -mass.

6.1 Witnesses and controlled density

Lemma 6.1 (Ordered witness scales). *Consider a contact path and a merger-free traversal of $[B, 2B]$ provided by Theorem 5.5, with $k \geq B^{-\varepsilon_0}$ throughout the traversal. Given a sufficiently small fixed number $\vartheta > 0$, there are at least $cB^{1/2}$ scales s_j , ordered in outward time, whose levels satisfy*

$$B + 1 < a(s_j) < 2B - 1, \quad a(s_{j+1}) - a(s_j) \geq B^{1/3}. \quad (6.1)$$

If more than one well remains, these scales satisfy

$$D_{s_j} \geq ce^{-a(s_j)}, \quad D_{\text{tr}}(s_j) \leq \vartheta e^{-a(s_j)}. \quad (6.2)$$

If only one well remains, they satisfy

$$D_{s_j} \geq B^{-\varepsilon_0} e^{-a(s_j)}. \quad (6.3)$$

The constants are independent of the contact and of the positive bottom scale. The large threshold for B may depend on ε_0 .

Proof. We first record precisely the differential information used here. By [Theorems 5.2](#) and [5.4](#), $0 \leq Z \leq A \leq I$, $A' - Z' \geq 0$, and

$$|a'| + |I'|/I + |Z'|/I \leq C, \quad |(Z'/I)'| \leq C \quad (6.4)$$

almost everywhere. The last estimate follows also directly by differentiating the Gaussian source: its first two logarithmic derivatives are bounded by Gaussian averages of fixed polynomial weights times τ , and the dilation estimate bounds these by CI/s^2 . The constant background in the source is negligible since $a \leq T/2$. At a reset $I = A$ we have $I' = A'$ almost everywhere on the reset set. Off that set $I' = KI$ and $a' = 2 - K < 0$. The weighted scalar estimate is

$$|M|_s \leq C_0(D_s + \Lambda_s) + a^{-P}I/s^2, \quad (6.5)$$

where P can be chosen as large as needed. With several wells,

$$Z' + c_0 \frac{(\log a)^2}{a} s^2 D_{\text{tr}}(s) \leq a^{-P}I. \quad (6.6)$$

Here $D_{\text{tr}}(s)$ is the Gaussian average over the enlarged transition set of [Theorem 5.4](#). This set includes all regions where derivatives of either Γ or V_b differ from their near-well values, in particular the cutoff annuli of V_b inside the quadratic neighborhoods of Γ ; clipping tails are estimated separately. These inequalities retain the cumulative contact condition $Z \geq 0$; no sign for Z' at a later scale is asserted.

Here is a time-bin estimate that does not require monotonicity of a or of I . Fix an endpoint t_1 of an interval on which $a \geq A_* > 2$, and put $M_* = \max_{t \leq t_1} I(t)$ on that interval. For nonnegative integers j and integers $n \geq \lfloor A_* \rfloor$, the simultaneous conditions

$$n \leq a(t) < n+1, \quad e^{-j-1}M_* < I(t) \leq e^{-j}M_*$$

confine t to an interval of length at most one, because

$$t = \frac{a(t) + \log I(t)}{2}. \quad (6.7)$$

The total length of the set of such times, even if it has many components, is therefore at most one. Summing the integral over these sets gives

$$\int_{t \leq t_1} a(t)^{-P} I(t) dt \leq CM_* \sum_{j \geq 0} e^{-j} \sum_{n \geq \lfloor A_* \rfloor} n^{-P} \leq CA_*^{1-P} M_*. \quad (6.8)$$

For an interval confined to $[B, 2B]$, the same estimate has CB^{1-P} on the right.

Suppose first that the selected traversal has several wells. All earlier stages also had several wells, since the construction only removes wells. Its initialization gives $Z(t_0) \leq a_0^{-P}I(t_0)$. By [\(6.6\)](#) and [\(6.8\)](#), the positive variation of Z up to any endpoint is at most ϵM_* , where $\epsilon \rightarrow 0$ as $a_0 \rightarrow \infty$. Since $A - Z$ is nondecreasing, for $u \leq t$ we have

$$A(t) \geq A(u) - Z(u) \geq A(u) - Z(t_0) - \int_{t_0}^t (Z')_+ dv.$$

The future envelope has the elementary property

$$\max_{t_0 \leq u \leq t} I(u) \leq \max \left\{ I(t), \max_{t_0 \leq v \leq t} A(v) \right\}. \quad (6.9)$$

Indeed a term realizing $I(u)$ either comes from an $A(v)$ with $v \leq t$, when its exponential weight is at most one, or comes from a later v or from the artificial endpoint, when the same term at

t is larger by $e^{K(t-u)}$. Approximating a supremum proves the assertion if it is not attained. Combining these facts, and taking a_0 large, yields

$$I(t) \geq \frac{1}{2} \max_{t_0 \leq u \leq t} I(u). \quad (6.10)$$

This argument is unchanged at a merger: I, A, Z do not jump, and (6.6) holds on both adjacent multiple-well stages.

Choose a traversal $[t_b, t_e]$ with $a(t_b) = B$, $a(t_e) = 2B$, and $B \leq a \leq 2B$ between them. Let $M(t) = \max_{t_b \leq v \leq t} I(v)$ and $I_b = I(t_b)$. On this interval

$$\int_{t_b}^t (Z')_+ dv \leq CB^{1-P}M(t), \quad Z(t) \leq I_b + CB^{1-P}M(t), \quad I(t) \geq M(t)/2. \quad (6.11)$$

Fix a small constant $\delta_1 > 0$ and set $h_B = \delta_1(\log B)^2/B$. We bound the levels whose first increasing passage has $Z'/I < -h_B$.

Before M first reaches $B^{10}I_b$, split the range of M into successive factors of two. There are $O(\log B)$ such pieces. Applying (6.8) on each piece shows that

$$\int \frac{(Z')_+}{M} dt \leq CB^{1-P} \log B.$$

Integration by parts against the continuous nondecreasing function M gives

$$\int \frac{-Z'}{M} dt = \frac{Z(t_b)}{M(t_b)} - \frac{Z(t)}{M(t)} - \int \frac{Z}{M^2} dM \leq 1.$$

Consequently $\int (Z')_-/M \leq 2$ for large B . In view of (6.10), the time spent with $Z'/I < -h_B$ is at most C/h_B . By $|a'| \leq C$, the image of these times in the level variable has length at most

$$CB/(\log B)^2 = o(B). \quad (6.12)$$

Constants here may depend on the fixed δ_1 .

After this growth of M , (6.11) implies $Z(t) \leq C(B^{-10} + B^{1-P})I(t)$. If $Z'/I < -h_B$ at a time with $B+1 < a < 2B-1$, (6.4) keeps $Z'/I < -h_B/2$ on a following interval of length ch_B . The interval stays in the traversal, and I changes by at most a fixed factor there. It would decrease Z by at least $ch_B^2 I(t)$, exceeding the available value of $Z(t)$ for $P > 12$ and large B . This contradicts $Z \geq 0$. Thus no such bad time occurs after the stated growth.

For completeness, first increasing passages can be selected without assuming that a itself is increasing. Use the running maximum of a starting at t_b . It is Lipschitz. Except for a null set of levels, a first passage is a differentiability point with $a' > 0$: the image under a Lipschitz function of its nondifferentiability set is null, and the image of its zero-derivative set is null by the one-dimensional area formula. First passages so obtained are ordered. Since off resets $a' = 2 - K < 0$, they are resets. The excluded levels are bounded by (6.12).

At every remaining passage $I = A$, so $M_s = I/s^2 = e^{-a}$. The lower bound $Z'/I \geq -h_B$ and the definition of the source imply

$$C_Q \Lambda_s \leq (2 + h_B)e^{-a}.$$

Apply (6.5). Taking C_Q sufficiently large compared with C_0 , and then B large, forces $D_s \geq ce^{-a}$. Moreover (6.6) gives

$$D_{\text{tr}}(s) \leq C \left(\delta_1 + \frac{B^{1-P}}{(\log B)^2} \right) e^{-a}.$$

Choose δ_1 first so that the last bound is at most ϑe^{-a} . This proves (6.2) on all but $o(B)$ of the first-passage levels in the band.

Now suppose there is one well. If no scale in a traversal of a level interval of length $B^{1/4}$ satisfied (6.3), then throughout it $s^2 D_s < B^{-\varepsilon_0} I$. At a reset the Gaussian horizontal identity gives

$$I' = 2I - s^2 D_s + s^2 R_s + o(I) \geq I.$$

Also (6.5), together with $M_s = e^{-a}$, forces $\Lambda_s \geq ce^{-a}$. Thus $Z' \leq -I$ when C_Q is sufficiently large. At a nonreset, the same scalar estimate and $C_Q > 2C_0$ instead give

$$Z' \leq Cs^2 D_s + o(I) \leq CB^{-\varepsilon_0} I.$$

These estimates explicitly include the possible geometry contribution. In the flat case a reset with such small tilt is already excluded by the scalar estimate.

Put $W = Z/I$. It is nonnegative and initially at most one. At resets $W' \leq -1$; at nonresets $W' \leq CB^{-\varepsilon_0}$. If the two time sets have lengths $\mathcal{R}, \mathcal{N}$, integration gives

$$\mathcal{R} \leq 1 + CB^{-\varepsilon_0} \mathcal{N}.$$

On the other hand $a' \leq 1$ at resets, while $a' = 2 - K$ at nonresets. A net increase of $B^{1/4}$ therefore requires

$$\mathcal{R} \geq B^{1/4} + (K - 2)\mathcal{N},$$

a contradiction for large B .

In the multiple-well case select good first-passage levels greedily with separation $2B^{1/3}$; removing an interval of length at most $2B^{1/3}$ per choice leaves at least $cB^{2/3}$ choices. In the one-well case use ordered, disjoint level intervals of length $B^{1/4}$ separated by $2B^{1/3}$, and choose a witness in each corresponding first-passage traversal. These traversals are ordered in time. Discarding witnesses if necessary gives (6.1) and the asserted number. $\square$

Lemma 6.2 (A controlled-density fraction). *Fix a finite family of nested dyadic grids with the property that every sufficiently small ball is contained in a cube of one of the grids with comparable diameter. The grids may be taken half-open, so their cubes form genuine partitions. At each witness of Theorem 6.1 there is a cube J of one of these grids such that, with $\delta = e^{-a(s)}$,*

$$B_s(x) \subset \text{int } J, \quad \ell(J) \leq Cs\sqrt{\log B}, \quad cB^{-2\varepsilon_0}\delta \leq \frac{\nu(J)}{|J|} \leq B^{\varepsilon_0}\delta. \quad (6.13)$$

There are fixed constants $C_*, c_* > 0$, independent of ε_0 , such that

$$\nu \left(\left\{ y \in J : \frac{\nu(R)}{|R|} \leq B^{C_*}\delta \text{ for every dyadic } R \subset J \text{ containing } y \right\} \right) \geq B^{-c_*\varepsilon_0} \nu(J). \quad (6.14)$$

Here $|J|$ denotes coordinate Lebesgue volume. The metric volume is uniformly comparable to it.

Proof. A finite family as stated can be constructed by using, in each coordinate, the three dyadic grids whose shifts alternate between the one-third positions at successive generations; this standard Euclidean construction is described in [13, p. 2]. At a scale between a fixed multiple of the ball diameter and twice that multiple, one of the three boundary sets in each coordinate is farther than the ball diameter from the center. Taking their product gives the required cube. We shall apply this property to a larger concentric ball, so the first inclusion in (6.13) has room to spare.

Let V_b be the normal multiplier of Theorem 5.4, and put $u = (y - x)/s$. In the scale- s height coordinate $z' = z/s$, define $v(z') = V_b(sz')/s$. Thus $D_{z'}v(z') = DV_b(sz')$, equal to the identity when sz' is near a represented well. In all integrals below, v means $v(z/s) = V_b(z)/s$,

while the measures remain in the section coordinates. Dividing the normal first-variation identity in [Theorem 5.4](#) by s^2 , with its Gaussian cutoff, therefore gives

$$\mathcal{F} := \int G_s v \cdot P_{vh} u \, d\mu \geq cB^{-\varepsilon_0} \delta. \quad (6.15)$$

For multiple wells the discarded terms are bounded by a fixed multiple of $D_{\text{tr}}(s)$, and ϑ in [Theorem 6.1](#) is chosen small enough to absorb them. For one well there is no transition contribution. Clipping and chart tails have the arbitrarily small bounds in [Theorem 5.4](#), and the curved first-variation errors are $o(B^{-\varepsilon_0} \delta)$. This last assertion follows from $k \geq B^{-\varepsilon_0}$, the moment barrier, the dilation bound, and the physical exponential smallness in [Theorem 3.1](#).

We give the moment and tail details needed to use this flux. Fix the witness scale s , and keep its wells, clipping scale, and multiplier $V_b = V_b(s)$ fixed while the ball radius varies. Choose a fixed $p > 2$ in the range (2.17), and fix an exceptional proportion small enough for the concentric dyadic comparisons below, independently of T , a , and R . For $R \geq 1$ in any fixed polynomial range in a , the dilation bound and a Gaussian lower bound give

$$(Rs)^{-m} \tau(B_{cRs}(x)) \leq d_R := C(1+R)^{K-2} \delta = o(1)$$

on the required fixed enlargement. Its $|M|$ part identifies Q by the smooth mass average in [Theorem 2.5](#). Choose the joint majority-and- L^p witness $\mathcal{B}_R$ in [Theorem 2.7](#), with that exceptional proportion. The bound $d_R = o(1)$ is uniform in the stated range, so it lies below the corresponding fixed small-density threshold. Its moment bound is

$$\left((Rs)^{-m} \int_{\pi^{-1}(B_{Rs}(x))} \text{dist}(z, \mathcal{B}_R)^p \, d\mu \right)^{1/p} \leq C(1+R)^{K/2} s \sqrt{\delta}.$$

This integral includes exceptional fibers and points with zero projection Jacobian.

Take the analogous witness $\mathcal{B}_1$ at scale s . For each of its entries, the matched projection entry on every good fiber is within $Cs\sqrt{\delta}$. The area formula and the uniform upper bound for J_π give these entries a fixed amount of G_s -weighted μ -mass. Using $|DV_b| \leq C$, $|V_b|^2 \leq \Gamma_s$, and the barrier $U \leq s^2 \delta / k$ therefore gives

$$\max_{\xi \in \mathcal{B}_1} |V_b(\xi)|^2 \leq CU + Cs^2 \delta \leq Ck^{-1} s^2 \delta.$$

This estimate is made one entry at a time, so repeated entries require no disjoint allocation of mass. Match the centers of $\mathcal{B}_R$ to those of $\mathcal{B}_1$ through concentric dyadic dilations. The majority sets have fixed positive relative overlaps, and the sum of the local matching bounds is at most $C(1+R)^{K/2} s \sqrt{\delta}$. For the fixed multiplier,

$$|v(z/s)| \leq \frac{C}{s} \text{dist}(z, \mathcal{B}_R) + \max_{\xi \in \mathcal{B}_R} \frac{|V_b(\xi)|}{s}.$$

The last three bounds and the bounded normalized projected mass give

$$\left((Rs)^{-m} \int_{\pi^{-1}(B_{Rs}(x))} |v|^p \, d\mu \right)^{1/p} \leq C(1+R)^C k^{-1/2} \sqrt{\delta}. \quad (6.16)$$

All these bounds use displacements from the fixed wells. Gaussian summation of (6.16) and of the energy dilation bound gives

$$\left(\int G_s |v|^p \, d\mu \right)^{1/p} \leq B^{c_1 \varepsilon_0} \sqrt{\delta}, \quad \int G_s (1 + |u|^2) e \, d\mu \leq C\delta. \quad (6.17)$$

The constants c_1 and C are fixed. To control more distant tails uniformly even when s is very small, note that clipping gives $|v| \leq C$ and the projected mass identity and dilation bound give $(\pi_{\#}\mu)_{\sqrt{2}s} \leq C$. Gaussian comparison therefore gives

$$\int_{|u|>R} G_s |v|^p d\mu \leq C e^{-R^2/4}.$$

At $R = C\sqrt{a}$, with C sufficiently large, this is smaller than $B^{-N}\delta^{p/2}$ for any prescribed fixed N . Likewise Gaussian comparison, with a slightly larger fixed dilation to absorb $|u|^2$, bounds the energy tail by $C e^{-cR^2}\delta$. These estimates cover the part beyond the polynomial dilation range and the fixed chart cutoff; they introduce no negative power of s .

Take $R_B = C_1\sqrt{\log B}$, with C_1 fixed sufficiently large. Annular Gaussian summation using (6.16) makes both the flux tail outside $B_{R_B s}(x)$ and the tilt tail there smaller than $o(B^{-\varepsilon_0}\delta)$. Choose J to contain this ball in its interior, with $\ell(J) \leq C R_B s$. The remaining tilt has Gaussian integral at least $c B^{-\varepsilon_0}\delta$. Since $G_s \leq C s^{-m}$,

$$\nu(J) \geq c B^{-\varepsilon_0}\delta s^m.$$

The upper density bound follows from the energy dilation estimate on a ball containing J . All resulting factors are fixed powers of $\log B$, and hence are at most B^{ε_0} after increasing the threshold for B . This proves (6.13).

Let H be the union of the maximal dyadic subcubes of J with density larger than $B^{C_*}\delta$. They are disjoint, and the upper bound in (6.13) gives

$$\frac{|H|}{|J|} \leq B^{\varepsilon_0 - C_*}. \quad (6.18)$$

It is essential here to estimate mass as well as volume. The identity $\pi_{\#}\mu = Q \text{vol}_S + M$ gives

$$\frac{\mu(\pi^{-1}H)}{|J|} \leq C Q \frac{|H|}{|J|} + \frac{|M|(J)}{|J|} \leq C B^{\varepsilon_0 - C_*} + B^{\varepsilon_0}\delta. \quad (6.19)$$

In particular the mass defect has not been discarded. Writing $\sigma = \frac{1}{2} - \frac{1}{p} > 0$, the Gaussian-weighted mass of $\pi^{-1}H$ is at most $B^{c_2\varepsilon_0 - C_*} + B^{c_2\varepsilon_0}\delta$, after absorbing the volume ratio $|J|/s^m$ into a power of B^{ε_0} . Hölder's inequality with exponents $p, 2, 1/\sigma$, together with $|P_{vh}u| \leq \sqrt{e}|u|$ and (6.17), bounds the absolute flux on this set by

$$B^{c_3\varepsilon_0}\delta (B^{c_2\varepsilon_0 - C_*} + B^{c_2\varepsilon_0}\delta)^{\sigma}. \quad (6.20)$$

Choose C_* large, independently of sufficiently small ε_0 , and then B large. Expression (6.20) is less than one quarter of the lower bound in (6.15). The strict inequality $p > 2$ is used exactly at this step.

There remains flux at least $c B^{-\varepsilon_0}\delta$ over $(J \setminus H) \cap B_{R_B s}(x)$. Cauchy–Schwarz, the second-moment consequence of (6.17), and $|u|^2 \leq R_B^2$ on this core imply

$$c B^{-2\varepsilon_0}\delta^2 \leq B^{c_4\varepsilon_0}\delta C R_B^2 s^{-m} \nu(J \setminus H).$$

Thus $\nu(J \setminus H) \geq B^{-c_5\varepsilon_0}\delta s^m$. Compare with $\nu(J) \leq B^{\varepsilon_0}\delta|J|$ and absorb the remaining fixed logarithmic factors. This proves (6.14) with a fixed c_* . The set $J \setminus H$ is precisely the set in that formula, with the harmless choice of strict or non-strict threshold indicated there. $\square$

6.2 Packing and contact charging

Lemma 6.3 (Dyadic packing). *Let ν be a finite positive measure on the union X of disjoint roots of a dyadic forest. For a cube R put $d(R) = \nu(R)/|R|$. Suppose that a family of marked cubes has densities in $[\lambda_-, \lambda_+]$, where $0 < \lambda_- \leq \lambda_+ < \infty$, and that every marked cube P satisfies*

$$\nu\{x \in P : d(R) \leq Ld(P) \text{ for every descendant } R \ni x\} \geq \zeta\nu(P), \quad (6.21)$$

where $L \geq 1$ and $0 < \zeta < 1$. Let E_N be the union of all full cubes F admitting a marked chain

$$W_1 \supsetneq W_2 \supsetneq \cdots \supsetneq W_N = F, \quad d(W_{j+1}) \geq Sd(W_j).$$

For $R_* > e$ and $S > LR_*$ one has

$$\nu(E_N) \leq \left[(1 - \zeta)^{N-1} + \frac{e}{R_*} \left(1 + \left\lfloor \log \frac{\lambda_+}{\lambda_-} \right\rfloor \right) \right] \nu(X). \quad (6.22)$$

In particular, the full innermost witness cubes furnished by [Theorems 6.1 and 6.2](#), for a fixed band and grid, have union E satisfying

$$\nu(E) \leq C \exp(-B^{1/8})d \quad (6.23)$$

when ε_0 is sufficiently small.

Proof. We first prove the estimate for downward retracing of density. Let

$$J_* = 1 + \lfloor \log(\lambda_+/\lambda_-) \rfloor, \quad \lambda_j = \lambda_- e^j \quad (0 \leq j < J_*).$$

For each j , stop at the first cubes, descending from the roots, whose density is at least λ_j . These high stopping cubes H are disjoint; a root is included if it already qualifies. Within each H stop at the first strict descendants R with $d(R) \leq e\lambda_j/R_*$. These low stopping cubes are also disjoint, and

$$\sum_R \nu(R) \leq \frac{e\lambda_j}{R_*} \sum_R |R| \leq \frac{e\lambda_j}{R_*} |H| \leq \frac{e}{R_*} \nu(H).$$

Now suppose $P \supsetneq P'$ are any two band cubes, meaning their densities belong to $[\lambda_-, \lambda_+]$, with $d(P') \leq d(P)/R_*$. Choose j so that $\lambda_j \leq d(P) < e\lambda_j$. The cube P lies below a high stopping cube H , and P' lies inside a low stopping cube below that H . Hence the union E_{ret} of all such low cubes has

$$\nu(E_{\text{ret}}) \leq (eJ_*/R_*)\nu(X). \quad (6.24)$$

This is a bound in ν -mass. Moreover, if the ancestry of a terminal cube has such a retrace before reaching that cube, the *entire* terminal cube is in E_{ret} .

We next construct a global forest of paying cubes. Its first generation consists of the first marked cubes below the roots. For each paying cube P , stop at the first strict descendants C with $d(C) > Ld(P)$. By (6.21),

$$\sum_C \nu(C) \leq (1 - \zeta)\nu(P).$$

Below each such C take the first marked cubes, including C itself if marked, as the next paying generation. All cubes in one paying generation are disjoint. Induction gives

$$\nu(\text{union of paying generation } j) \leq (1 - \zeta)^{j-1} \nu(X). \quad (6.25)$$

Consider a terminal F with a chain $W_1, \dots, W_N$ and with no band retrace in its ancestry before F . The first paying cube on its ancestry contains W_1 . Suppose the j th paying cube P_j contains W_j . Both cubes are band cubes. Absence of a retrace implies $d(W_j) > d(P_j)/R_*$ when they differ; the lower bound needed below also holds when they coincide. Consequently

$$d(W_{j+1}) \geq Sd(W_j) > Ld(P_j).$$

The next density crossing, and then the next first marked cube, therefore occur at or before W_{j+1} . The $(j+1)$ st paying cube contains W_{j+1} . It follows that F is wholly contained in a paying cube of generation N . Thus

$$E_N \subset E_{\text{ret}} \cup (\text{union of paying generation } N).$$

Equations (6.24) and (6.25) prove (6.22). All families are countable; half-open cubes give the statement also for measures charging dyadic boundaries. Finite-depth truncation followed by monotone convergence is an alternative justification of every stopping construction above.

We apply Theorem 6.3 to all cubes which can occur in Theorem 6.2, marking a cube if it admits at least one of the witness labels. A label need not be selected coherently between different contact paths. Each such cube satisfies (6.21) with the uniform choices

$$\begin{aligned} \lambda_- &= cB^{-2\varepsilon_0}e^{-2B}, & \lambda_+ &= B^{\varepsilon_0}e^{-B}, \\ L &= \max\{1, c^{-1}B^{C_*+2\varepsilon_0}\}, & \zeta &= B^{-c_*\varepsilon_0}. \end{aligned}$$

Indeed $Ld(J) \geq B^{C_*}e^{-a}$ for any of its labels. Take $R_* = \exp(B^{1/4})$ and $S = \exp(B^{1/3}/2)$. The inequality $S > LR_*$ holds for large B , and $J_* = O(B)$.

There is a fixed number of grids. A fixed fraction of the ordered witnesses on each contact path belongs to one grid. Retain that subsequence and reverse its order, so that cubes are listed from outer to inner. By (6.4) and (6.1), consecutive radii differ by at least $\exp(cB^{1/3})$. The first two properties in (6.13) therefore give strict nesting, with the same contact point in every cube. The density bounds give the factor S for consecutive densities for large B . There are still $N \geq cB^{1/2}$ cubes. Start the dyadic forest at its maximal cubes contained in the working chart; all witness cubes lie in this forest, and its roots are disjoint. Its total ν -mass is at most d . Thus (6.22) is at most

$$\left[\exp\{-cB^{1/2-c_*\varepsilon_0}\} + CB \exp(-B^{1/4}) \right] d.$$

Choose ε_0 so small that $c_*\varepsilon_0 < 1/4$. This proves (6.23) after increasing the threshold for B . The estimate concerns the union of full innermost cubes, not just the contact points or almost every point on selected infinite chains. $\square$

Lemma 6.4 (Charging the contact measure). *Let h and ν be finite positive measures on $\mathbb{R}^m$, and let $\mathcal{C}$ be a set of contact centers. For each of its contacts suppose there are radii $0 < u_x \leq s_x$ with*

$$\nu_{s_x}(x) \geq c\theta, \quad \nu_{\sqrt{2}s_x}(x) \leq C\theta, \quad h_{u_x}(x) \leq C\theta, \quad (6.26)$$

where $\theta > 0$ and the constants are fixed. There is a fixed L_0 such that, if $B_{L_0s_x}(x) \subset E$ for every contact, then

$$h^*(\mathcal{C}) \leq C\nu(E). \quad (6.27)$$

Here E is Borel and h^* is outer measure. The statement also holds for a submeasure of a contact measure whose spatial projection is dominated by h .

Proof. Outside $B_{L_0 s_x}(x)$ the Gaussian kernels satisfy

$$G_{s_x}(y-x) \leq 2^{m/2} e^{-L_0^2/4} G_{\sqrt{2}s_x}(y-x).$$

Choose L_0 sufficiently large. By (6.26), at least $c\theta/2$ of the s_x -Gaussian integral is then inside this ball. Since $G_{s_x} \leq C s_x^{-m}$, the threshold balls $B_x = B_{L_0 s_x}(x)$ satisfy

$$\nu(B_x) \geq c\theta s_x^m. \quad (6.28)$$

Apply the usual decreasing-radius selection to these balls; see [19, Chapter 1, Lemma 3.4] for the blocking-ball property used below. For clarity, grouping radii in successive factors of two and taking maximal disjoint families gives disjoint selected balls $B_i = B_{R_i}(x_i)$ such that every original ball meets a selected ball with radius at least half its own. Assign each contact to one such blocking selected ball; this gives both

$$|x - x_i| < 3R_i, \quad u_x \leq s_x \leq 2R_i/L_0.$$

The radius condition is part of the assignment, and does not follow merely from coverage of centers by enlarged balls. The radii are bounded, since $c\theta \leq \nu_{s_x}(x) \leq C_m s_x^{-m} \nu(\mathbb{R}^m)$.

Within one assigned family apply the same selection to the derivative balls $B_{u_x}(x)$. Obtain disjoint balls $D_{ij} = B_{u_{ij}}(x_{ij})$ whose fivefold enlargements cover its centers. All the original D_{ij} lie in $B_{(3+2/L_0)R_i}(x_i)$, so comparison of ordinary volumes gives $\sum_j u_{ij}^m \leq C R_i^m$. Positivity of the Gaussian on each fivefold ball and (6.26) give

$$h(5D_{ij}) \leq C\theta u_{ij}^m.$$

Therefore the h outer mass of the contacts assigned to i is at most $C\theta R_i^m \leq C\nu(B_i)$, by (6.28). The B_i are disjoint and contained in E , which proves (6.27). This argument uses no absolute continuity of h , and remains valid if the derivative radii are much smaller than the threshold radii. $\square$

Completion of the proof of Theorem 3.1. Fix Q and assume the assertion for all smaller positive integers. Choose C_Q large enough for Theorems 4.3, 5.2, 5.4 and 6.1 and the bottom-contact estimate in Theorem 5.1. The lower-integer constants are already fixed. If $Q = 1$, use only the one-well statements; no separated estimate, function b , or multiple-well construction is used.

Put $F = F_Q(T)$, $\eta = e^{-2F}$, and $a_0 = F^2$, as in Theorem 5.2. Suppose the localized source has mass at least ηd . For each positive bottom scale use the entrance boundary K_σ , the supported contact set $\mathcal{S}_\sigma$, and the Borel threshold event $\mathcal{G}_\sigma$ defined in Section 5. Write $\hat{h} = \hat{h}_\sigma$ and $h = h_\sigma = (\pi_X)_\# \hat{h}$; both have mass $\int f$. By Theorem 5.1, the $\hat{h}$ -mass of $K_\sigma \setminus \mathcal{G}_\sigma$ is at most

$$o(1) + C e^{-a_0} d \quad (6.29)$$

as the bottom tends to zero. We estimate the contacts in $\mathcal{G}_\sigma \cap \mathcal{S}_\sigma$, whose $\hat{h}$ -mass equals $\hat{h}(\mathcal{G}_\sigma)$, uniformly in that bottom.

At such a contact let $\underline{s}$ be its derivative scale and $s_0 \geq \underline{s}$ a threshold equality on the outward vertical. In particular this includes a contact on a vertical side below a grazing equality. The continuous stopping rule and localization give

$$\tau_{s_0}(x) = e^{-a_0}, \quad \tau_v(x) \leq e^{-a_0} \quad (\underline{s} \leq v \leq \rho), \quad h_{\underline{s}}(x) \leq C e^{-a_0}.$$

The harmless cutoff errors can be absorbed in fixed constants in these relations. There is room for the fixed dilation $\sqrt{2}s_0 < \rho$: for $s \in [\rho/\sqrt{2}, \rho]$, the total mass bound gives $\tau_s \leq C \rho^{-m} d = o(e^{-a_0})$, using $a_0 = F_Q(T)^2 = o(T)$ and $|\log \rho| = O(F_Q(T) + \log T)$. Thus (6.26) holds with $\theta = e^{-a_0}$, $u_x = \underline{s}$, and $s_x = s_0$.

For this periodized application, $u_x \leq s_x = s_0 < \rho$, and every fixed dilation of the balls used in the proof has a Euclidean lift inside the fixed coordinate chart for large T . The upper Gaussian comparison holds termwise over lifts outside the torus metric ball; at these radii $G_s \leq Cs^{-m}$, and the lower bound on fivefold balls comes from the nearest lift. The two Euclidean selections and their volume comparisons therefore apply in that chart, with the radii already bounded by ρ .

Each such contact has a band and a grid supplying the ordered witnesses of [Theorems 6.1](#) and [6.2](#). For a band i and grid g , let $A_{i,g} \subset \mathcal{G}_\sigma \cap \mathcal{S}_\sigma$ be the contacts admitting an eligible chain for that pair, and let $E_{i,g}$ be the union of all its eligible full innermost cubes. The finitely many sets $A_{i,g}$ cover $\mathcal{G}_\sigma \cap \mathcal{S}_\sigma$; they may overlap and need not be measurable. Each $E_{i,g}$ is Borel because the dyadic family is countable, and [Theorem 6.3](#) gives $\nu(E_{i,g}) \leq Ce^{-B_i^{1/8}}d$.

For a contact in $A_{i,g}$, choose any one eligible chain. Its threshold ball lies in the innermost cube. Indeed $a(s_0) = a_0 + O(1)$, the smallest witness has level at least $B_i \geq 2a_0$, and $|a'| \leq C$. Hence

$$\log(s_{\min}/s_0) \geq c(B_i - a_0 - O(1)) \longrightarrow \infty.$$

The cube contains $B_{s_{\min}}(x)$, so it contains $B_{L_0 s_0}(x)$ for the fixed constant in [Theorem 6.4](#).

For every subset $A \subset K_\sigma$, the induced outer measures obey

$$\hat{h}^*(A) \leq h^*(\pi_X A).$$

Indeed, a Borel spatial set $V \supset \pi_X A$ has $A \subset \pi_X^{-1}V$ and $\hat{h}(\pi_X^{-1}V) = h(V)$; take the infimum over such V . Apply [Theorem 6.4](#) to the center set $\pi_X A_{i,g}$, choosing for each center any one contact in that class and its radii. Its threshold ball lies in $E_{i,g}$, and the Gaussian hypotheses were verified above for the full projection h . Thus repeated centers introduce no multiplicity factor, and $h^*(\pi_X A_{i,g}) \leq C\nu(E_{i,g})$. The covering proof uses no measurable choice of contacts or chains. Outer subadditivity now gives

$$\begin{aligned} \hat{h}(\mathcal{G}_\sigma) &= \hat{h}(\mathcal{G}_\sigma \cap \mathcal{S}_\sigma) \\ &\leq \sum_{i,g} \hat{h}^*(A_{i,g}) \leq \sum_{i,g} h^*(\pi_X A_{i,g}) \\ &\leq C \sum_{i,g} \nu(E_{i,g}) \leq C \sum_{i=1}^Q e^{-B_i^{1/8}}d. \end{aligned} \tag{6.30}$$

The band choice in [Theorem 5.5](#) ensures $B_1^{1/10} \gg F^3$. Consequently the right side of (6.30), and also $Ce^{-a_0}d$, are $o(\eta d)$ as $T \rightarrow \infty$. First take T large with all fixed parameters chosen as above, and then take the bottom small in (6.29). These bounds contradict $\hat{h}(K_\sigma) = h(X) \geq \eta d$.

Thus the localized source has mass less than ηd . Restoring its subtractive margin, whose integral is at most $C\eta d$, proves

$$\int w(L - C_Q \Lambda) \leq Ce^{-2F_Q(T)}d \leq e^{-F_Q(T)}d$$

for sufficiently large T , uniformly over the stated nonnegative unit Lipschitz weights. This proves the induction step and hence [Theorem 3.1](#) for every positive integer Q . $\square$

7 A center adapted to the multiplicity defects

We now use the signed estimate to construct a single reference graph. Its height error will be supported in a set of vanishing density. Two features of the construction are needed later:

neighboring minimal fits agree to *second* order in the height error, and failure of a fit forces a definite amount of lower multiplicity in every fixed interior subball.

Minimal-graph refinement and Jacobi regularization of an average appear in Brena–De Lellis–Franceschini [4, Sections 2 and 6.3–6.4]. Almgren developed the center-manifold approach to higher-multiplicity regularity [3]. De Lellis–Spadaro construct a center manifold from smoothed local averages and Whitney gluing [9, Sections 1 and 4]. Here the fits are selected using multiplicity defects, and we prove their quadratic compatibility before gluing them.

7.1 Minimal fits and separation

We specify the points at which the construction is made. Translate such a point to the origin, and write $P_0 = \mathbb{R}^m \times \{0\}$ and π_0 for its orthogonal projection. Define the dilated varifold explicitly by $V_r(\Phi) = r^{-m} \int \Phi(X/r, T_X E) d\mu(X)$. The hypotheses are

$$V_r \longrightarrow Q|P_0|, \quad r^{-m} \mu(\mathbf{B}_r(0) \cap \{\theta \neq Q\}) \longrightarrow 0, \quad Q \in \mathbb{N} \setminus \{0\}, \quad (7.1)$$

where the first convergence includes the tangent-plane variable. Thus dilated masses and tangent projections converge to those of the multiplicity- Q plane. As usual, θ is the integer multiplicity at almost every point of the rectifiable set.

These hypotheses also give a support statement. In a sufficiently small punctured ambient ball,

$$|X_v| \leq \varepsilon(|X|)|X_h| \quad (X \in \text{spt } \mu), \quad \varepsilon(s) \longrightarrow 0. \quad (7.2)$$

Indeed, if $|X_v| \geq \varepsilon_0|X|$ along a sequence $X \rightarrow 0$, the balls $\mathbf{B}_{c\varepsilon_0|X|}(X)$ have mass at least $c'\varepsilon_0^m|X|^m$ by [Theorem 2.1](#). After dilation by $|X|^{-1}$ these balls stay a fixed positive distance from P_0 , contrary to (7.1). Passing from $|X|$ to $|X_h|$ changes the function ε only by a factor tending to one.

Choose a cylinder strictly inside that ambient ball, with the support separated from its vertical boundary, and dilate it to contain the fixed base ball $B_{300} \subset P_0$. All varifolds below are restrictions to this cylinder, and all tests have horizontal support away from its side. The mass of each fixed interior ball is bounded above by a constant depending on the fixed data, including Q , times its radius to the power m : apply monotonicity at that center and a fixed larger radius. By choosing the initial dilation sufficiently small, the height on the fixed cylinder and the weak mass error can be made arbitrarily small. In addition the measure

$$\lambda = (\pi_0)_\#(\mu_\perp \{\theta \neq Q\}) \quad (7.3)$$

satisfies, for all radii needed in the fixed chart,

$$\lambda(B_s(0)) \leq \eta s^m, \quad \lambda(B_s(0)) = o(s^m) \quad (s \downarrow 0). \quad (7.4)$$

Here $\eta > 0$ can be prescribed before making the dilation. To see this, (7.2) puts the relevant part of the cylinder above $B_s(0)$ inside $\mathbf{B}_{2s}(0)$, and then one uses the second limit in (7.1). We will choose one sufficiently small η below and keep the resulting data fixed.

Lemma 7.1 (Minimal fits and their compatibility). *Fix a finite derivative order $J \geq 6$. There are an integer $k_0 > J + 4$, integers q, N , and constants $C, m_* > 0$, depending only on the dimensions, Q , and these fixed orders, with the following property. For $0 < m_0 < m_*$ and sufficiently small initial height and mass error, one can make the following construction simultaneously for every $x \in B_5$.*

At the scales $\ell_j = 2^{-j}$ construct smooth minimal graphs $S_{x,j} = \text{graph } f_{x,j}$ over $B_{80\ell_j}(x)$, and stop at the first scale for which

$$\sup\{|X_v - f_{x,j}(X_h)| : X \in \text{spt } \mu, X_h \in B_{64\ell_j}(x)\} \leq m_0 \ell_j^{k_0} \quad (7.5)$$

fails. If there is no failure, all scales are constructed. The graph at the first failed scale is included in the construction. On fixed interior enlargements of its test ball, every constructed graph satisfies the height upper bound $Cm_0\ell_j^{k_0}$, the tilt and mass-defect bounds

$$\ell_j^{-m}(D + |M|)(B_{70\ell_j}(x)) \leq Ca_j^2, \quad a_j = m_0\ell_j^{k_0-1}, \quad (7.6)$$

and the following centering estimate. For each smooth vector test ζ supported in B_{70} ,

$$\left| \ell_j^{-m} \int \frac{X_v - f_{x,j}(X_h)}{\ell_j} \cdot \zeta \left(\frac{X_h - x}{\ell_j} \right) d\mu(X) \right| \leq Ca_j^2 \|\zeta\|_{H^q}. \quad (7.7)$$

The same assertion holds in normal coordinates over $S_{x,j}$, with base volume in the definition of the distribution.

Every graph has physical derivatives of orders $1, \dots, J$ bounded by Cm_0 in the fixed initial coordinates. If two constructed fits have comparable scales and a common interior region containing a ball of radius comparable to ℓ , then, on a fixed smaller ball in that region,

$$\|D^i(f - f')\|_{L^\infty} \leq Cm_0^2 \ell^{2k_0-1-i}, \quad 0 \leq i \leq J. \quad (7.8)$$

The constants in this last assertion can depend on the fixed scale ratio and the fixed interior room. No measurability of $x \mapsto f_{x,j}$ is required.

Proof. We first give a local correction argument, keeping all its domains a fixed positive distance apart. Work in units of one fitting scale and normal coordinates over a smooth minimal input S' . Suppose that its slope and a sufficiently large finite number of its scaled derivatives are small, the mass is bounded, the selected integer is Q , and $|z| \leq Ca$. Here and throughout this proof constants may incorporate fixed factors such as 2^{k_0} .

For the ambient normal displacement field $Z(X) = X - \pi(X)$, direct differentiation gives

$$\text{tr}(PDZ) = e - P_{hh} : A_z(I - A_z)^{-1} = e - H_{S'} \cdot z + O(|A|^2|z|^2 + |A|e|z|). \quad (7.9)$$

Testing stationarity with $\chi^2 Z$, as in the classical height-to-tilt estimate [2, Lemma 31], absorbing the last term, and using $|P_{vh}|^2 \leq e$ gives

$$\int \chi^2 e \, d\mu \leq C \int (|\nabla \chi|^2 + |A|^2 \chi^2) |z|^2 \, d\mu \leq Ca^2.$$

Here $H_{S'} = 0$ is essential: it removes the term linear in height. [Theorem 2.5](#) then gives $|M| \leq Ca^2$ on smaller balls. The integer is the previously selected Q : moving a normalized smooth mass test between the old and new projections changes its average by $O(a)$, while the old mass defect is $O(a^2)$. For small a its average therefore remains within $1/4$ of Q . The exceptional projection counts occupy only $O(a^2)$ of each such ball. Apply [Theorem 2.5](#) successively through overlapping balls if the smaller region is not concentric.

Let $u = Q^{-1}\pi_\#(z\mu)$, interpreted as a distributional normal section using $\text{vol}_{S'}$. Normal stationarity and horizontal stationarity with a linear height multiplier imply

$$\left| \int_{S'} \left(\nabla^\perp u : \nabla^\perp v - \text{tr}(A_u A_v) \right) d\text{vol}_{S'} \right| \leq Ca^2 \|v\|_{C^N} \quad (7.10)$$

for compactly supported smooth normal fields v in a smaller region. Derivatives of u in this formula are distributional. For completeness, the trace of the normal test $v \circ \pi$ is

$$-P_{hh} : A_v(I - A_z)^{-1} + P_{vh} : \nabla^\perp v (I - A_z)^{-1}.$$

Its terms of first order in (z, P_{vh}) are $-\text{tr}(A_z A_v) + P_{vh} : \nabla^\perp v$; the constant trace is $-H_{S'} \cdot v = 0$. The horizontal lift multiplied by $z \cdot w$ identifies $\nabla^\perp u$ with $Q^{-1}\pi_\#(P_{vh}\mu)$ to first order. All

remainders contain e , $|z|\sqrt{e}$, or $|z|^2$, against bounded derivatives of the test. Their mass is $O(a^2)$. This proves (7.10), including the normal-connection terms. Its bilinear form can also be read directly from the quadratic part of the graph area integrand, whose differential blocks are $(I - A_v, \nabla^\perp v)$.

Here is the local elliptic argument that turns (7.10) into an exact minimal correction. Choose an integer $q > m/2 + 1$ sufficiently large that the right side of (7.10) is bounded by the H^{q+2} norm of v , and choose a positive integer $s > m/2 + J + 3$. Fix a nested chain between the available radius 128 and the required radius 80, for example with intermediate radii 120, 112, 104, 96, 88, subdividing these intervals into finitely many further radii for the Sobolev bootstrap. Successive restrictions below use this fixed chain; all cutoffs and extensions have a positive buffer. For any nested pair of these working balls the Jacobi operator $\mathcal{J}$ has a local right inverse $\mathcal{R}$ with

$$\|\mathcal{R}F\|_{H^{t+2}} \leq C_t \|F\|_{H^t}, \quad -q - 2 \leq t \leq s, \quad \mathcal{J}\mathcal{R}F = F \quad \text{on the smaller ball.} \quad (7.11)$$

Only finitely many coefficient derivatives are involved. One explicit construction removes boundary compatibility issues as follows. Embed the working balls in a fixed flat torus, extend the small coefficient perturbations of $-\Delta$ by a fixed cutoff, and choose a smooth scalar bump b , supported outside the larger ball, with integral one. The operator $-\Delta + b \int_{\mathbb{T}^m}(\cdot)$ is invertible: integrate the equation to obtain the mean of the unknown, solve the remaining mean-zero equation by Fourier series, and restore that mean. The Fourier multiplier gains two derivatives on every integer Sobolev scale. Its sufficiently small coefficient perturbation is invertible on each of the finitely many scales in (7.11) by a Neumann series. Multiply F by a cutoff equal to one on the desired smaller ball and apply this inverse. On that ball the added bump term is zero, so the asserted equation is exact.

Apply this construction to the residual $\mathcal{J}u$. On a smaller ball, $v_0 = u - \mathcal{R}(\mathcal{J}u)$ satisfies $\mathcal{J}v_0 = 0$ and $u - v_0 = O(a^2)$ in H^{-q} . Moreover $u = O(a)$ in that norm, since it is a measure of mass $O(a)$. Interior regularity gives $\|v_0\|_{H^{s+2}} \leq Ca$ after another restriction. This last estimate follows directly from (7.11): apply the equation to nested cutoffs of v_0 ; each commutator has order at most one, so each step improves the available Sobolev order by one. Iteration reaches $s + 2$ and then Sobolev embedding gives every fixed derivative bound required here.

Write the minimal graph equation for a normal section as $\mathcal{J}v + \mathcal{N}(v) = 0$. In H^{s+2} on these fixed balls, the product rule and $s > m/2$ give

$$\|\mathcal{N}(v)\|_{H^s} \leq C \|v\|_{H^{s+2}}^2, \quad \|\mathcal{N}(v) - \mathcal{N}(w)\|_{H^s} \leq C (\|v\|_{H^{s+2}} + \|w\|_{H^{s+2}}) \|v - w\|_{H^{s+2}}.$$

Extend v_0 by a cutoff and solve $w = -\mathcal{R}\mathcal{N}(v_0 + w)$ on a ball of radius Ca^2 in H^{s+2} , putting the nonlinear cutoff equal to one on the ultimate working region. This is a contraction for small a . On that region $v = v_0 + w$ defines an exact minimal graph and $v - u = O(a^2)$ in H^{-q} . Interior differentiation of the equation gives smoothness and any further fixed derivative estimates.

Converting this normal graph to a graph over P_0 gives the desired centered fit. Vertical displacement at the fixed horizontal base is a smooth linear transformation of z , up to $O(|z|^2)$. Moving the argument of a smooth test from one projection to the other costs $O(a^2)$ when multiplied by the height. Finally, evaluating the graph correction against $\pi_\# \mu$ instead of $Q \text{vol}_{S'}$ costs $O(a^3)$, by the mass-defect bound and its $O(a)$ size. The first-order term is $Q(u - v)$, proving (7.7). The same argument after changing to the new normal coordinates proves the normal version. The new graph has height error Ca and tilt Ca^2 ; another application of Theorem 2.5 gives its mass estimate.

We apply this procedure at $j = 0$ using the horizontal plane as input, with initial errors small enough that (7.5) holds there. At $j > 0$ use the preceding successful fit. Its test ball has radius $128\ell_j$, whereas the new graph is required only on $B_{80\ell_j}(x)$. Thus all the

fixed intermediate balls just used fit inside the available region. Its height error in the new scaled units is at most $2^{k_0} m_0 \ell_j^{k_0-1}$. This proves the stated bounds at every constructed scale, including the first failure. The correction at scale ℓ has physical derivatives of order i bounded by $C m_0 \ell^{k_0-i}$. Summing over the earlier dyadic scales is finite for $i \leq J < k_0$. This maintains the small physical C^J bounds. The order choices here are noncircular: first fix J and the finite test order in (7.10), then q, s and the finite coefficient order required by (7.11), then k_0 and finally the smallness threshold for m_0 . The required coefficient order may exceed J . Its uniform *scaled* derivative bounds follow by interior bootstrapping of each exact minimal input from its small slope and scaled $C^{2,\alpha}$ bound, using the room between its domain and the new working region. The plane supplies this bound at initialization. Under the next halving of scale the second-derivative norm contracts by $1/2$ and its Hölder seminorm by $2^{-1-\alpha}$, while the correction adds only $C a_j$ and the slope increments $C a_j$ are summable. This closes the inductive bound before the higher-order bootstrap is applied. In particular the scaled C^{q+2} bounds used for changes of projection, multiplication of Sobolev tests, and the overlap argument below do not require physical derivatives through order $q+2$ to be included in J . Every threshold and constant is thus uniform along the construction.

It remains to prove the quadratic compatibility. Scale a common interior ball to unit size and put $a = m_0 \ell^{k_0-1}$. For two fits f_1, f_2 , the difference $d = f_1 - f_2$ solves a homogeneous uniformly elliptic linear system: integrate the derivative of the minimal-graph flux along the segment between their gradients. Its coefficients have the bounded derivatives and small ellipticity perturbation already established. The height bounds on the support and the projection-count conclusion of Theorem 2.5 first give $\|d\|_{L^1} \leq C a$ on an interior ball. Indeed, on a typical normal fiber of the first fit, choose a support point. Both fits are within $C a$ of that point at its horizontal coordinate. Moving back to the horizontal coordinate of the normal base point costs $O(a)$ by the small slopes, so the difference there is $C a$. The exceptional base volume is $O(a^2)$, and the small slopes bound the difference there by a fixed constant once it is bounded at one point. Interior estimates for this homogeneous system therefore give $\|d\|_{C^{q+2}} \leq C a$ on a smaller ball.

Subtract the two fixed-base moment estimates. This gives $\int d\zeta \, d((\pi_0)_\# \mu) = O(a^2) \|\zeta\|_{H^q}$, increasing q once if necessary. To compare this with base volume, first replace $\pi_0 X$ by $\pi_0 \pi_{S_1} X$ in the test $d\zeta$. The displacement is $O(a)$ and $d\zeta$ has derivatives $O(a)$, so the error is $O(a^2)$. Now Theorem 2.5 changes the measure to $Q \text{vol}_{S_1}$ with error $O(a^3)$. Thus, writing J_{f_1} for the area density of the graph over P_0 , $J_{f_1} d = O(a^2)$ in a fixed negative Sobolev norm. Multiplication by $J_{f_1}^{-1}$ preserves that bound. The same homogeneous interior estimates, now starting from this negative norm, give $\|d\|_{C^J} \leq C a^2$. Returning to physical coordinates gives (7.8). $\square$

Lemma 7.2 (A failed fit forces separation). *Fix $0 < c_0 < 1/10$ and a tilt threshold $\varepsilon_0 > 0$. By reducing the admissible upper bound for m_0 in Theorem 7.1, there is a constant $c > 0$ with the following property. If the first failure at x has scale ℓ , then in every base ball of radius $c_0\ell$ whose center lies in $B_{30\ell}(x)$, normal projection onto the failed fit has a subset of base volume at least $c\ell^m$ on which the total integer count is Q and two entries have separation at least $cm_0\ell^{k_0}$. Those two entries have tilt at most ε_0 relative to the fit and each has multiplicity strictly less than Q . The word “entry” refers to a point where the projection Jacobian is nonzero, with its integer multiplicity; no smoothness of a sheet is asserted. In particular no failure occurs when $Q = 1$.*

Proof. Suppose the assertion fails along a sequence $m_0 \downarrow 0$. Dilate each failed fitting ball by ℓ^{-1} and put $a = m_0\ell^{k_0-1}$. Relative to its failed fit, $|z| \leq Ca$ and $D + |M| \leq Ca^2$ on fixed interior balls. The reference graphs converge smoothly to a plane. Identify their bases by their graph coordinates, including their volume densities in the transported measures. After taking a subsequence on the connected interior disk B_{70} ,

$$a^{-2}\pi_{\#}(|z|^2\mu) \rightharpoonup \nu, \quad a^{-2}\pi_{\#}(B\mu) \rightharpoonup \beta, \quad a^{-2}D \rightharpoonup d, \quad a^{-2}M \rightharpoonup \mathbf{m}, \quad d = \text{tr } \beta. \quad (7.12)$$

Here ν, d are positive, β is a positive semidefinite matrix-valued measure, and $\mathbf{m}$ is signed. The bound $|z|/a \leq C$ and $\pi_{\#}\mu = Q \text{vol}_S + O(a^2)$ in total variation imply $\nu \leq C \, dy$ on every compact interior set.

Failure means that the fixed-base vertical error exceeds a at a support point over B_{64} . Since the failed graph has uniformly small slope, its nearest normal distance is comparable to that vertical error. Thus $|z| \geq ca$ at this point. The change of projection moves its base point by $O(a)$, so a fixed interior ball about it still lies over B_{70} . By (7.9) and the vanishing curvature of the scaled fits, the tangential Laplacian of $|z|^2$ is bounded below by $-o(a^2)$. More explicitly, absorb the $e|A||z|$ term in the nonnegative e term and retain $-C|A|^2|z|^2$. Adding $o(a^2)|X - X_1|^2$ gives a nonnegative subharmonic function on a fixed interior ambient ball about that point. Its mean-value inequality from Theorem 2.1 shows that $\nu(B_{70}) \geq c > 0$.

The signed estimate gives the crucial additional inequality

$$2\mathbf{m} \leq d. \quad (7.13)$$

We check its physical-scale hypothesis even if ℓ stays bounded away from zero. Before applying Theorem 3.1, dilate the original physical space by $\lambda_0 = m_0^\sigma$, where $0 < \sigma < 1/(J-1)$ and J is increased beforehand if further fixed coordinate derivatives are required. A physical derivative of order $i \leq J$ of a fit is now bounded by $Cm_0\lambda_0^{1-i}$, hence uniformly bounded. A comparable test ball has physical radius $l \asymp \lambda_0\ell$, and its physical height is at most $C\lambda_0\ell a$. With $d_0 = Ca^2$ and $T = -\log d_0$, choose

$$0 < \kappa < \min \left\{ \frac{\sigma}{2}, \frac{1}{2(k_0-1)} \right\}. \quad (7.14)$$

Then $l + \sup |z_{\text{phys}}| \leq d_0^\kappa$ for all sufficiently small m_0 , uniformly in $0 < \ell \leq 1$. The fits are minimal, so their Λ consists only of physical height squared and $\Lambda/a^2 \leq C\lambda_0^2\ell^2\pi_{\#}\mu \rightarrow 0$ locally. Apply Theorem 3.1 on finitely many interior balls to an arbitrary nonnegative smooth test, rescaling its size and Lipschitz constant by fixed factors. Divide by a^2 and pass to the limit. Its error is $O(e^{-F_Q(T)})$, which tends to zero, proving (7.13).

We next prove that ν is nonzero in *every* open interior subball. This is a finite-radius unique-continuation argument for the limiting measures, and requires no regularity theorem for a multiple-valued limiting function. Fix an interior center y_0 and an integer $p \geq 4$, and set

$$\phi_s(y) = s^{-m} \left(1 - \frac{|y - y_0|^2}{s^2} \right)_+^p, \quad \psi_s(y) = 2p s^{-m} \left(1 - \frac{|y - y_0|^2}{s^2} \right)_+^{p-1}.$$

In this paragraph only, define

$$\mathcal{H} = \int \psi_s \, d\nu, \quad \mathcal{A} = s^2 \int \phi_s \, d\mathbf{m}, \quad \mathcal{D} = s^2 \int \phi_s \, dd, \quad \mathcal{R} = \int \psi_s \, d\beta[y - y_0, y - y_0].$$

Dots denote differentiation in $\log s$. Stationarity gives

$$\dot{\mathcal{A}} = 2\mathcal{A} - \mathcal{D} + \mathcal{R}, \quad \dot{\mathcal{H}} = 2\mathcal{D}, \quad \mathcal{R}\mathcal{H} \geq \mathcal{D}^2, \quad 2\mathcal{A} \leq \mathcal{D}. \quad (7.15)$$

Here are the identities before the limit. The horizontal test $(y - y_0)\phi_s$ gives the first equation. The normal test $z\phi_s$ identifies $s^2 \int \phi_s e/a^2 \, d\mu$ with

$$\int \psi_s \frac{z \cdot P_{vh}(y - y_0)}{a^2} \, d\mu + o(1).$$

The horizontal test $(y - y_0)\psi_s|z|^2/a^2$ gives twice this flux for the derivative of the height integral. Terms containing both tilt and $|z|^2/a^2$ have mass $O(a^2)$, and the geometric errors tend to zero by (7.9) and Theorem 2.2. Finally $P_{hv}P_{vh} = B - B^2 \leq B$ and Cauchy–Schwarz bound the square of the flux by the product defining $\mathcal{R}\mathcal{H}$. The last inequality is (7.13).

Suppose there is a positive first radius s_0 at which this height becomes nonzero. Put $t = \log s$, $t_0 = \log s_0$, and choose a fixed $t_1 > t_0$ with all the balls strictly interior. On $(t_0, t_1]$ put $P = \mathcal{A}/\mathcal{H}$ and $n = \mathcal{D}/\mathcal{H}$. Then

$$n \geq 0, \quad 2P \leq n, \quad \dot{P} \geq 2P - n + n^2 - 2nP. \quad (7.16)$$

For $P \geq 1$ the last expression is nonnegative on $n \geq 2P$. Consequently $P(t) \leq \max(1, P(t_1))$ going inward. On the zero-height core, the second identity in (7.15), applied also to smaller balls and other centers, gives $d = 0$. To pass the total-variation estimate of Theorem 2.5 to this core, reuse the auxiliary physical dilation $\lambda_0 = m_0^\sigma$ above: it leaves normalized M, D unchanged and makes $a^{-2}\Lambda \rightarrow 0$. For a ball B with $3B$ compactly inside the core, take $\chi \in C_c(3B)$ equal to one on $2B$. For every $\zeta \in C_c(B)$ with $|\zeta| \leq 1$, that estimate gives

$$|\mathbf{m}(\zeta)| \leq C \lim_i \int \chi a_i^{-2} (D_i + \Lambda_i) = C \int \chi \, dd = 0.$$

Hence $\mathbf{m} = 0$ in the core, without a boundary-continuity assumption on the limiting measures. The compact kernels converge uniformly as $t \downarrow t_0$ and vanish on the boundary of the core, so $\mathcal{A}(t) \rightarrow 0$ even if $\mathbf{m}$ charges that boundary. Moreover

$$(e^{-2t}\mathcal{A})^\cdot = e^{-2t}(\mathcal{R} - \mathcal{D}) \geq -\frac{1}{4}e^{-2t}\mathcal{H},$$

because $n^2 - n \geq -1/4$. The function $\mathcal{H}$ is nondecreasing. Integrating from t_0 therefore gives the explicit lower bound

$$P(t) \geq -\frac{e^{2(t-t_0)} - 1}{8}.$$

With P bounded on this finite interval, (7.16) implies $\dot{P} \geq n^2/2 - C$. Integration gives a uniform bound for $\int_{t_0+\epsilon}^{t_1} n^2 \, dt$. Cauchy–Schwarz and $(\log \mathcal{H})^\cdot = 2n$ now preclude $\mathcal{H}(t) \rightarrow 0$ at t_0 , a contradiction. If ν vanished in an open ball but were nonzero elsewhere in the connected interior disk, choose a point of the zero set sufficiently close to its boundary. Its distance to $\text{spt } \nu$ is positive and smaller than the available interior radius, producing precisely the excluded s_0 .

We also need strong, rather than merely distributional, information about the first moment. Divide the first moment by a and write $U_\alpha = \pi_\#((z_\alpha/a)\mu)$. The horizontal height-multiplier test has, in the limiting flat coordinates, the distributional form

$$\partial_i U_\alpha = F_{i\alpha} + \partial_j E_{ij\alpha}, \quad \|F\| \leq C, \quad \|E\| = o(1). \quad (7.17)$$

In the flat case one may take $F_{i\alpha} = \pi_{\#}(P_{v_{\alpha}h_i}\mu/a)$ and $E_{ij\alpha} = \pi_{\#}((z_{\alpha}/a)B_{ij}\mu)$. Cauchy–Schwarz bounds the first mass by C and the second by $C \int e \, d\mu = O(a^2)$. In the actual coordinates, multiply the identity by the smooth inverse base metric and volume density. The additional terms have the same form: coefficients converge smoothly, flux masses remain bounded, and the terms containing height times tilt have mass tending to zero.

Take the divergence in (7.17) and solve on a smaller ball by a cutoff Newtonian potential. First derivatives of the Newtonian kernel convolved with bounded measures form a precompact family in local L^1 : their local kernels are integrable and have a uniform L^1 translation modulus. The second-derivative potential of E tends to zero in measure by the weak $(1, 1)$ estimate proved in Theorem 2.4. The remaining harmonic distributions are compact on interior balls by their bounded negative norms. Thus U is precompact locally in measure after replacing it by its projection-count density. This replacement is legitimate: the area formula and the Jacobian estimate give

$$\left\| U - \left(\sum_{X \in \pi^{-1}(y)} \theta(X) z(X)/a \right) d\text{vol}_S(y) \right\| = o(1).$$

One can make the potential argument for functions throughout by mollifying with a radius chosen separately for each member of the sequence so that the count density changes by $o(1)$ in L^1 . The above small total-variation difference remains $o(1)$ after mollification. The densities are uniformly integrable, since $|z|/a \leq C$ and their total counts differ from Q by $o(1)$ in L^1 . Finally (7.7) says that their distributional limit is zero. Precompactness in measure and uniform integrability therefore show that the normalized count first moment tends to zero in measure (indeed in local L^1).

On each fixed subball, the limiting second moment is positive. Fibers of total count different from Q have vanishing count mass by Theorem 2.5; points with tilt exceeding a number tending slowly to zero have vanishing mass by (7.6). Discard both. On a remaining fiber list its Q normalized heights with repetitions as $w_1, \dots, w_Q$, and set $\bar{w} = Q^{-1} \sum_i w_i$. The identity

$$\sum_{i=1}^Q |w_i|^2 = \sum_{i=1}^Q |w_i - \bar{w}|^2 + Q|\bar{w}|^2$$

and $|w_i| \leq C$ show that a positive amount of the second moment is variance. Hence a subset of fixed positive base volume has variance bounded below, and therefore has two heights separated by a fixed positive amount. Their distinct spatial entries each have multiplicity less than Q . Restoring lengths gives the asserted separation. For $Q = 1$ the variance is identically zero, so failure is impossible. If uniform constants for the prescribed comparable subballs failed, their centers and radii would have a convergent subsequence in the compact permitted range; the preceding argument on an interior subball of its limit gives the same contradiction. $\square$

7.2 The center and weighted estimates

Proposition 7.3 (The glued center and its influence region). *Fix the desired output order $J \geq 6$, apply Theorem 7.1 with derivative order $J + 1$, and then choose its corresponding k_0 and initial parameters. There is a C^J graph S on an interior neighborhood of 0, with C^J norm at most Cm_0 , through 0 and tangent to P_0 there, and a nonnegative Lipschitz function t on the fixed base, such that*

$$|H_S| + |\nabla H_S| \leq Cm_0^2 t^{2k_0-4}, \quad |z| \leq Cm_0 t^{k_0} \quad \text{on } \text{spt } \mu. \quad (7.18)$$

In the first inequality t is evaluated at the horizontal coordinate of the point of S ; in the second it may be evaluated either at $\pi_0 X$ or at $\pi_0 \pi_S X$, with a change of constant. Writing

$\mathcal{I} = \{t > 0\}$, one has

$$t(0) = 0, \quad \sup_{B_r} t = o(r), \quad |\mathcal{I} \cap B_r| = o(r^m). \quad (7.19)$$

In particular the support height is zero off the influence region. On S itself its normal-projection version has the same vanishing volume density. If ρ is distance to 0 on S , then $|z| \leq C\rho^2$ on the support in a sufficiently small tube.

Proof. Let ℓ_x be the first failed scale at $x \in B_5$, and put $\ell_x = 0$ if there is no failure. Define

$$t(y) = \sup_{x \in B_5} (\ell_x - |x - y|)_+, \quad y \in B_2. \quad (7.20)$$

Every function in this supremum is 1-Lipschitz, so t is 1-Lipschitz and $\mathcal{I}$ is the union of the failure balls.

We describe the gluing before proving the density properties. For $\epsilon > 0$ use $t_\epsilon = \max(t, \epsilon)$. Choose a covering by balls of radius $ct_\epsilon(y_i)$, with $c > 0$ a sufficiently small fixed constant, by first taking disjoint balls of a smaller such radius and then enlarging them. The Lipschitz bound makes radii of intersecting enlarged balls comparable; the covering has bounded overlap. Smooth bumps on these balls, divided by their sum, give a partition $\{\chi_i\}$ with $|D^j \chi_i| \leq C_j t_\epsilon(y_i)^{-j}$. At y_i choose a dyadic constructed scale L_i between $C_0 t_\epsilon(y_i)$ and $2C_0 t_\epsilon(y_i)$, capped at the initial scale 1. The constant C_0 is fixed and the covering radii can be reduced to leave all required overlap room. Since $\ell_{y_i} \leq t(y_i)$, these scales have been constructed. Where the cap is active, $L_i \asymp t_\epsilon(y_i)$ still holds, with a constant depending on C_0 . Let $f_\epsilon = \sum_i \chi_i f_{y_i, L_i}$.

On an overlapping cell of size L , fix one of its fits f_* and write $f_\epsilon - f_* = \sum_i \chi_i (f_i - f_*)$. By (7.8), every derivative of order $j \leq J+1$ in this identity is at most $Cm_0^2 L^{2k_0-1-j}$, by the choice of derivative order $J+1$ in Theorem 7.1. Each f_i is minimal. The mean-curvature operator and its first derivative involve at most three derivatives of the graph. Substitution therefore gives

$$|H_{\text{graph}} f_\epsilon| + |\nabla H_{\text{graph}} f_\epsilon| \leq Cm_0^2 t_\epsilon^{2k_0-4}.$$

The support height is at most $Cm_0 t_\epsilon^{k_0}$, by the height bounds for the fits. Uniform C^{J+1} bounds give, along a sequence $\epsilon \downarrow 0$, a local C^J limit f . Passing to the limit proves (7.18), including at $t = 0$. Uniform small slope supplies a normal tube on each smaller chart. The change in horizontal argument under nearest projection is at most $Cm_0 |z| \leq Cm_0^2 t_\epsilon^{k_0}$, which is a small fraction of t when $t > 0$. The Lipschitz bound on t therefore proves the assertion about the projected argument, including the zero set by a limit.

We next use multiplicity defects to control the tents. Choose the comparable subball in Theorem 7.2 strictly inside the original failure ball. Small slope and the much smaller height displacement then imply, in the fixed horizontal projection,

$$\lambda(B_\ell(x)) \geq c\ell^m \quad \text{for every failure } (x, \ell). \quad (7.21)$$

This charges inside the original ball, a fact needed in the disjoint covering below. The change from normal to horizontal projection has a uniformly bounded Jacobian and displaces the chosen smaller subball by much less than its interior margin.

Fix a large finite K , greater than every fixed enlargement factor that will occur in the following local arguments. At initialization choose $\eta < c/(K+1)^m$ in (7.4). If $|x| \leq K\ell$, then

$$c\ell^m \leq \lambda(B_\ell(x)) \leq \lambda(B_{|x|+\ell}(0)) \leq \eta(K+1)^m \ell^m,$$

a contradiction. The enclosing ball is inside the fixed chart; one uses its actual radius $|x| + \ell$, not an unnecessarily larger ball. Thus $|x| > K\ell$ for all relevant failures, and $t(0) = 0$. If $B_{A\ell}(x)$ meets $B_r(0)$, for a fixed $A < K$, then

$$\ell \leq \frac{r}{K-A}, \quad B_\ell(x) \subset B_{CAr}(0), \quad C_A = 1 + \frac{A+1}{K-A}.$$

Combining (7.21) with the little- o part of (7.4) now gives

$$\sup\{\ell : B_{A\ell}(x) \cap B_r(0) \neq \emptyset, (x, \ell) \text{ a failure}\} = o(r). \quad (7.22)$$

All limits here hold for the one fixed choice of the initial data. This first excludes large intrusions using a uniform bound and only then uses differentiation to obtain the little- o conclusion.

For small r , all original failure balls meeting B_r lie in B_{2r} by (7.22). Select disjoint original balls B_i whose fivefold enlargements cover their union. The charges in (7.21) are disjoint, so

$$|\mathcal{I} \cap B_r| \leq C \sum_i |B_i| \leq C \sum_i \lambda(B_i) \leq C \lambda(B_{2r}) = o(r^m).$$

The same small-failure estimate gives $\sup_{B_r} t = o(r)$. The support contains 0, so its zero height at $t(0) = 0$ shows that S passes through 0. The superlinear height bound and (7.1) force $T_0 S = P_0$. Smooth change of coordinates transfers the volume statement to S , and the projected height bound gives $|z| \leq C m_0 o(\rho)^{k_0} \leq C \rho^2$. $\square$

We state the weighted consequences with the physical units retained. Let $\pi = \pi_S$, $z = X - \pi(X)$, and let ρ be geodesic distance on S to 0 inside a normal coordinate ball. In these weighted estimates and the final height argument, B_r denotes the geodesic ball $\{y \in S : \rho(y) < r\}$. Local covering balls may be taken in fixed smooth charts; their comparison with metric balls changes only fixed constants. Choose a fixed integer $p \geq 4$, increasing it if needed in subsequent compact-kernel tests, and put

$$\phi_r = (1 - \rho^2/r^2)_+^p, \quad \psi_r = 2p(1 - \rho^2/r^2)_+^{p-1}, \quad \mathbf{u} = \nabla_S \rho, \quad K_z = (I - A_z)^{-1}. \quad (7.23)$$

Weighted integrals from this point onward include r^{-m} ; when there is any possibility of confusion we write this factor explicitly. The heights z are not divided by r . The symbols B, e, M, D, L, Λ now always refer to this center. Define

$$b = B[K_z \mathbf{u}, K_z \mathbf{u}], \quad h_r = 1 - \frac{b}{|K_z \mathbf{u}|^2}, \quad (7.24)$$

$$H(r) = r^{-m} \int \psi_r |z|^2 h_r \, d\mu, \quad \delta(r) = r^{-1} \sup\{|z| : X \in \text{spt } \mu, \rho(\pi X) < r\}.$$

In particular H already contains the radial projection factor. One has $0 \leq h_r \leq 1$, since $0 \leq B \leq I$. At a fixed support point this factor has no dependence on the radius r ; the subscript only distinguishes it from other height variables. On the central fiber it may be assigned any value, because $z = 0$ there.

Proposition 7.4 (Weights up to the boundary of the ball). *Either the support lies on S in a neighborhood of 0, or there exist $r_0 > 0$, $c, C > 0$, and a finite J_0 such that, for every $0 < r < r_0$,*

$$0 < \delta(r) \leq Cr, \quad c\delta(r)^{J_0} \leq \frac{H(r)}{r^2} \leq C\delta(r)^2. \quad (7.25)$$

In the latter alternative, throughout $\rho < r$ one also has

$$|H_S| + |\nabla H_S| \leq C\delta(r)r, \quad (7.26)$$

and

$$r^{-m} \int \phi_r (|H_S|^2 + |\nabla H_S|^2) \, d\mu \leq Cr^{-m} \int \phi_r |z|^2 \, d\mu. \quad (7.27)$$

The constants may depend on the fixed center construction. One may take any fixed $J_0 \geq 2 + (m + p - 1)/k_0$ after changing c .

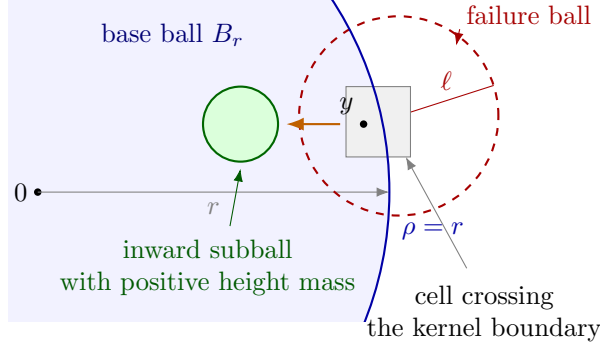


Figure 2: An inward subball controls the boundary weights on the reference center. A failure at scale ℓ supplies separated projection entries on every sufficiently small comparable subball in its comparison region. Moving such a subball inward by $O(\ell)$ gives positive height mass where the polynomial kernel has a controlled lower bound. This pays for a local cell even when that cell crosses $\rho = r$. The picture is schematic; no circular shape of a failure region is assumed.

Proof. We give the geometric estimate that pays for points close to the boundary $\rho = r$. If $t(y) > 0$, choose a failure (x, ℓ) for which

$$\ell - |x - y| > t(y)/2. \quad (7.28)$$

Then $\ell > t(y)/2$, $|x - y| < \ell$, and the Lipschitz bound gives $t(v) \leq (2 + A)\ell$ when $|v - y| \leq A\ell$. An upper comparison of ℓ with $t(y)$ is neither asserted nor needed. On this whole neighborhood the support has height at most $Cm_0\ell^{k_0}$ over S , and its mass over any ball of radius comparable to ℓ is at most $C\ell^m$ by the ambient upper mass bound.

In each fixed small comparable subball within the permitted distance from x , [Theorem 7.2](#) gives mass at least $c\ell^m$ with distance from S at least $cm_0\ell^{k_0}$ and with tilt relative to S less than, say, $1/4$. To verify this transfer, the normal fiber of the failed fit is transverse to the small-slope graph S . It has only one intersection with that graph in the working tube, and distance to S is comparable to distance along the fiber to that intersection. Of two separated entries at least one is a fixed fraction of their separation from S . Their tilt relative to S is still small, since both reference graphs have small slope. Restrict the subball slightly first: all changes of projection argument are $o(\ell)$ and so preserve its interior margin. On this mass $h_r \geq 3/4$ after reducing the tilt threshold in [Theorem 7.2](#) once.

For y with $\rho(y) < r$, the witnessing scale satisfies $\ell = o(r)$ by (7.22). There is a ball of radius $c_1\ell$ at distance $O(\ell)$ from y , lying inside $\rho < r$, on which

$$\inf_C \phi_r \geq c \sup_C \phi_r, \quad \inf_C \psi_r \geq c \sup_C \psi_r, \quad \inf \psi_r \geq c(\ell/r)^{p-1}. \quad (7.29)$$

Here C is a cell of side $c_2\ell$ containing y , and the suprema concern its intersection with $\rho < r$. If y is already a fixed multiple of ℓ inside the boundary, take a nearby subball there. Otherwise move its center inward along the radial geodesic by $C_1\ell$ and take c_1 much smaller than C_1 . The boundary distance on the new ball is at least a fixed multiple of the largest boundary distance on the cell and at least $c\ell$. Since $1 - \rho^2/r^2$ is comparable to $(r - \rho)/r$ near the boundary, this gives (7.29); in the central half ball all weights are bounded below. The displacement is bounded by a fixed small multiple of ℓ , so the new ball is among those allowed in [Theorem 7.2](#). These assertions are unchanged by the smooth base-coordinate conversions. Thus they also apply to cells straddling $\rho = r$.

The separated mass on this inward ball gives

$$\delta(r)r \geq cm_0\ell^{k_0}. \quad (7.30)$$

If the support fails to lie on S in every neighborhood, then $\delta(r) > 0$ for every small r . Choose a point with height at least half the defining supremum. Its height is nonzero, so it has a witness as in (7.28); (7.18) and (7.30) give

$$cm_0\ell^{k_0} \leq \delta(r)r \leq Cm_0\ell^{k_0}.$$

The inward ball has mass at least $c\ell^m$, height at least $cm_0\ell^{k_0}$, and $h_r \geq 3/4$. Consequently

$$\frac{H(r)}{r^2} \geq c\delta(r)^2(\ell/r)^{m+p-1} \geq c(m_0)\delta(r)^{2+(m+p-1)/k_0}.$$

In the last step use $\ell/r \geq c(m_0)\delta(r)^{1/k_0}r^{1/k_0-1}$ and $r \leq 1$. The upper bound in (7.25) follows from bounded normalized mass and $h_r \leq 1$. The bound $\delta(r) \leq Cr$ follows from Theorem 7.3. This proves (7.25).

For any point with $t > 0$, its witness and (7.30) give

$$|H_S| + |\nabla H_S| \leq Cm_0^2\ell^{2k_0-4} \leq Cm_0\ell^{k_0-4}\delta(r)r \leq C\delta(r)r.$$

At $t = 0$ the left side is zero. This proves (7.26).

Finally group points with $t > 0$ according to the dyadic scale of a witnessing failure. At scale $\ell = 2^{-j}$ cover the group by grid cells of side $c_2\ell$, taking each grid cell only once. For each occupied cell choose a witness from that group and an inward ball as above. On the cell the curvature-square integrand is at most $Cm_0^4\ell^{4k_0-8}$. Its mass is at most $C\ell^m$. On the corresponding inward ball the integral of $\phi_r|z|^2$ is at least $cm_0^2\ell^{2k_0+m}\sup_C \phi_r$ by (7.29). Hence

$$\int_C \phi_r(|H_S|^2 + |\nabla H_S|^2) d\mu \leq Cm_0^2\ell^{2k_0-8} \int_{B_C} \phi_r|z|^2 d\mu.$$

The displacement from a cell to its inward ball is $O(\ell)$, so these balls have bounded overlap at each fixed dyadic scale. Sum first over the cells and then over j . The geometric series $\sum_j 2^{-j(2k_0-8)}$ converges because $k_0 > 4$. Points with $t = 0$ contribute nothing, and division by r^m gives (7.27). $\square$

Corollary 7.5 (Regularity when height vanishes). *If the support lies on the center S in a neighborhood of the origin, then the origin is regular.*

Proof. Write the restricted varifold as $\vartheta \mathcal{H}^m \llcorner S$, allowing the integer density ϑ to be zero off its original support. This is the standard constancy argument; compare [19, Chapter 8, Theorem 4.1]. Tangential first variation gives $\int_S \vartheta \operatorname{div}_S X = 0$ for all compactly supported tangent fields. Thus ϑ is constant almost everywhere on a connected smaller disk. The constant is Q , by (7.1). Normal first variation gives $H_S = 0$, and the small-gradient minimal graph equation bootstraps S to a smooth graph. The origin is therefore regular. $\square$

It remains to exclude the positive-height alternative, which is the purpose of the inward comparison.

8 Inward comparison and the conclusion

We use the center supplied by Theorems 7.3 and 7.4 at a point with the density and tangent properties specified there. If the varifold lies on the center in a neighborhood of the point, Theorem 7.5 already gives regularity. We therefore assume the positive-height alternative of Theorem 7.4 and derive a contradiction. In particular, none of the arguments below assumes height doubling at the outset. The inward subballs in Theorem 7.4 controlled the polynomial weights at each fixed radius. We now use those bounds in a separate signed-mass comparison near the same boundary: a radial test handles a thin outer collar, and the signed-excess theorem applies on interior balls.

8.1 Signed comparison and frequency identities

The comparison of height with energy follows Almgren's frequency method [11, Section 3.4]; cutoff frequency on a center manifold is developed in [10, Section 3]. We derive the identities needed here from stationarity. The signed-excess estimate supplies the additional inequality that will yield the inward height comparison.

Keep the physical normal displacement z and the radial functions

$$\phi_r = (1 - \rho^2/r^2)_+^p, \quad \psi_r = 2p(1 - \rho^2/r^2)_+^{p-1},$$

where $p \geq 4$ is the fixed integer chosen in Theorem 7.4. For clarity, write

$$\int^{(r)} f \, d\nu = r^{-m} \int f \, d\nu.$$

This convention applies both to measures on S and to measures on the varifold; base functions in a varifold integral are pulled back by nearest projection. All integrals below have this normalization unless explicitly written otherwise. Abbreviate ϕ_r, ψ_r to ϕ, ψ at the radius under consideration. With $\mathbf{u} = \nabla_S \rho$, set

$$J_z = I - A_z, \quad K_z = J_z^{-1}, \quad v = K_z \mathbf{u}, \quad b = B[v, v], \quad h_r = 1 - \frac{b}{|v|^2}.$$

Here A_z is the shape operator in the normal direction z ; it is distinct from the scalar A defined next. Recall that $0 \leq h_r \leq 1$ and that h_r has no dependence on the radius at a fixed point and tangent plane. On the central fiber its value is immaterial, since $z = 0$ there. Define

$$\begin{aligned} H &= \int^{(r)} \psi |z|^2 h_r \, d\mu, & A &= r^2 \int^{(r)} \phi \, dM, \\ E &= r^2 \int^{(r)} \phi \, dD, & R &= \int^{(r)} \rho^2 \psi b \, d\mu, \\ P &= A/H, & n_* &= E/H, & q_* &= R/H. \end{aligned} \tag{8.1}$$

The scalar ratios n_*, q_* are distinct from the ambient dimension parameters. The comparison we need is $2A \leq E$ up to small errors: after division by H , its coefficient one gives $n_* \geq 2P$ up to those errors. To see its role, omit the errors from the frequency inequalities proved below. With a dot denoting differentiation in $\log r$, their leading terms give, for $P \geq 0$,

$$q_* \geq n_*^2, \quad \dot{P} \geq 2P - n_* + q_* - 2Pn_* \geq (n_* - 1)(n_* - 2P).$$

Thus the signed comparison makes this leading derivative nonnegative when P is large. We first preserve its coefficient through the boundary of the ball; the bootstrap will then estimate the actual errors in this calculation.

Put

$$\delta = r^{-1} \sup_{\rho < r} |z|, \quad T = \log(r^2/H), \quad g = T^{-20}.$$

By Theorem 7.4, after reducing the radius,

$$0 < \delta \leq Cr, \quad c\delta^{J_0} \leq H/r^2 \leq C\delta^2, \quad c|\log \delta| \leq T \leq C|\log \delta|. \tag{8.2}$$

The finite exponent J_0 and all constants may depend on the fixed center. We distinguish the unit error $\eta_1(r) = C_1 \delta(r)^\alpha$ in the collar estimate from the common error $\eta(r) = C_\eta \delta(r)^\alpha$ in the frequency argument. Here $\alpha > 0$ is fixed independently of the multiplier and chosen sufficiently small for the finitely many power errors below, and C_1 depends only on the fixed center data.

We choose C_η to dominate the remaining error coefficients and the fixed multiples of η_1 arising from the actual collar applications. Their multiplier norms are verified below to have one bound independent of r ; no factor unbounded as $r \downarrow 0$ is absorbed into C_η . Decreasing α and increasing the constants a finite number of times gives common bounds for all the inequalities below. In particular, with $s = -\log r$,

$$T \geq 2s - C, \quad \eta \leq C \exp(-cT). \quad (8.3)$$

The weights first give the following comparisons:

$$\begin{aligned} \int^{(r)} \phi |z|^2 d\mu &\leq C(H + \delta^2 E), \\ \int^{(r)} \psi |z|^2 d\mu &\leq C\{H + \delta^2(E + R)\}, \\ r^2 \int^{(r)} \phi d\Lambda &\leq Cr^2(H + \delta^2 E). \end{aligned} \quad (8.4)$$

Indeed, $\phi \leq C\psi$, $1 - h_r \leq Ce$, and $|z| \leq \delta r$ prove the first inequality. On $\rho < r/2$ the two weights are comparable, so the same argument applies to the second. On $r/2 \leq \rho < r$ use $1 - h_r \leq Cb$ and $r^2\psi b \leq C\rho^2\psi b$ instead. The last inequality follows from the weighted curvature estimate in [Theorem 7.4](#) and the definition of Λ .

Lemma 8.1 (The signed estimate up to the outer collar). *For $a \in C^1(B_r)$ put*

$$\|a\|_{1,r} = \sup_{B_r} (|a| + r|\nabla_S a|).$$

There is a radius $r_c > 0$, depending only on the fixed center data, such that for every $0 < r < r_c$ the first inequality below holds for every $a \in C^1(B_r)$ with finite norm, and the second holds independently of a :

$$\begin{aligned} r^2 \left| \int^{(r)} a \phi dM \right| &\leq \|a\|_{1,r} [C(E + r^2 H) + \eta_1 R], \\ 2A &\leq E + Cg(E + H) + Cr^2 H + \eta_1 R. \end{aligned} \quad (8.5)$$

The constants C, C_1 and the radius r_c are independent of the multiplier. For a family a_r , the coefficient of R in the first line is $\|a_r\|_{1,r} \eta_1(r)$, so each use of its smallness must control this product. The second line uses $a = 1$ only in its thin-part proof. The coefficient of E in the second line is exactly one before the displayed small errors are added.

Proof. For the absolute estimate it suffices to treat $\|a\|_{1,r} \leq 1$ uniformly: apply that case to $a/\|a\|_{1,r}$ when the norm is positive, and the zero case is immediate. The signed estimate will use the member $a = 1$ of this unit class. Choose a small fixed $\beta > 0$, with $\beta < 1/4$, and put $\sigma = \delta^\beta$; we may take $0 < \alpha \leq 2\beta$ in η_1 . Choose a radial cutoff χ equal to one when $0 \leq 1 - \rho/r \leq \sigma^2$, equal to zero when $1 - \rho/r \geq 2\sigma^2$, and with derivative bounded by $C/(\sigma^2 r)$. Write $\phi_{\text{thin}} = \chi\phi$ and $\phi_{\text{bulk}} = (1 - \chi)\phi$.

We first treat the bulk, including the overlap with the thin part. Fix first an enlargement factor $c_{\text{out}} > 8$ sufficient for [Theorems 2.5](#) and [3.1](#) and for a Caccioppoli cutoff equal to one on $B_{8\ell}$ and supported in $B_{c_{\text{out}}\ell}$. Then choose the fixed c_0 small and put $\ell = c_0\sigma^2 r$. Cover the bulk by balls of radius ℓ with centers x_i in its support. These centers have boundary depth at least $\sigma^2 r$, so c_0 can be chosen so that all $B_{c_{\text{out}}\ell}(x_i)$ stay inside B_r . Choose the cover with bounded overlap of these enlargements and so that the intermediate balls on which [Theorem 2.5](#) selects

integers have overlaps of fixed positive relative volume along the connecting chains. Choose a smooth nonnegative partition of unity $\{\vartheta_i\}$ subordinate to the original balls, with

$$|D^j \vartheta_i| \leq C_j \ell^{-j} \quad (0 \leq j \leq k),$$

where D denotes base-coordinate derivatives and $k = k(m)$ is the finite test order in [Theorem 2.4](#). Put $\phi_i = \phi(x_i)$. On each enlargement ϕ is comparable to ϕ_i . The partitioned weights, divided by ϕ_i and by a fixed constant, have Lipschitz norm at most one in units of ℓ .

The mass on each enlarged ball is at most $C\ell^m$. To see this without assuming a local graphical representation, observe that $\delta r = o(\ell)$; thus the part of the tube over the ball lies in an ambient ball of radius $C\ell$ centered at a support point. The upper bound follows by monotonicity, comparing with an ambient ball of radius comparable to r and using the fixed normalized mass bound at the origin. Empty balls need no estimate.

For the local tilt estimate, use ordinary, unnormalized physical integrals. Let ξ be a base cutoff equal to one on $B_{4\ell}$, supported in $B_{\text{coul}\ell}$, and satisfying $|\nabla_S \xi| \leq C/\ell$. The analogous cutoff equal to one on $B_{8\ell}$ has the same derivative bound and outer support. Testing stationarity with $\xi^2 z$ and using the normal-displacement trace (7.9) gives

$$\int \xi^2 e \, d\mu \leq C \int [(|\nabla_S \xi|^2 + |\text{II}|^2 \xi^2) |z|^2 + \xi^2 |H_S| |z|] \, d\mu.$$

Indeed, the derivative of the cutoff contributes at most $C\xi|\nabla_S \xi||z|\sqrt{e}$; Young's inequality absorbs this term and the small $|\text{II}||z|e$ term from the trace. The displayed $|H_S||z|$ term is retained because the center need not be minimal. The bounds $|z| \leq \delta r$ and $|H_S| + |\nabla H_S| \leq C\delta r$, together with the local mass bound, give $\ell^{-m} D(B_{4\ell}) \leq C[(\delta r/\ell)^2 + (\delta r)^2]$. The definition of Λ gives $\ell^{-m} \Lambda(B_{4\ell}) \leq C(\delta r)^2$ separately. Hence

$$\ell^{-m} (D + \Lambda)(B_{4\ell}) \leq C(\delta/\sigma^2)^2 \leq C\delta^{2-4\beta}. \quad (8.6)$$

The same estimate holds on $B_{8\ell}$, using its cutoff in $B_{\text{coul}\ell}$. These enlargement factors were fixed before c_0 . The coordinates have the required fixed geometry bounds, and the terms in Λ still use physical units.

Apply [Theorem 2.5](#) first to select the local integers. On the intermediate balls just chosen, the scalar exceptional sets have relative volume $O(\delta^{2-4\beta})$. For small r they cannot fill the chosen overlaps of fixed positive relative volume, so their integers agree. The bulk is connected; call the common integer q_r . To identify it, rescale one fixed smooth nonnegative test supported in $B_{1/2}$ to $B_{r/2}$ and normalize its reference-volume integral to one. Call it φ_r ; then $|D^j \varphi_r| \leq Cr^{-m-j}$ for $0 \leq j \leq k$. Partition this test over the cover. After pullback to a unit ℓ -cell, each product $\vartheta_i \varphi_r$ has C^k norm at most Cr^{-m} , since $\ell \leq r$. The coordinate-volume factor in the scalar proof has fixed scaled derivative bounds through order k , so multiplying by it preserves this estimate. The distributional estimate (2.14) used in the proof of [Theorem 2.5](#), with (8.6), bounds each cell's error relative to q_r by $C\delta^{2-4\beta} \ell^m r^{-m}$. Bounded overlap gives $\sum_i \ell^m \leq Cr^m$, so its projected-mass average differs from q_r by at most $C\delta^{2-4\beta}$. Tangent convergence and the smooth rescaling of the fixed center make that same average $Q + o(1)$. Integrality therefore gives $q_r = Q$ for all sufficiently small r . Exact agreement on overlaps involves no error accumulated along the chain of balls. This also identifies the integer on the balls used in the overlap and in the thicker error collar below. Consequently [Theorem 2.5](#) bounds $|M|$ on each smaller ball by the enlarged $D + \Lambda$ mass. In particular, applying it with fixed interior room and using (8.6) also on the larger ball gives

$$\begin{aligned} |M|(B_{4\ell}(x_i)) &\leq C(D + \Lambda)(B_{8\ell}(x_i)), \\ \ell^{-m} \tau(B_{4\ell}(x_i)) &\leq C\delta^{2-4\beta}. \end{aligned} \quad (8.7)$$

All these balls lie inside B_r by the choice of c_0 . Summation with comparable weights gives

$$r^2 \int^{(r)} \phi_{\text{bulk}} d|M| \leq Cr^2 \int^{(r)} \phi d(D + \Lambda) \leq C(E + r^2 H), \quad (8.8)$$

using (8.4) to absorb a small multiple of E . For every member of the unit multiplier class this also gives $r^2 |\int^{(r)} a \phi_{\text{bulk}} dM| \leq C(E + r^2 H)$.

For the signed estimate, invoke [Theorem 3.1](#) afresh on each normal chart at physical radius ℓ . The measures M, D, Λ and the base tests are geometric, so they give the required data in the unit coordinates, with the measures divided by ℓ^m . The physical condition below belongs to this new theorem instance; it does not change the inherited geometry parameter in an off-center rescaling of a prior instance. On a cell use

$$d_i = \max\{C\ell^{-m}\tau(B_{4\ell}(x_i)), \delta^{2J_0+4}\}, \quad T_i = -\log d_i,$$

with the fixed enlargement and constant sufficient for [Theorem 3.1](#). Equation (8.7) shows that $c|\log \delta| \leq T_i \leq (2J_0 + 4)|\log \delta|$, where the upper bound follows from the floor in d_i . Moreover

$$\ell + \sup |z| \leq C\delta^{2\beta} \leq \exp(-\kappa T_i)$$

for some fixed $\kappa > 0$ and all small r ; choose, for example, $\kappa(2J_0 + 4) < 2\beta$. Thus both smallness hypotheses of [Theorem 3.1](#) hold even if δ is smaller than every fixed power of r .

Apply that result to $w_i = \vartheta_i \phi_{\text{bulk}} / (C\phi_i)$ in the coordinates scaled by ℓ . They are supported in B_1 in those coordinates and satisfy $0 \leq w_i \leq 1$ and $\text{Lip } w_i \leq 1$. Restoring the physical measure factor ℓ^m and then the normalization r^{2-m} gives

$$\begin{aligned} 2r^2 \int^{(r)} \phi_{\text{bulk}} dM &\leq r^2 \int^{(r)} \phi_{\text{bulk}} dD + C_Q r^2 \int^{(r)} \phi_{\text{bulk}} d\Lambda \\ &\quad + Cr^{2-m} \sum_i \phi_i \ell^m d_i e^{-F_Q(T_i)}. \end{aligned} \quad (8.9)$$

The last sum includes the enlarged cells, even where ϕ_{bulk} vanishes in the overlap with the thin part. On these cells use the local estimate (8.7): ϕ is comparable to ϕ_i throughout $B_{8\ell}(x_i)$, and these balls have bounded overlap inside B_r . Therefore

$$\begin{aligned} \sum_i \phi_i \ell^m d_i &\leq C \sum_i \phi_i \tau(B_{4\ell}(x_i)) + C\delta^{2J_0+4} \sum_i \phi_i \ell^m \\ &\leq C \sum_i \phi_i (D + \Lambda)(B_{8\ell}(x_i)) + C\delta^{2J_0+4} \sum_i \phi_i \ell^m \\ &\leq Cr^m \int^{(r)} \phi d(D + \Lambda) + Cr^m \delta^{2J_0+4}. \end{aligned}$$

Here the last floor sum is bounded by the ordinary volume of the bounded-overlap cover. Since $T_i \geq c|\log \delta|$ and $\exp[-F_Q(c|\log \delta|)]$ is smaller than any fixed negative power of T , the remainder in (8.9) is at most

$$Cg \left(r^2 \int^{(r)} \phi d(D + \Lambda) + r^2 \delta^{2J_0+4} \right).$$

The floor term $r^2 \delta^{2J_0+4}$ is at most $C\delta^{J_0+4}H$ by (8.2). The weighted comparison (8.4) controls the remaining integral by $C(E + r^2 H)$ and bounds the $C_Q \Lambda$ term in (8.9) by $Cr^2 H + o(g)E$, since $r^2 \delta^2 = o(g)$. This proves the second line of (8.5) for the bulk, with its leading tilt term unchanged.

It remains to estimate the signed mass on the thin part. A direct total-variation estimate here would unnecessarily add a full-tilt term. Instead solve a radial divergence equation. In geodesic polar coordinates write the volume element as $j(\rho, \omega) d\rho d\omega$. Throughout this thin calculation $\|a\|_{1,r} \leq 1$, and all constants are uniform over that class. Initially put

$$F(\rho, \omega) = -j(\rho, \omega)^{-1} \int_{\rho}^r a(t, \omega) \phi_{\text{thin}}(t) j(t, \omega) dt, \quad X = F\mathbf{u}.$$

Then $\text{div}_S X = a\phi_{\text{thin}}$. Multiply X by a radial cutoff equal to one at depths $r - \rho \leq \sigma r$ and zero at depths $r - \rho \geq 2\sigma r$. On this latter collar the integrated numerator has size $O(r\sigma^{2p+2})$ relative to j . The cutoff divergence error q_c consequently satisfies

$$\text{div}_S X = a\phi_{\text{thin}} + q_c, \quad |q_c| \leq C\sigma^{p+1}\phi,$$

and is supported where $\sigma r \leq r - \rho \leq 2\sigma r$. The same integration, also differentiated in angular directions, gives

$$|X|/r + |\nabla_S X - a\phi_{\text{thin}}\mathbf{u} \otimes \mathbf{u}| \leq C\sigma^2\phi. \quad (8.10)$$

For example, at depth $dr \leq 2\sigma^2 r$ the primitive divided by r is $O(d^{p+1})$, whereas $\phi \asymp d^p$; beyond that depth its size is $O(\sigma^{2p+2})$. On the cutoff collar the derivative loss σ^{-1} still leaves $\sigma^{p+1}\phi$. This proves (8.10); bounded derivatives of the polar volume factor only change its constants. A radial spatial derivative of the primitive uses the value of a at its lower endpoint; an angular derivative uses only its first spatial derivative, bounded by $1/r$. The scale parameter r is held fixed in this calculation. Near $\rho = r$, the preceding estimates give $|F| \leq Crd^{p+1}$ and $|\nabla_S(F\mathbf{u})| \leq Cd^p$. Thus X and its first derivative vanish at the boundary, so extending X by zero gives a C^1 compactly supported base field using only values of a in B_r . The inner cutoff removes the polar origin. When $m = 1$ the construction is made separately on the two radial intervals and has no angular term.

Use the lifted field of Theorem 2.2. The $q_c M$ term is estimated by the already proved bulk total-variation bound: for small σ , its support has depth between σr and $2\sigma r$, where $\chi = 0$ and $\phi_{\text{bulk}} = \phi$. Its contribution is at most $C\sigma^{p+1}(E + r^2 H)$. Put $W = \int^{(r)} \phi |z|^2 d\mu \leq C(H + \delta^2 E)$. Equation (8.10) gives $|X| \leq C\sigma^2 r \phi$ and $|\nabla_S X| \leq C\phi$. After multiplication by r^2 , the three error types in Theorem 2.2 are therefore bounded by

$$C\delta r E, \quad C\sigma^2 r^3 W, \quad C\sigma^2 r^2 \sqrt{EW},$$

using the weighted curvature-square bound for the term containing $\nabla^\perp H_S$. Young's inequality puts these terms into $Cr^2 H + C(\delta r + \sigma^2)E$. The stress left after its radial part costs $C\sigma^2 E$. Since $p \geq 4$, $r \leq 1$, and $0 < \alpha \leq 2\beta$, all these small coefficients of E are at most η_1 after fixing C_1 . The main stress term gives

$$r^2 \left| \int^{(r)} a\phi_{\text{thin}} dM \right| \leq Cr^2 \int^{(r)} \phi_{\text{thin}} B[\mathbf{u}, \mathbf{u}] d\mu + \eta_1 E + Cr^2 H. \quad (8.11)$$

Since $\rho \geq r/2$ on the thin part,

$$\phi_{\text{thin}} \leq C\sigma^2(\rho/r)^2\psi, \quad B[\mathbf{u}, \mathbf{u}] \leq 2b + C|z|^2 e.$$

Thus the first term of (8.11) is at most $C\sigma^2 R + C\delta^2 r^2 E \leq \eta_1(R + E)$, after increasing C_1 . Adding the unit bulk bound gives

$$r^2 \left| \int^{(r)} a\phi dM \right| \leq C(E + r^2 H) + \eta_1 R \quad (\|a\|_{1,r} \leq 1).$$

Homogeneity now gives the first line of (8.5) on the same radius interval. For the signed estimate use $a = 1$, add twice its thin-part bound to the bulk signed estimate, and use $\int \phi_{\text{bulk}} dD \leq \int \phi dD$. Because $\eta_1 = o(g)$ by (8.2), its E term is absorbed into CgE after reducing the fixed radius. Enlarge C_1 once to cover the fixed factors in both conclusions. The coefficient one has been preserved as claimed. $\square$

Lemma 8.2 (Frequency inequalities). *With a dot denoting differentiation with respect to $\log r$, the quantities in (8.1) satisfy*

$$n_* \leq \sqrt{q_*} + Cr^2 + \eta n_*, \quad (8.12)$$

$$\left| \frac{\dot{A}}{H} - (2P - n_* + q_*) \right| \leq C\{r\sqrt{q_*} + r^2(1 + n_*)\} + \eta(n_* + q_*), \quad (8.13)$$

$$-m \leq \frac{\dot{H}}{H} \leq 2n_* + Cr + \eta(n_* + q_*). \quad (8.14)$$

Proof. Test normal stationarity with $z\phi$. The exact normal-displacement trace formula (7.9) and (8.4) give

$$\left| E - \int^{(r)} \rho \psi z \cdot P_{vh} v \, d\mu \right| \leq Cr^2 H + \eta E. \quad (8.15)$$

In particular the $H_S \cdot z$ term is bounded using the weighted curvature-square estimate, rather than a height-independent error. For the tangent projection P_T , the algebra is exact: $P_T^2 = P_T$ and $(P_T)_{hh} = I - B$ imply

$$|P_{vh} v|^2 = v \cdot (B - B^2)v = b - |Bv|^2 \leq b - \frac{b^2}{|v|^2} = bh_r.$$

Cauchy–Schwarz therefore bounds the absolute value of the flux in (8.15) by $\sqrt{RH}$. Division by H proves (8.12).

For the horizontal identity put $Z_0 = \rho \mathbf{u}$ on S and let $\bar{Z}_0(q) = J_z Z_0(\pi q)$ be its unweighted lift. We first compute the full derivative of the kernel in the test $(\phi \circ \pi) \bar{Z}_0$. Since $D\pi = K_z P_h$ and $\nabla_S \phi = -(\rho/r^2)\psi \mathbf{u}$,

$$\nabla(\phi \circ \pi) = -\frac{\rho}{r^2} \psi K_z \mathbf{u}.$$

Both this vector and $\bar{Z}_0 = \rho J_z \mathbf{u}$ are horizontal. The product rule and $(P_T)_{hh} = I - B$ therefore give exactly

$$\begin{aligned} P_T : D((\phi \circ \pi) \bar{Z}_0) &= \phi P_T : D\bar{Z}_0 - \frac{\rho^2}{r^2} \psi (K_z \mathbf{u}) \cdot (I - B)(J_z \mathbf{u}) \\ &= \phi P_T : D\bar{Z}_0 - \frac{\rho^2}{r^2} \psi + \frac{\rho^2}{r^2} \psi B[K_z \mathbf{u}, J_z \mathbf{u}]. \end{aligned}$$

The identity coefficient is one because J_z, K_z are self-adjoint inverses and $(K_z \mathbf{u}) \cdot (J_z \mathbf{u}) = |\mathbf{u}|^2 = 1$. The radial terms are understood to be zero on the central fiber, as their factors of ρ^2 require.

For the remaining derivative, the pointwise block calculation in the proof of Theorem 2.2 gives

$$\begin{aligned} P_T : D\bar{Z}_0 &= \text{div}_S Z_0 - B : \nabla_S Z_0 + \varepsilon_Z, \\ |\varepsilon_Z| &\leq C \left\{ |z|e|\nabla_S Z_0| + \left[(|\nabla^\perp H_S| + |z|)|z| + \sqrt{e}|z| \right] |Z_0| \right\}. \end{aligned}$$

This is the estimate for the unweighted lift, so the product formula multiplies its entire remainder by ϕ . Subtract the reference identity $Q \int_S \operatorname{div}_S(\phi Z_0) d\operatorname{vol}_S = 0$, in which $\operatorname{div}_S(\phi Z_0) = \phi \operatorname{div}_S Z_0 - \rho^2 \psi / r^2$. Since $\dot{A} = (2 - m)A + \int^{(r)} \rho^2 \psi dM$, we obtain, with

$$\tilde{R} = \int^{(r)} \rho^2 \psi B[K_z \mathbf{u}, J_z \mathbf{u}] d\mu,$$

the exact identity

$$\begin{aligned} \dot{A} &= 2A - E + \tilde{R} + r^2 \int^{(r)} \phi(\operatorname{div}_S Z_0 - m) dM \\ &\quad - r^2 \int^{(r)} \phi B : (\nabla_S Z_0 - I) d\mu + r^2 \int^{(r)} \phi \varepsilon_Z d\mu. \end{aligned} \tag{8.16}$$

In geodesic normal coordinates y , Gauss' lemma gives $Z_0 = y^i \partial_i$. If g_{ij}^S are the metric components, then

$$f := \operatorname{div}_S Z_0 - m = y \cdot D \log \sqrt{\det(g_{ij}^S)}.$$

The logarithmic volume factor has vanishing first derivative at 0 and bounded second derivative in the fixed chart. Consequently

$$|f| \leq C\rho^2, \quad |\nabla_S f| \leq C\rho, \quad \|a_r\|_{1,r} \leq K_{\text{geo}}, \quad a_r = f/r^2,$$

for one fixed K_{geo} . Also $\nabla_S Z_0 = I + O(\rho^2)$. The number $K_{\text{app}} = \max\{1, K_{\text{geo}}\}$ bounds both actual collar multipliers: a_r here and $a = 1$ in the bootstrap. Choose C_η in the convention above to dominate $K_{\text{app}} C_1$ and the finitely many other error prefactors in the frequency estimates. Then $\eta \geq K_{\text{app}} \eta_1$ has a fixed prefactor, so (8.3) applies. The first line of (8.5) and positivity of B now give

$$\begin{aligned} r^2 \left| \int^{(r)} \phi(\operatorname{div}_S Z_0 - m) dM \right| &\leq Cr^2(E + r^2 H) + r^2 \eta R, \\ r^2 \left| \int^{(r)} \phi B : (\nabla_S Z_0 - I) d\mu \right| &\leq Cr^2 E. \end{aligned}$$

For the pointwise remainder use $|Z_0| \leq r$, $|\nabla_S Z_0| \leq C$, and $\sup |z| \leq \delta r$. The first and last lines of (8.4), followed by Cauchy–Schwarz, bound its three terms respectively by

$$\begin{aligned} r^2 \int^{(r)} \phi |\varepsilon_Z| d\mu &\leq C\delta r E + Cr^3(H + \delta^2 E) + Cr^2 \sqrt{E(H + \delta^2 E)} \\ &\leq Cr^2(E + H) + \eta E. \end{aligned}$$

Here the term involving $\nabla^\perp H_S$ uses the weighted curvature-square bound inside Λ ; the last term comes from $\sqrt{e}|z||Z_0|$.

It remains to compare the radial stress $\tilde{R}$ with R . Positivity of B , its trace e , and $|J_z - K_z| \leq C|z|$ imply $|B[v, (J_z - K_z)\mathbf{u}]| \leq C|z|\sqrt{be}$. Thus $|\tilde{R} - R|$ is at most

$$Cr\sqrt{R} \left(\int^{(r)} \psi |z|^2 e d\mu \right)^{1/2}, \tag{8.17}$$

by Cauchy–Schwarz. Since $e \leq m$, the second line of (8.4) gives

$$|\tilde{R} - R| \leq Cr\sqrt{RH} + C\delta r\sqrt{R(E + R)} \leq Cr\sqrt{RH} + \eta(E + R).$$

Combining these estimates in (8.16) and dividing by H proves (8.13). The differentiated kernel has been exhausted by $\tilde{R}$; every other remainder in that identity carries the weight ϕ .

Finally, set

$$Y = \frac{\rho K_z \mathbf{u}}{|K_z \mathbf{u}|^2}.$$

This field is defined independently of the varifold tangent plane. Since $\nabla \rho = K_z \mathbf{u}$, it satisfies $Y \cdot \nabla_T \rho = \rho h_r$. Testing stationarity with $\psi |z|^2 Y$ consequently cancels the differentiated kernel exactly and gives

$$\dot{H} = \int^{(r)} \psi \left\{ |z|^2 (\text{tr}_T DY - m h_r) + \frac{2\rho z \cdot P_{vh} K_z \mathbf{u}}{|K_z \mathbf{u}|^2} \right\} d\mu. \quad (8.18)$$

There is no derivative of the measurable tangent projection in this calculation. Also $\nabla |z|^2 = 2z$ in the ambient Euclidean space, so normal-frame connection terms do not arise from this factor.

The center estimates give $t(y) = o(\rho(y))$ and hence $|z| \leq C\rho^2$ on the support near the origin. Although derivatives of $\mathbf{u}$ have size $O(\rho^{-1})$, differentiation of Y thus gives, on the support,

$$DY = P_h + O(\rho + |z|/\rho) = P_h + O(r).$$

It follows that $\text{tr}_T DY - m h_r = m(1 - h_r) - e + O(r)$. The positive $m(1 - h_r)$ contribution is bounded by $C\delta^2(E+R)$, by the proof of (8.4). The $O(r)$ contribution is at most $CrH + \eta(E+R)$, and the term $-e|z|^2$ can be discarded for an upper bound. Replacing the denominator in the second term of (8.18) by one costs at most $C\delta r\sqrt{RH} \leq CrH + \eta R$. Equation (8.15) now proves the upper bound in (8.14).

To justify the test at the central fiber, first remove $\rho < \epsilon$ by a smooth radial cutoff χ_ϵ , equal to zero for $\rho \leq \epsilon$, equal to one for $\rho \geq 2\epsilon$, and satisfying $|\nabla \chi_\epsilon| \leq C/\epsilon$. Since $|Y| \leq C\rho$, $|Y||\nabla \chi_\epsilon| \leq C$ on the transition annulus. Its extra term is therefore bounded, at fixed r , by a constant times $r^{-m} \int_{\rho < 2\epsilon} |z|^2 d\mu$, which tends to zero. The preceding support bound and local mass bounds also justify all other limiting terms. The lower bound in (8.14) follows directly from differentiating $r^{-m}\psi_r$: its derivative in $\log r$ is at least $-mr^{-m}\psi_r$, and the remaining factor $|z|^2 h_r$ is nonnegative and independent of r . $\square$

8.2 Doubling and vanishing height

Proposition 8.3 (The inward bootstrap). *On all sufficiently small radii, P is bounded above and below. For each fixed $\lambda \in (0, 1)$ there are constants $C_\lambda, c_\lambda > 0$ such that*

$$c_\lambda H(r) \leq H(\lambda r) \leq C_\lambda H(r), \quad \int_{\lambda r}^r q_*(v) \frac{dv}{v} \leq C_\lambda. \quad (8.19)$$

There is a sequence $r_j \downarrow 0$ on which both $n_(r_j)$ and $q_*(r_j)$ are bounded.*

Proof. All quantities are locally absolutely continuous as functions of $\log r$ where differentiated; $H > 0$, and inequalities for derivatives may be read almost everywhere. Put $v_0 = \dot{A}/H$ and $h_0 = \dot{H}/H$ temporarily. The first line of (8.5) with $a = 1$, followed by (8.12), gives

$$|P| \leq C(n_* + r^2) + \eta q_* \leq C(\sqrt{q_*} + r^2) + C\eta q_*.$$

It follows, for a sufficiently large fixed K and $|P| \geq K$, that

$$\frac{q_*}{|P|} \geq c \min\{|P|, \eta^{-1}\}, \quad v_0 \geq cq_*. \quad (8.20)$$

For the first assertion, either $C\sqrt{q_*}$ or $C\eta q_*$ is at least a fixed fraction of $|P|$. The second follows from (8.13), increasing K and then shrinking the radius so that q_* absorbs $2|P|$, n_* , and every error.

If $P \leq -K$, the lower bound $h_0 \geq -m$ gives

$$\dot{P} = v_0 - Ph_0 \geq cq_* + mP > 0.$$

Such an excursion persists as the radius decreases: in inward time $s = -\log r$, $P_s < 0$ throughout this region. There also $A < 0$ and $\dot{A} > 0$, so A decreases inward and stays below a fixed negative number. This contradicts $A \rightarrow 0$, which follows from its factor r^2 and bounded normalized mass. We conclude that $P \geq -K$ on the small-radius interval.

There are arbitrarily small radii at which $P \leq T^2$. Otherwise, eventually $P > T^2$, and (8.20) and (8.3) imply $v_0/P \geq cT^2$. Define

$$Z = \log(r^2/A) = T - \log P.$$

Inward differentiation gives $Z_s = v_0/P - 2 \geq cT^2 - 2$. Since $T \geq 2s - C$, Z becomes positive and unbounded. Also $Z \leq T$ in this regime, whence eventually $Z_s \geq c'Z^2$. Integrating $(1/Z)_s \leq -c'$ forces a blow-up at finite inward time, although A, H are defined and positive throughout every finite subinterval of this regime. This is impossible.

We next prove the differential estimate

$$\dot{P} \geq -C(rP + gP^2) \quad \text{when } K \leq P \leq T^4. \quad (8.21)$$

Combining (8.13) and (8.14) gives

$$\dot{P} \geq 2P - n_* + q_* - 2Pn_* - C\{r\sqrt{q_*} + r^2(1 + n_*) + rP\} - C\eta(1 + P)(n_* + q_*).$$

If $q_* \geq LP^2$ for a sufficiently large fixed L , then $n_* \leq C\sqrt{q_*} + Cr^2$ and $\eta T^4 = o(1)$ show that the right-hand side is positive. Otherwise $n_* \leq CP$, and squaring (8.12) gives

$$q_* \geq n_*^2 - Cr^2P - C\eta P^2.$$

The independent signed second line of (8.5), with $\eta_1 \leq \eta$, gives

$$n_* \geq 2P - d_0, \quad d_0 = C(gP + r^2 + \eta P^2).$$

For large K and small r , this also gives $n_* > 1$. The principal terms factor as

$$2P - n_* + n_*^2 - 2Pn_* = (n_* - 1)(n_* - 2P) \geq -CPd_0.$$

Thus $\dot{P} \geq -C(rP + gP^2 + \eta P^3)$. The last term is absorbed by gP^2 , since $P \leq T^4$ and $\eta T^4 \leq g$ after shrinking the radius. This proves (8.21).

It remains to justify that this restricted estimate gives a bound on every smaller radius. Choose an arbitrarily small moderate radius $r_0 = e^{-s_0}$, so $P_0 \leq T_0^2$. Take s_0 so large that

$$\int_{s_0}^{\infty} C\{e^{-s} + T(s)^{-16}\} ds \leq \varepsilon_0 < 1.$$

This is possible by (8.3). On each component where $P \geq K$, before a possible exit through $P = T^4$, (8.21) yields

$$(\log P)_s \leq C(r + gP) \leq C(r + T^{-16}).$$

Moreover $Z_s > 0$ there by (8.20), increasing K if necessary. If the first component starts at s_0 , these inequalities give

$$P \leq e^{\varepsilon_0} P_0 \leq e^{\varepsilon_0} T_0^2, \quad T = Z + \log P \geq T_0 - \log P_0 + \log K \geq T_0/2.$$

For large T_0 these bounds exclude $P = T^4$. If a component starts later, at s_i , then $P(s_i) = K$, and the same argument gives $P \leq e^{\varepsilon_0} K$ and $T \geq T(s_i)$ there, again excluding the upper exit. This proves the finite uniform bound

$$-K \leq P(r) \leq e^{\varepsilon_0} \max\{K, P_0\} \quad (0 < r \leq r_0).$$

Write this bound as $|P| \leq B_0$. After B_0 is fixed, choose $0 < \bar{r}_0 \leq r_0$ so that $C\eta(1+B_0) \leq 1/4$ for the constants in [Theorem 8.2](#) whenever $0 < r \leq \bar{r}_0$; this absorbs its ηq_* terms after quotient differentiation. For $P \geq 0$ use the upper height derivative bound in $\dot{P} = v_0 - Ph_0$; for $P < 0$ use its lower bound. In both cases [Theorem 8.2](#) and $n_* \leq C\sqrt{q_*} + Cr^2$ imply

$$\dot{P} \geq cq_* - C_{B_0}. \quad (8.22)$$

Integrating over a logarithmic interval of fixed length bounds $\int q_* d\log r$. Cauchy–Schwarz then bounds $\int n_* d\log r$, and integration of (8.14) proves both sides of the height comparison in (8.19). In each interval $[2^{-j-1}\bar{r}_0, 2^{-j}\bar{r}_0]$ choose a radius where q_* is bounded by its logarithmic average. Equation (8.12) bounds n_* at those radii, proving the last assertion. $\square$

Proposition 8.4 (Vanishing height at a typical point). *The positive-height alternative at the origin is impossible. Consequently every point satisfying the typical-point hypotheses (7.1) is regular.*

Proof. Take the radii r_j from [Theorem 8.3](#), and write $r = r_j$. On $\rho < 3r/4$ the weights ϕ, ψ have positive lower bounds depending only on p . Equations (8.4), (8.12), and the boundedness of n_*, q_* give

$$r^{-m} \int_{\rho < 3r/4} |z|^2 d\mu + r^{2-m}(D + \Lambda)(B_{3r/4}) \leq CH(r). \quad (8.23)$$

Here a height integral is over the part of the separated tube above the indicated base ball, whereas D, Λ are projected measures. On the other hand, applying (8.19) with $\lambda = 1/4$ and using $\psi \leq 2p$, $h_r \leq 1$ at radius $r/4$ gives

$$r^{-m} \int_{\rho < r/4} |z|^2 d\mu \geq cH(r) > 0. \quad (8.24)$$

Fix an exceptional fraction $\delta_{\text{cfg}} = 1/4$ and a finite exponent $p_* > 2$ satisfying $m(1/2 - 1/p_*) < 1$. Use the enlargement c fixed from the local data in [Theorem 2.7](#), with the required scalar-selection room, and then fix $d_{\text{cfg},0} = d_0(\delta_{\text{cfg}}) > 0$. Then choose $\epsilon > 0$ so small that $B_{r/4}$ has a finite cover by balls of radius ϵr whose required enlargements lie in $B_{r/2}$. The number of balls and ϵ are fixed independently of r .

Apply [Theorem 2.5](#) with room between $3r/4$ and $r/2$; its integer is again Q . Its variation estimate and (8.23) give, for every enlarged ball $B_{\epsilon r}(x)$ of this cover,

$$(\epsilon r)^{-m} \tau(B_{\epsilon r}(x)) \leq C_\epsilon H(r)/r^2 \longrightarrow 0,$$

uniformly across the cover. The normalized smooth mass averages used in the local scalar selections differ from Q by at most the same bound. Thus, after discarding finitely many selected radii, the scalar smallness and smooth-average conditions select Q on every required local ball, and its displayed budget is below $d_{\text{cfg},0}$. Applying [Theorem 2.7](#) with the fixed choices above now supplies on each covering ball a configuration with the prescribed majority guarantee and a configuration-distance moment whose normalized L^{p_*} norm is at most $C_\epsilon \sqrt{H(r)}$.

Every entry of each such configuration has size at most $C_\epsilon \sqrt{H(r)}$. Indeed, on a fixed majority of its base ball the entries match the actual heights within this error. The area formula, the bounded projection Jacobian, and (8.23) show that a subset of this majority has

sum of squared actual heights at most $C_\epsilon H(r)$. Every configuration entry has positive integer weight, proving the claim. Adding these configuration heights to the distance moment and summing the fixed finite cover yields

$$r^{-m} \int_{\rho < r/4} |z|^{p_*} d\mu \leq CH(r)^{p_*/2}. \quad (8.25)$$

Let $\mathcal{I} = \{t > 0\}$ on the center. By [Theorem 7.3](#), its relative base volume tends to zero and the height vanishes above $S \setminus \mathcal{I}$. The same is true for normalized projected mass, since [Theorem 2.5](#) gives

$$\begin{aligned} r^{-m} \pi_{\#} \mu(\mathcal{I} \cap B_{r/4}) &\leq Q r^{-m} \text{vol}_S(\mathcal{I} \cap B_{r/4}) + r^{-m} |M|(B_{r/4}) \\ &\leq o(1) + CH(r)/r^2 = o(1). \end{aligned} \quad (8.26)$$

Hölder's inequality with (8.25) and (8.26) now gives

$$r^{-m} \int_{\rho < r/4} |z|^2 d\mu \leq CH(r) [r^{-m} \pi_{\#} \mu(\mathcal{I} \cap B_{r/4})]^{1-2/p_*} = o(H(r)),$$

contradicting (8.24). Thus the varifold lies on the center near the origin, and [Theorem 7.5](#) gives regularity. $\square$

Completion of the proof of [Theorem 1.1](#). The weighted rectifiable tangent and density theorem [[19](#), Chapter 3, Definition 1.7, Remark 1.8, and Theorem 1.9], together with differentiation of the integer multiplicity sets, imply that at μ -almost every point x there is a positive integer Q and an approximate tangent plane such that the dilated varifolds tend to the multiplicity- Q plane and

$$\frac{\mu(\{y : \theta(y) \neq Q\} \cap \mathbf{B}_r(x))}{r^m} \longrightarrow 0.$$

The tangent statement includes convergence of the tangent projections; it follows by differentiating the measurable tangent-plane map on a countable rectifiable covering. These are precisely the typical-point hypotheses used in the center construction. No regularity conclusion is used in choosing the point. [Theorem 8.4](#) therefore shows that $\mu(\text{Sing}(V)) = 0$.

For completeness, this implies the requested Hausdorff-measure conclusion even for support points outside the chosen representative of the rectifiable set. Let $N \subset \text{spt } \mu$ be a μ -null set in a compact subset of U , and let O be an open neighborhood of N with arbitrarily small $\mu(O)$. For each $x \in N$ take arbitrarily small balls $\mathbf{B}_r(x) \subset O \cap U$. The support-point monotonicity bound in [Theorem 2.1](#) gives $\mu(\mathbf{B}_r(x)) \geq \omega_m r^m$. The disjoint-ball covering lemma selects pairwise disjoint such balls whose fivefold enlargements cover N . Their radii can all be taken below any prescribed bound, and therefore

$$\mathcal{H}_\epsilon^m(N) \leq C_m \sum_i r_i^m \leq C_m \sum_i \mu(\mathbf{B}_{r_i}(x_i)) \leq C_m \mu(O).$$

Let $\mu(O) \downarrow 0$ and then $\epsilon \downarrow 0$. An exhaustion of U proves $\mathcal{H}^m \ll \mu$ on $\text{spt } \mu$. Applying this to the singular set proves $\mathcal{H}^m(\text{Sing}(V)) = 0$. $\square$

9 The round-sphere consequence

Proof of [Theorem 1.2](#). For $(x, P) \in G_m(S^{m+n})$, define the cone by

$$\int \varphi d\mathcal{C}(V) = \int_{G_m(S^{m+n})} \int_0^\infty \varphi(rx, \mathbb{R}x \oplus P) r^m dr dV(x, P).$$

The polar area formula [19, Chapter 3, §2, (2.4)–(2.6)] gives an integral $(m+1)$ -varifold in $\mathbb{R}^{m+n+1}$ with the same integer multiplicities away from the vertex. The vertex has zero mass, and

$$\|\mathcal{C}(V)\|(B_R(0)) = \frac{R^{m+1}}{m+1} \|V\|(S^{m+n}).$$

The radial and angular first-variation calculation below adapts the hypersurface calculation in [17, Lemma 6.1]; we include its derivation in the present dimensions. For a compactly supported C^1 ambient field X vanishing near the vertex, write $X(rx) = f(r, x)x + Y_r(x)$ with $Y_r(x) \perp x$. Then

$$\operatorname{div}_{\mathbb{R}x \oplus P} X(rx) = \partial_r f + \frac{m}{r} f + \frac{1}{r} \operatorname{div}_P^S Y_r.$$

After multiplication by r^m , the radial terms integrate to the endpoint values of $r^m f$, and the angular term vanishes by spherical stationarity for every r . Thus the cone is stationary off the vertex. For a general compactly supported field, a radial cutoff between radii ε and 2ε has first-variation error $O_{X,V}(\varepsilon^{m+1} + \varepsilon^m)$ by the mass formula. This tends to zero because $m \geq 1$, so the cone is stationary everywhere.

The polar map identifies the punctured support of the cone with $(0, \infty) \times \operatorname{spt} \|V\|$. A regular link patch gives a smooth embedded local cone with the same constant multiplicity, and cone stationarity makes it minimal. Conversely, suppose the cone is locally $q|M|$ near rx , with M smooth embedded and minimal. Its tangent planes contain the radial line almost everywhere, hence everywhere by continuity. The radial vector field is therefore tangent to M , and the radius function is a submersion on M . The dilation flow identifies a smaller patch of M with a radius interval times a smooth embedded spherical section. Comparing the polar varifold formulas on this product neighborhood identifies V there with q times that section, including its tangent planes; spherical stationarity makes the section minimal. Consequently

$$\operatorname{Sing} \mathcal{C}(V) \setminus \{0\} = \{rx : r > 0, x \in \operatorname{Sing}_S V\}.$$

Apply Theorem 1.1 with dimensions $(m+1, n)$ to obtain $\mathcal{H}^{m+1}(\operatorname{Sing} \mathcal{C}(V)) = 0$. Let $A = \operatorname{Sing} \mathcal{C}(V) \cap \{1 < |z| < 2\}$ and $\rho(z) = |z|$. Regularity is local, so A is Borel. The general Hausdorff slicing inequality [19, Chapter 2, §1, Theorem 1.10] applied to ρ gives

$$\int_1^2 \mathcal{H}^m(A \cap \rho^{-1}(r)) \, dr = 0.$$

For $1 < r < 2$, the slice is $r \operatorname{Sing}_S V$. Choosing a radius with zero slice measure and using homogeneity gives $\mathcal{H}^m(\operatorname{Sing}_S V) = 0$ in the chordal metric. The round and chordal metrics are locally bi-Lipschitz equivalent, so they have the same Hausdorff null sets. Homotheties give the assertion for every positive radius. $\square$

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